63 Hemodialysis

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CHAPTER OUTLINE

Hemodialysis (HD) sustains life for more than 1 million patients throughout the world. Without it, most would die within a few weeks.[1,2] This vital aspect of the treatment highlights the need among caregivers to gain an in-depth understanding of all aspects of HD, including its target, the uremic syndrome (see Chapter 52). This chapter reviews the history of dialysis, the epidemiology of the HD patient population, the physical, chemical, and clinical principles of HD as they relate to the treatment of patients with uremia, and the complications associated with this treatment.

HD has been applied routinely to preserve life in patients with end-stage kidney disease (ESKD) only for the past 40 years. Several early pioneers laid the foundation. Graham (1805–1869), a Scottish professor of chemistry, invented the fundamental process of separating solutes in vitro using semipermeable membranes and coined the word "dialysis."[3] In 1916, Abel dialyzed rabbits and dogs with a so-called *vividiffusion* device using celloidin membranes and a leech extract, hirudin, as an anticoagulant.[4] He was the first to dialyze a living organism and to use the term "artificial kidney." In 1924, in Germany, Haas was the first to dialyze a human[5] but was only marginally successful because of toxicity from his crude anticoagulant.

In 1944, Willem Kolff succeeded in using extracorporeal dialysis to support patients with acute kidney failure.[6] His success was partly attributable to the invention of cellophane, the discovery of antibiotics, and the availability of heparin. Kolff was often called the "father of hemodialysis," and his method became the standard for temporary replacement of kidney function in patients with short-lived acute renal failure.[7,8] However, HD could not support patients with prolonged or permanent loss of kidney function because of the difficulty with vascular access, subsequently solved by the creation of the arteriovenous (AV) fistula (see "Vascular Access").

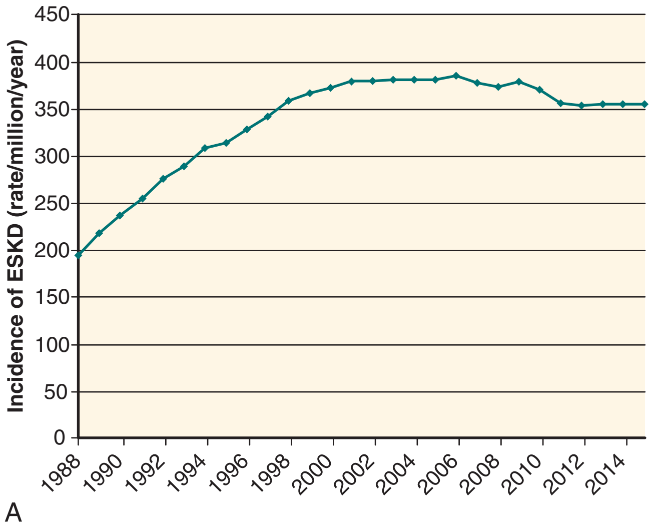
Although it had become technically feasible, HD remained expensive and inefficient and was offered only to those who were free of comorbid conditions, gainfully employed, and better educated. Because dialysis was so successful in preventing death from kidney failure, Congress, after much debate, passed a law in 1973 approving public funding for dialysis and kidney transplantation, regardless of the patient's means, education, employment, or comorbidities.[9] This law paved the way to life-sustaining kidney replacement for all US patients.

THE HEMODIALYSIS POPULATION

INCIDENCE AND PREVALENCE

According to the US Renal Data System (USRDS), 124,114 patients developed ESKD in the United States in 2015, with an unadjusted incidence rate of 378/million population. Of these, 2.5% were transplanted preemptively, 87.7% were started on HD, and 9.6% on peritoneal dialysis.[10] Fig. 63.1A shows that the adjusted incidence rate of ESKD in the United States steadily increased from 1987 to 2002, most likely as a result of aging of the population and increasing acceptance of dialysis for older patients as part of their Medicare entitlement. This leveled off in 2002 and has declined since 2006, with the largest decrease of 3.8% in 2011, and has leveled off again since 2013 (see Fig. 63.1B).

In contrast, both the prevalent count and prevalence rate of ESKD continue to rise in the United States (see Fig. 63.1C).[10] At the end of 2015, 703,243 patients had ESKD, with an adjusted prevalence of 2203/million population. Of these patients, 30% had functioning transplants, and the remainder were maintained by dialysis, with 11.7% of patients undergoing dialysis on home modalities (10% on peritoneal dialysis; 1.7% on home hemodialysis). Although the prevalent number of patients with ESKD continues to grow, the annual growth rate in the prevalent count of 3.3% to 3.6% and in adjusted prevalence of 1.2% to 1.6% since 2011 represent the lowest in 35 years.



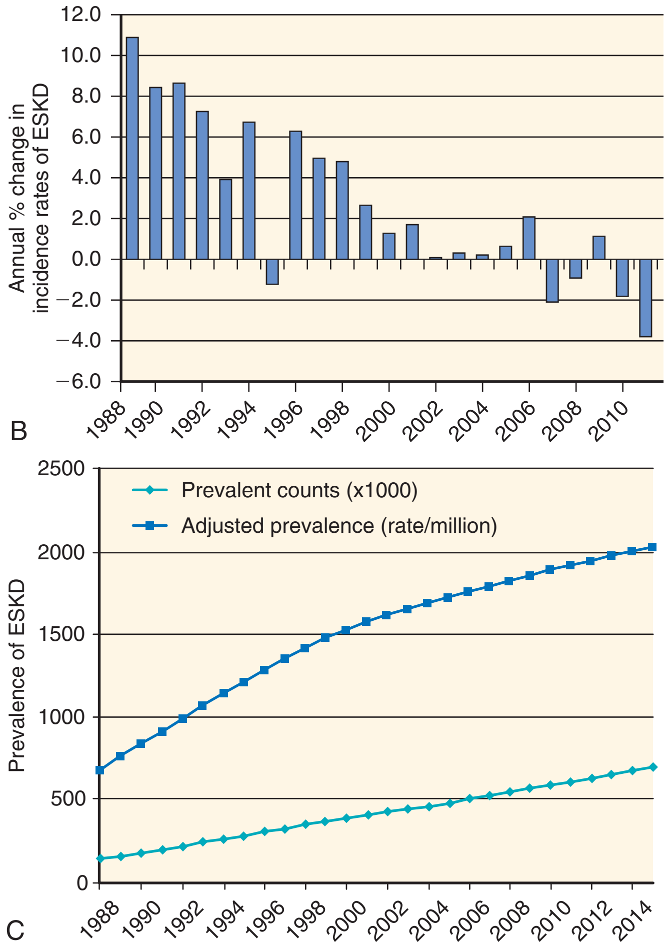


Fig. 63.1 (A) Adjusted incidence rates of end-stage kidney disease (*ESKD*) in the United States with time. (B) Annual percentage change in adjusted incidence rates and prevalence of ESKD with time. Incidence rate of ESKD leveled off in 2002, declined in 2006, and then has leveled off again since 2013. Prevalence of ESKD decreased to their lowest rates from 2011 to 2015. (C) Prevalent counts (×1000) and adjusted prevalence of ESKD in the United States with time. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

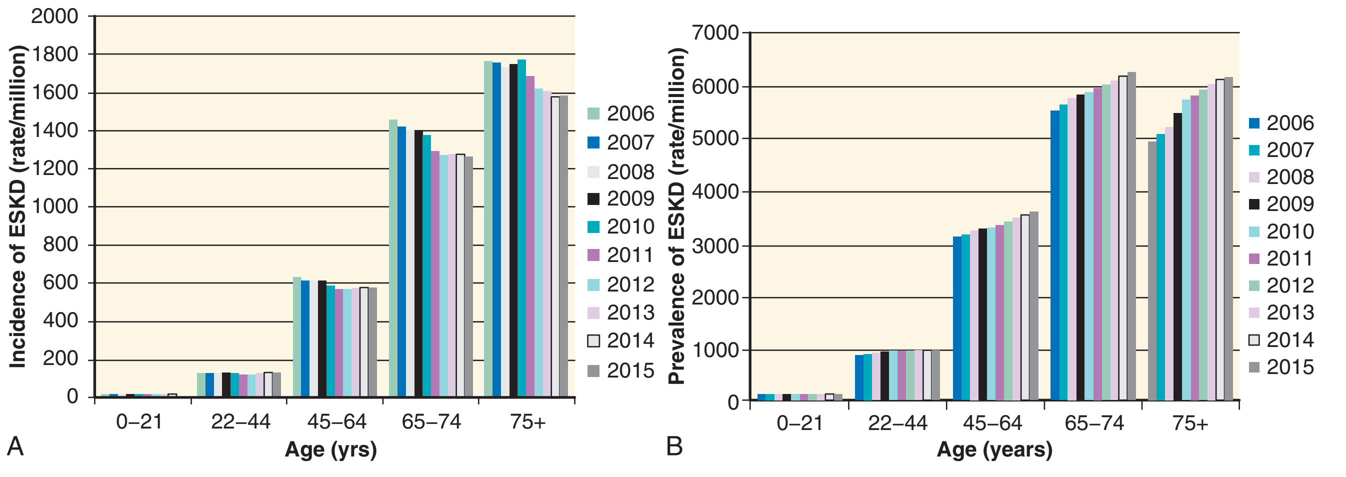


Fig. 63.2. Incidence (A) and prevalence (B) of end-stage kidney disease (rate/million per year) with age from 2006 to 2015. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

Both the prevalence and incidence of ESKD vary widely with age (Fig. 63.2), gender (Fig. 63.3), and race and ethnicity (Fig. 63.4), with a predilection for older age, men, African-Americans, Native Hawaiians, and Pacific Islanders.[10] Over the past few years, the incidence of ESKD has declined in persons older than 45, African-Americans, Hispanics, and American Indians and Alaskan natives, but has continued to rise in Native Hawaiians and Pacific Islanders (see Figs. 63.2A and 63.4A). In contrast, the prevalence of ESKD has continued to rise, suggesting improved survival in those on dialysis.

Worldwide, in 2014, Taiwan, the Jalisco region in Mexico, and the United States had the highest incidence rates for ESKD at 455, 421, and 370 per million population (pmp), respectively, followed by Thailand (299 pmp), Singapore (294 pmp), and Japan (285 pmp).[11] Thailand experienced the highest percentage rise in ESKD incidence rates from 2002 to 2014 at 1009%, followed by Bangladesh (643%), Russia (291%), the Philippines (190%), and Malaysia (162%). In contrast, the ESKD incidence rates decreased in Austria, Denmark, Sweden, Scotland, Finland, and Iceland by 3% to 14%. The prevalence rate for ESKD continues to rise in all countries, with the highest rates in Taiwan at 3219 pmp (2014 data), followed by Japan (2505 pmp) and the United States (2076 pmp). The median increase in prevalence rates from 2002 to 2014 was 48%, ranging from 18% to 1092%, with the largest proportionate increase in prevalence rates in the Philippines, Thailand, and the Jalisco region of Mexico (343%–1092%). However, the worldwide prevalence of ESKD varies greatly, with the lowest rates reported in the Philippines, Russia, South Africa, Indonesia, and Bangladesh at less than 300 pmp. The broad ranges of reported ESKD prevalence likely result from variable access to treatment and to preventive and maintenance health care.

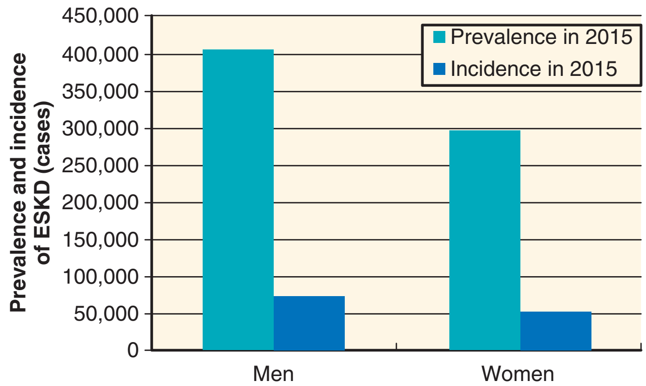


Fig. 63.3. Incidence and prevalence of end-stage kidney *(ESKD)* disease with gender. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

CAUSES OF END-STAGE KIDNEY DISEASE

The causes of ESKD in the United States are listed in Table 63.1. Since 1980, the percentage of patients with diabetic kidney disease starting dialysis has increased from near 0% to 45% of patients starting dialysis in 2015 (Fig. 63.5), primarily because of increased acceptance of patients with diabetes into dialysis programs, but with a potential contribution from improved survival of these patients so that they live long enough to develop ESKD. Although the incidence rate of ESKD from diabetes in the United States has declined since 2006 (172.9 pmp in 2006 compared with 162.7 pmp in 2015; see Fig. 63.5), diabetes remains the most common cause of ESKD in the United States and in many other countries,[11] exceeding 40% in Israel, the Republic of Korea, Hong Kong, Taiwan, the Philippines, Japan, the United States, New Zealand, Thailand, Chile, and Kuwait, and reaching as high

Table 63.1 Causes of End-Stage Kidney Disease in Incident and Prevalent Patients[a]

Primary Kidney Disease	Incident Patients No.	Incident Patients % of Total	Prevalent Patients No.	Prevalent Patients % of Total
Diabetes mellitus	56,218	45.4	267,956	38.2
Hypertension	34,727	28.1	178,875	25.5
Glomerulonephritis	9198	7.4	112,235	16.0
Cystic kidney disease	2833	2.3	33,194	4.7
Other	20,856	16.8	109,092	15.6
Total	123,832	100	701,352	100

[a]In the United States in 2015.

Data from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.

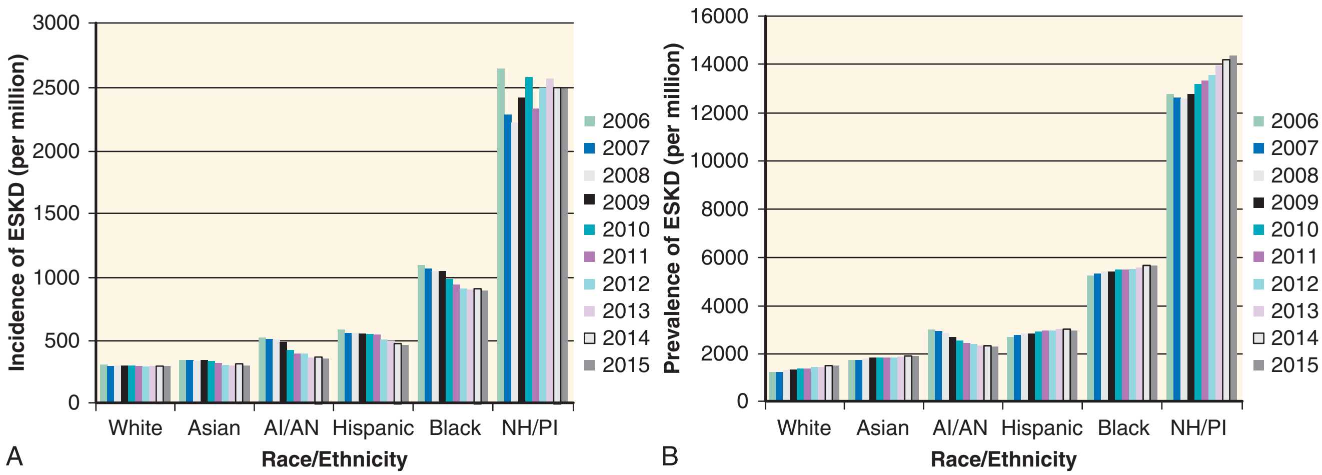


Fig. 63.4. Incidence (A) and prevalence (B) of end-stage kidney disease *(ESKD)* with race and ethnicity. *AI/AN,* American Indian or Alaska Native; *Black,* black or African-American; *NH/PI,* Native Hawaiian or Pacific Islander. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

as 58% to 66% in Singapore, Malaysia, and the Jalisco region of Mexico. Remarkably, diabetes accounted for less than 20% of new ESKD cases in Italy, Switzerland, the Netherlands, Belgium, Estonia, Norway, Romania, and Iceland.

MORTALITY

The survival of patients undergoing dialysis in the United States has slowly improved in the past 20 years, despite increasing comorbidities, with the largest gains in the last 10 years. The adjusted mortality rate for prevalent hemodialysis patients has decreased by 29% from 1996 to 2014, with a 4% reduction from 1996 to 2003 and 24% from 2004 to 2014 (Fig. 63.6).[11] Although 5-year survival rates have increased from 30% in the 1998 cohort to 42% in the 2010 cohort (Fig. 63.7), mortality remains very high. Patients starting hemodialysis in 2010 have a 78% 1-year survival, 66% 2-year survival, and 42% 5-year survival (see Fig. 63.7).[11] Compared with age-matched persons without kidney disease, patients undergoing dialysis have a markedly reduced life expectancy (Fig. 63.8A).[10] At age 60 years, a healthy person can expect to live for about 20 years, but the median life expectancy of a 60-year-old patient starting HD is only 4 to 5 years (see Fig. 63.8A). Remarkably, mortality rates of patients with ESKD undergoing dialysis is higher than that of patients with AIDS, most forms of cancer, and heart failure (see Fig. 63.8B). Patients older than 65 years were faring better in 2014 than in 1996, with a 39% decline in adjusted mortality rate, compared with a 37% drop in mortality rate for cancer and 21% for heart failure.[11]

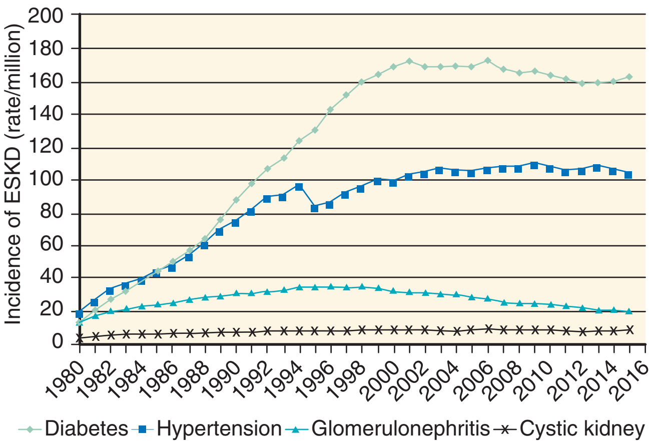
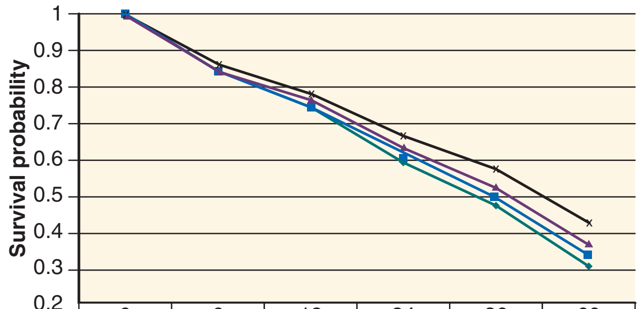


Fig. 63.5 Causes of end-stage kidney disease *(ESKD)* in the United States. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)



	0 months	6 months	12 months	24 months	36 months	60 months
1998	1	0.84	0.74	0.59	0.47	0.30
2002	1	0.84	0.74	0.60	0.49	0.33
2006	1	0.84	0.76	0.63	0.52	0.36
2010	1	0.86	0.78	0.66	0.57	0.42

Fig. 63.7 Probability of survival for hemodialysis patients. The likelihood for survival of hemodialysis patients has improved slightly over the past 10 years. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

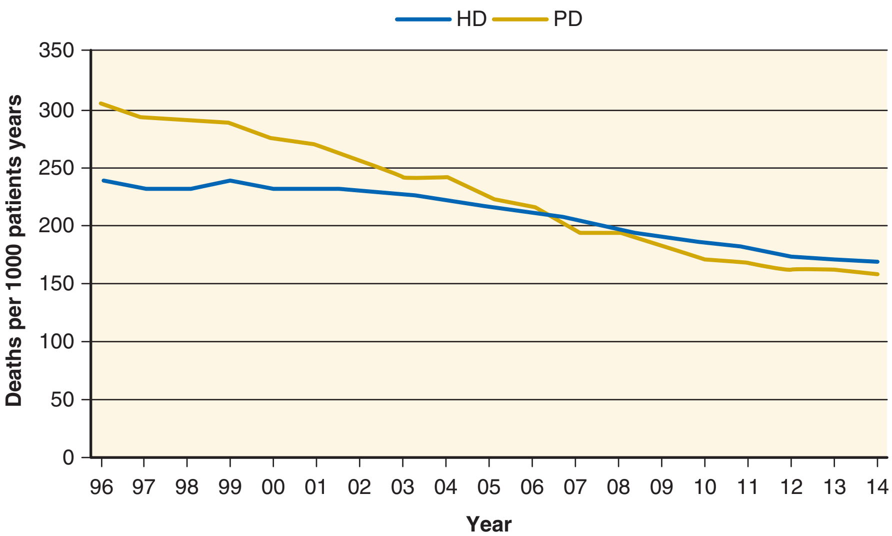


Fig. 63.6 Mortality rates for patients on hemodialysis. Mortality rates for patients on hemodialysis (HD) as well as peritoneal dialysis (PD) have declined steadily. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

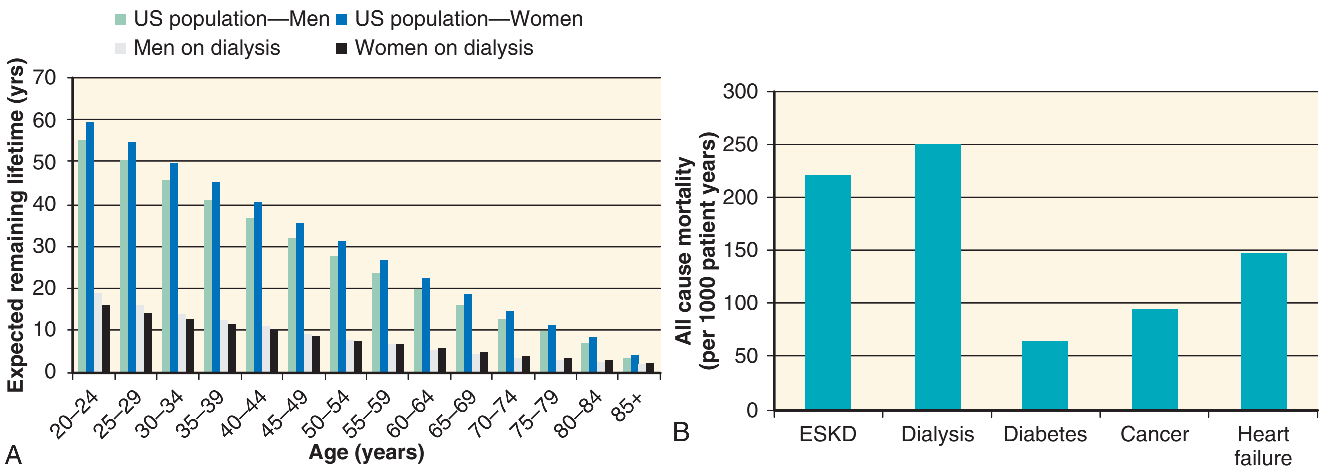


Fig. 63.8. Life expectancy and adjusted all-cause mortality rates for end-stage kidney disease *(ESKD)* patients in 2014. (A) Patients with ESKD or on dialysis have a markedly reduced life expectancy compared with the general population. (B) Their mortality rate is even higher than that of patients with diabetes, cancer, and heart failure. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

Table 63.2 Causes of Death for Prevalent Hemodialysis Patients (2013)

Cause of Death	% of Total
Cardiovascular disease	41%
• Arrhythmia, cardiac arrest	29%
• Myocardial infarction	4%
• Other	8%
Infection	8%
Withdrawal from dialysis	13%
Other causes	14%
Missing causes	24%

Data from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.

Causes of death are listed in Table 63.2. Deaths from cardiovascular disease were 41%, with 29% from arrhythmia or cardiac arrest and only 4% from myocardial infarction.[10] The cause(s) for the high burden of cardiovascular disease in patients undergoing dialysis remain a source of debate, but it appears that the dialysis process itself may play a significant role through the long interval between dialysis over the weekend, excessive rates of ultrafiltration during dialysis, and aggressive correction of electrolyte and acid-base abnormalities during dialysis[12] (see Chapter 54 and "Dialysate Composition," later). Infections accounted for another 8% of deaths.[10,13] Overall, voluntary withdrawal from dialysis accounted for another 13% of deaths, a 3% increase compared with 2009 to 2011 data. Patients who withdraw from dialysis tend to be older, women, and white and have higher rates of medical events and higher levels of morbidity in the period preceding withdrawal.[14] In addition, a nonuniform approach to advanced care planning in patients with multiple comorbidities and poorer quality of life and insufficient exploration of goals of care in this population before embarking on maintenance dialysis may contribute toward the rising incidence of voluntary withdrawal.[15,16]

TRANSITION FROM CHRONIC KIDNEY DISEASE, STAGE 5

The high mortality of patients undergoing HD results in part from many associated comorbid conditions, including cardiovascular disease, diabetes, and hypertension. These comorbid conditions extract a toll on patients with chronic kidney disease (CKD) before they develop ESKD and require dialysis. The relative risk of death is two to three times higher in patients with CKD, diabetes, and cardiovascular disease than in patients without these conditions; patients with CKD stage 4 or 5 experience the highest mortality risk.[11] A patient with stage 4 or 5 CKD who has an acute myocardial infarction (MI) has a 2-year mortality rate of 47% compared with 20% in the absence of CKD. Reducing or improving comorbidities, then, is critical in caring for patients with CKD (see Chapter 59).

In addition, close attention to replacing hormone deficiencies such as erythropoietin (see Chapter 55) and calcitriol (see Chapter 53) and preventing complications of uremia such as malnutrition (see Chapter 60) and CKD–mineral bone disorder (Chapter 53) are essential. Another critical element in preparing patients with CKD for ESKD care is preemptive planning for potential transplantation as well as for a permanent vascular access to support hemodialysis (see later), beginning when the estimated glomerular filtration rate (eGFR) has declined to 20 to 25 mL/min/1.73 m^2.

Psychological support is another important but often overlooked and poorly understood part of caring for a patient on hemodialysis. Depression, disabling fatigue, severe restless legs syndrome, insomnia, anxiety, and prolonged recovery time after dialysis are common in patients undergoing HD.[17–33] Their presence is associated with poorer quality of life, malnutrition, poor quality of sleep, low level of physical function, presence of inflammation, hospitalization, and death.[17,19,20,22,25,26,29,30,34–42]

Despite the profound impact of these factors on the quality of life and outcome of patients on dialysis, little is known about their true epidemiology, pathogenesis, and effective treatment. Diagnosing depression may be difficult in patients on dialysis because some of the symptoms used to make the diagnosis may be present due to uremia.[17–19] Once the diagnosis is made, patients and their providers may be resistant to treatment because of increased pill burden, potential toxicity of medications, and lack of mental health resources.[43] Some studies have suggested that certain selective serotonin reuptake inhibitors (SSRIs), intensive hemodialysis, cognitive-behavioral therapy, and may be effective in reducing depressive symptoms, anxiety, sleep disturbances, postdialysis recovery time, and quality of life, but others debate the effectiveness of SSRIs and intensive dialysis, likely because many studies are small and/or uncontrolled.[17,31,35,44–58] Dopamine agonists, gabapentin, and exercise may ameliorate restless legs syndrome.[26,42,59,60] Larger and more rigorous studies are needed and are underway.[45,61]

Because of the multifaceted care required before starting HD, timely referral to a nephrologist is essential. Studies have documented higher hospitalization rates,[62,63] more symptoms of depression,[64] worse anemia and biochemical disturbances (e.g., hypoalbuminemia, hyperphosphatemia, hypocalcemia, acidosis, hyperkalemia),[62,63,65–67] lesser use of peritoneal dialysis,[63] higher use of dialysis catheters,[63,68] and higher mortality[62,63,67–73] when nephrology referral is delayed, but other studies have found no difference in mortality risk and in ESKD risk.[71,74–76] Potential confounders may be the reasons underlying late referral such as patient nonadherence, multiple comorbidities, advanced age, and rapid decline in kidney function.[67,77,78] The widely varying definitions of early referral, ranging from more than 1 to 6 months before starting dialysis to CKD stage 3 (eGFR <60 mL/min/1.73 m^2), may have also contributed to the disparate findings. Studies of patients with nondialysis-requiring CKD followed by nephrologists have suggested that death occurs more frequently than ESKD in patients with CKD 3a (eGFR 45–60 mL/min), whereas the opposite is true in patients with CKD 5 (eGFR <15 mL/min).[63,67,68,71,73–76,79,80]

The competing risks of death or ESKD reach parity at CKD stage 3b (eGFR 30–45 mL/min/1.73 m^2) to CKD stage 4 (eGFR 15–30 mL/min/1.73 m^2).[67,79–81] In patients managed only by primary care physicians, the mortality rate for patients with CKD stage 3a is comparable to those with CKD stages 1 to 2 and rises progressively with more advanced stages of CKD.[81] Referral to nephrologists at CKD stage 3a or 3b was not associated with a change in the risk for death and/or progression to ESKD,[75,76] whereas referral at CKD stage 4 was associated with a lower mortality risk but not with risk of progression to ESKD.[67] It is unclear why nephrology care was not associated with the risk for ESKD, but these findings have suggested that the most appropriate early referral to nephrology for established CKD may be at stage 3b or 4.

Current clinical practice guidelines have suggested that dialysis be initiated when patients become symptomatic from uremia, often occurring at an eGFR between 5 and 10 mL/min/1.73 m^2.[82] In some patients, volume overload and hyperkalemia not responsive to conservative medical management with diuretics and dietary potassium restriction and/or unexplained progressive decline in nutritional status, despite aggressive dietary intervention, may dictate earlier initiation of dialysis. The goal for patients with kidney failure is a smooth transition from CKD to ESKD, avoiding the complications of overt uremia. The results of several studies have supported these clinical practice guidelines.[83–86] The IDEAL (Initiating Dialysis Early and Late) study,[83] the only randomized controlled trial, has reported comparable survival and clinical outcomes in patients randomized to early start (eGFR >10 mL/min/1.73 m^2 by the Cockcroft-Gault equation) versus late start of dialysis (eGFR = 5–7 mL/min/1.73 m^2). However, 75% of the patients randomized to a late start of dialysis initiated dialysis when the eGFR was above 7 mL/min/1.73 m^2 because of intervening symptoms attributed to uremia, volume overload, or nutritional decline. A potential weakness of the IDEAL study was use of the Cockcroft-Gault equation to estimate the GFR. However, secondary analysis of the trial data using the MDRD (Modification of Diet in Renal Disease) and the CKD-EPI (Chronic Kidney Disease–Epidemiology) formula to estimate GFR also demonstrated no significant effect of timing of dialysis initiation on survival.[87] Earlier initiation of dialysis is not cost-effective[88] and may be harmful,[89–91] possibly because of an accelerated loss of residual kidney function, more use of dialysis catheters, myocardial stunning, depression, and provider inexperience at a time when lifesaving benefits from dialysis are low.[91,92] Current guidelines have suggested a patient-centered approach in determining when to initiate dialysis, balancing the prevention of uremic complications with quality of life and patient preferences.[82,93–97]

VASCULAR ACCESS

Establishing and maintaining a reliable and trouble-free vascular access, which is often seen as the weakest and most troublesome aspect of hemodialysis, is paramount for the successful long-term application of HD. Vascular access planning, placement and trouble shooting require a multidisciplinary team approach, which traditionally has included surgeons and interventional radiologists, in addition to the nephrologist. Interventional nephrology, a procedurally oriented subspecialty in nephrology, has been increasing its presence in many health markets. The field of interventional nephrology has allowed nephrologists to intervene in dysfunctional HD vascular accesses directly (see Chapter 68). Nonetheless, a large proportion of patients with advanced CKD begin HD with a catheter. Therefore, much work is still needed to optimize vascular access in both incident and prevalent patients.

BACKGROUND

Innovative techniques in vascular surgery have allowed the widespread use of maintenance HD as a viable approach to the management of ESKD. Although external constant flow arteriovenous (AV) devices, such as the Quinton-Scribner shunt, have led to more reliable and repeatable access to the circulation, thrombosis and infection have limited the longevity of these devices, and the procedure permanently damaged major arteries and veins.[98] In 1966, Brescia and colleagues[99] developed the procedure to create an endogenous

AV fistula which allowed for long-term HD. This procedure was relatively simple, was totally subcutaneous, and preserved arterial flow to the hand.

The advent of procedures using AV grafts allowed patients who were not candidates for a fistula to receive a durable subcutaneous access.[100,101] The 1970s brought the development of vascular conduits made of expanded polytetrafluoroethylene (ePTFE) and biologically processed bovine carotid arteries, which greatly expanded AV access for patients whose natural veins were not suitable for an AV fistula. In recent years, the use of polyurethane self-sealing materials in AV grafts, as well as a hybrid graft with a catheter venous outflow device, have expanded vascular access options for patients with suboptimal native vasculature.[102,103]

The arm is clearly the preferred site for fistulas or grafts; however, in some patients, the leg becomes an alternative when necessary.[104] It is nearly always preferred to use the most distal vein feasible for an AV shunt to preserve more proximal vessels for future HD accesses. Nephrologists should champion a "preserve the vein" program in patients with advanced CKD who will likely require HD in the future. Such a program includes the avoidance of a peripherally inserted central catheter (PICC line) in patients who may later require HD. PICC lines are often placed from the antecubital vein through the cephalic or basilic vein and into the central venous system. Used to facilitate medication infusions or blood draws, these catheters are associated with a high rate of venous stenosis,[105] and loss of these veins essentially prohibits that arm from serving as a site (or sites) for future HD vascular access.

Historically, the use of synthetic graft material does not provide the same long-term success as a Brescia-Cimino or other autogenous AV fistula with respect to duration of function and incidence of complications (see later). However, recent studies using more contemporary patient cohorts have indicated that the gap in key clinical outcomes between AV fistulas and AV grafts may not be as wide as previously described.[106,107] Rather than a fistula-first approach for all patients, nephrologists should call for a catheter-last approach to HD access.

Procedures that allowed the safe placement of wide-bore catheters in large-diameter veins were developed in the 1970s. First, single-lumen and then double-lumen catheters led to the use of these devices to support patients on maintenance HD who lacked AV access. The creation of more flexible plastics and the advent of cuffed catheters inserted through subcutaneous tunnels have led to the use of so-called permanent catheters, offering longevity of use that make them practical for maintenance HD. In the United States, the availability of these catheters has allowed patients to undergo maintenance HD, despite inadequate vasculature to support endogenous AV access. However, there is growing concern that these catheters are used for longer periods of time than necessary and, in some cases, in lieu of AV fistulas or grafts. Around 80% of patients in the United States start dialysis with a catheter access and, even among patients with CKD seen by a nephrologist prior to the development of ESKD, a large proportion of patients begin with a catheter.[10] By comparison, other countries where HD serves most patients have much lower incident catheter rates.[108] However, there will always be patients undergoing HD in whom a catheter will be the appropriate or only feasible long-term vascular access.

Characteristics of an ideal HD vascular access and how the present types of accesses compare with each other are listed in Table 63.3. The fact that no single type of vascular access fulfills all these desired requirements is what makes a reliable vascular access the Achilles' heel of long-term HD.[109] There are well-recognized associations among type of vascular access, mortality, and cost of care in ESKD.[110] The 2010 Annual Report of the US Renal Data Service (USRDS) has outlined the annual direct vascular access event costs per patient by access type (Fig. 63.9A), as well as the total annual health care expenditures per patient per year based on the type of HD vascular access (see Fig. 63.9B). Although the annual direct vascular access–related costs of an AV graft are higher than those for both catheter and fistula, the total annual health care expense is highest for a patient with a catheter.

TYPES OF VASCULAR ACCESS

ARTERIOVENOUS FISTULAS

This preferred type of vascular access is created by connecting a vein to an artery, which requires that the two vessels be in proximity to each other. An ideal mature (ready to use)

Table 63.3 Characteristics of Ideal Hemodialysis Vascular Access Compared With Commonly Available Types

Desired Characteristic	Autogenous AV Fistula	AV Graft	Central Venous Catheter
High primary patency rate	★	★★	★★★
Instant usability	★	★★	★★★
Long survival	★★★	★★	★
Low thrombosis rate	★★★	★★	★
Low infection rate	★★★	★★	★
High blood flow rate on hemodialysis	★★★	★★★	★
Patient comfort	★	★	★★★
Patient bathing/hygiene	★★★	★★★	★
Minimize needles	★	★	★★★
Minimal cosmetic affect	★	★★	★★

AV, Arteriovenous.

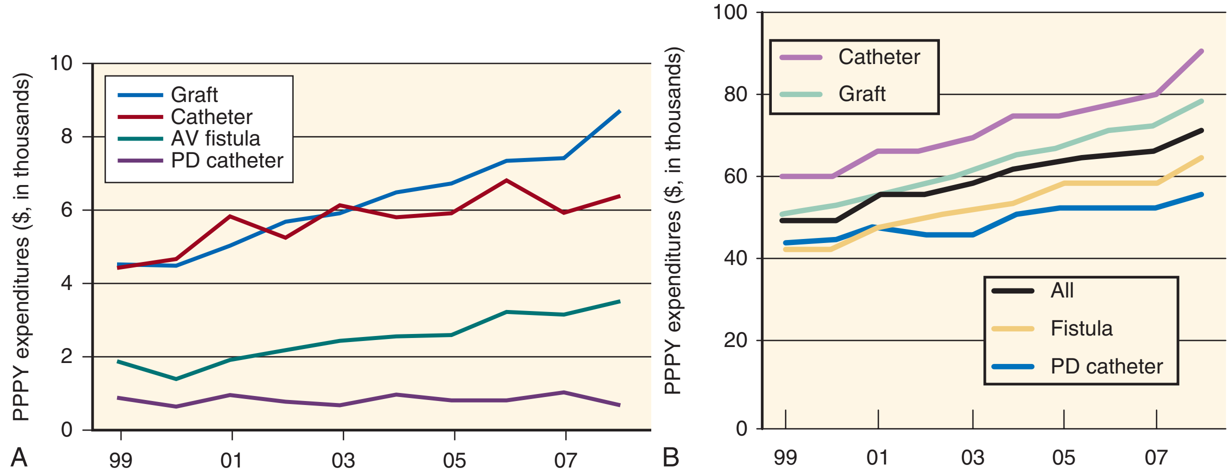


Fig. 63.9 Health cost of varying types of vascular access. (A) Direct expenditures of access events per person per year *(PPPY)* varies with the type of hemodialysis (HD) vascular access. The graft is most expensive to maintain, whereas the fistula is the least expensive form of vascular access. (B) Total health care expenditures PPPY vary with the type of HD vascular access. Patients with a catheter are the costliest, whereas the patient with a fistula is the least expensive. *AV,* Arteriovenous; *PD,* peritoneal dialysis. (Modified from US Renal Data System. 2017 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

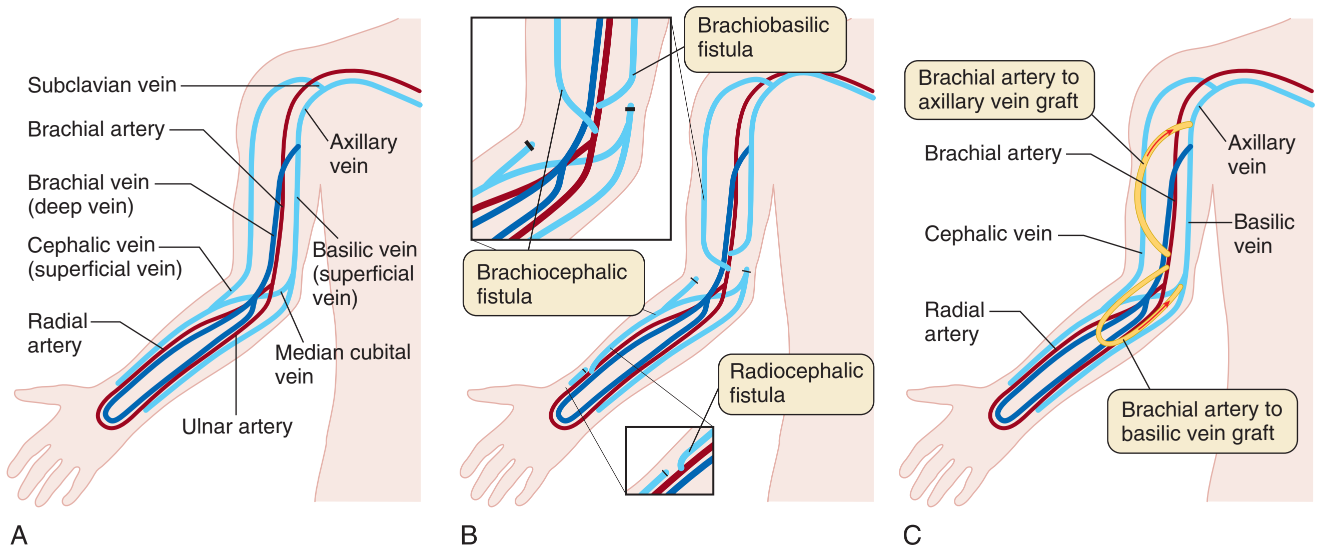


Fig. 63.10 (A) Vasculature of the right upper limb outlining superficial and deep veins and major arteries. (B) Location of anatamoses for typical wrist (radiocephalic) and upper arm (brachiocephalic and brachiobasilic) arteriovenous (AV) fistulae. (C) Location of typical AV grafts.

fistula should be easily accessible by physical examination and provide a robust blood flow of at least 400 mL/min, with enough distance between the cannulation of two needles to allow for dialysis machine blood pump flow rates of 400 mL/min without developing recirculation. The "rule of 6" serves as an easy way to remember physical examination guidelines to determine if an AV fistula is ready for use: (1) at least 6 mm in diameter; (2) at least 6 cm of overall needle-accessible length; and (3) no more than 6 mm below the skin surface.

At this point in time, fistula creation is a surgical procedure that can often be done with just local or regional block anesthesia. In general, both the artery and vein must have adequate lumens for the procedure to be successful. The original procedure described by Brescia and colleagues[99] was a side to side AV fistula using the radial artery and the cephalic vein at the wrist and has become the preferred access when these vessels are adequate. Recent studies in upper arm AV fistulas have suggested similar patency rates of side to side versus end to side anastomoses, but with a higher incidence of arterial steal phenomenon with the side to side technique.[111] An additional advantage of the end to side procedure is the avoidance of venous hypertension that can occur distally with the side to side approach. Data supporting the benefit of a fistula have created some concern that perhaps radiocephalic fistulas are being attempted in patients with known higher failure rates, such as women and persons with diabetes and obesity.[112] Typical vascular anatomy of the arm and the location of commonly placed fistulas are shown in Fig. 63.10A and B.

Systematic use of preoperative ultrasonographic imaging to examine vascular anatomy is often used in the preoperative evaluation to decide on the type and location of fistula placement and may increase the success rate of AV fistulas.[113,114] Commonly and inappropriately termed "vein mapping," this study should interrogate veins and arteries. Commonly used lower limits for artery and vein internal diameters are 2 mm and 2 to 3 mm, respectively. However, even in situations where blood vessel size appears to be appropriate, maturation of the AV fistula—the increase in vein blood flow rate, size, and wall strength that allows it to be used for HD—may still not occur. There is now better understanding that issues such as intimal hyperplasia not readily seen by imaging techniques or arterial lesions that limit augmentation of flow after fistula creation are probably additional factors in the unsuccessful AV fistula.[115,116] Surgical technique is also a well-recognized variable in AV fistula outcomes.[117]

The upper arm veins provide alternatives when a distal fistula has failed or when distal vessels are inadequate. With incident patients now older and with more extensive comorbidities, more first-time AV fistulas are being placed using the upper arm veins when vein mapping reveals small lower arm wrist vessels. The cephalic and basilic veins are considered superficial veins of the arm and lend themselves to fistula creation. The basilic vein courses up the arm and perforates the deep fascia to join the brachial vein somewhere between the midbiceps and the axilla. The basilic vein fistula often uses the brachial artery as its inflow and typically requires an additional procedure of transposition because of its location along the medial side of the biceps, making it an awkward location for cannulation in its native position. The brachiobasilic fistula can be performed in stages, which is particularly advantageous in children, with anastomosis first followed by the transposition surgery after the vein has matured.[118] Even in adults, this procedure appears to have advantages over an AV graft in terms of longevity.[119] The basilic vein has also been transposed to the forearm, with successful outcomes.[120] Alternatives to using the cephalic or basilic vein as outflow for an AV fistula are the brachial vein and median antecubital vein. The former is anastomosed to the brachial artery and requires a second procedure to superficialize and transpose, whereas the latter (variants of the Gracz fistula) has potential dual outflow through both the cephalic and basilic veins.[121] These AV fistulas have purported advantages and disadvantages that are beyond the scope of this chapter.

Focus on the superiority of fistulas versus other forms of vascular access appears to have successfully increased the prevalence of fistulas in the United States. In Europe, the prevalence of fistulas for maintenance HD has long been greater than in the United States. Data from the Dialysis Outcomes and Practice Patterns Study (DOPPS) have demonstrated a marked difference in the training of surgeons in Europe compared with those in the United States.[122] The much greater volume of procedures completed by European surgeons in training may lead to more expertise or comfort in performing fistula procedures.

The dogmatic placement of fistulas in all patients, however, has its disadvantages. A recent meta-analysis of over 12,000 patients has found that the failure rate for fistulas may be increasing, casting concern about overly aggressive "fistula-first" approaches.[123] The major disadvantages of the AV fistula are the high failure to mature rate—around 25%—and the need for surgical or percutaneous balloon angioplasty to assist in maturation in an additional 25% of patients.[115,124] These high incidences and the tradition of waiting months for the AV fistula to mature prolong the duration of and need for catheters. Although some data have shown that early cannulation of the AV fistula is associated with higher failure rates,[125] lack of change on physical examination of the fistula within the first 4 to 6 weeks after surgery may also be prognostic of nonmaturation and dictate sooner interventions.[126]

ARTERIOVENOUS GRAFTS

Vascular access using ePTFE conduits had become the most prominent type of AV vascular access in the United States in the 1980s and 1990s, at least partly due to the ease of placement, the short time required between the placement of an AV graft and initiation of cannulation, and possibly factors related to reimbursement. Placement of an AV graft is perceived as advantageous by many nephrologists and surgeons because it does not require maturation of the native vein and allows for a forearm site for access placement (see Fig. 63.10C). Although the recognized low primary failure rate for grafts is an early advantage, the primary patency rate at 12 months is only around 50%.[127] The graft material has a higher risk for development of infection compared with the native vessel of an AV fistula. Other approaches to placement are available when vascular anatomy or previous access prohibits placement in the forearm. The thigh often becomes the only remaining site for graft placement but poses a higher risk of infection; however, an observational report has suggested that a thigh graft may be a better alternative than a tunneled catheter.[128] Regardless of location, most ePTFE grafts require about 4 weeks of time after placement before cannulation is recommended. This allows time for the graft material to scar or incorporate into the surrounding soft tissue and for endotheliazation of the intraluminal surface.

More recent alternative graft materials include polyurethane, which has a self-sealing characteristic at the puncture site when the dialysis needle is removed. This property obviates the need for graft incorporation into the surrounding soft tissue, allowing for early cannulation after its surgical placement, and plays a role in limiting or avoiding a dialysis catheter. Although seemingly ideal, the clinical experience with these types of grafts has been mixed, with some reports of increased infections, although some retrospective series have found noninferior results to traditional grafts.[129,130]

A new hybrid device for patients with limited venous outflow options uses a typical graft at the point of the arterial anastomosis, transitions to a large-bore, single-lumen catheter at the venous outflow end, and is tunneled up the arm and placed into the central venous circulation, much like a typical catheter, to achieve central venous drainage. This Hemodialysis Reliable Outflow (HeRO) device (Merit Medical, South Jordan, UT) has a potential role in some patients as an AV access of last resort.[131]

A complex biologic response develops within the lumen of the graft and in the distal native vein after an AV graft is surgically placed. Possibly stimulated by areas of turbulent blood flow or changes in wall shear stress, a host of immune responses occur. The resulting hyperplasia leads to a high incidence of stenosis, which is the leading cause of graft failure.[132] The vast majority of stenoses that develop are at the venous anastomosis or beyond, within the native outflow vein. As discussed later, the high incidence of stenosis certainly is a rationale for monitoring graft flow in an effort to allow preemptive intervention before thrombosis. The documented inflammatory response and hyperplasia in grafts have focused attention on devices and drugs that may prevent or reduce the eventual stenosis. In addition to angioplasty, stents are frequently used as a tool to prevent restenosis and maintain patency. Whether drug-eluting stents will have a role in AV graft survival has yet to be determined. Use of drug elution from the graft material itself is also under investigation in animals and may offer promise in the future.[133] One study has redefined the concept of grafts by constructing totally biologic grafts, using artificial materials as the initial skeleton for a conduit over which autologous fibroblasts are grown. Then the lumen is seeded with autologous endothelial cells.[134] This technology may eventually offer biologic grafts that are less vulnerable to thrombosis and infection than conventional ePTFE grafts.

An access plan that is often overlooked, but has potential of combining the merits of the graft and fistula, is the creation of a secondary AV fistula after graft placement. In this scenario, an AV graft is placed to minimize or avoid catheter time. However, when the patient encounters the first episode of graft dysfunction, the outflow vein is evaluated for conversion to a traditional AV fistula. Theoretically, the outflow vein will already have arterialized due to its exposure to high flow and pressure from the prior graft and possibly allow for cannulation at the time of, or soon after, the creation of the new surgical anastomosis to the artery.[135]

Central Venous Catheters

Technology has greatly advanced the use of external connections for HD, providing HD with venous access alone and eliminating the need to sacrifice an artery, as required with the use of AV shunts. Initially, large single-lumen catheters supported HD through single veins. Later, improved plastic materials allowed the construction of large double-lumen catheters that greatly improved HD efficiency over single-lumen catheters. Further advances resulted in catheters made of softer materials that are also more resistant to degradation by cleansing solutions such as alcohols. Most importantly, the development of the subcutaneous cuff provided catheters that could be used for markedly longer durations. Clearly, this technology has been a blessing and a curse because the ease of insertion and duration of use have resulted in the cuffed catheter's being a viable vascular access for patients on maintenance HD.

Some patients would face death from kidney failure without the availability of this catheter technology. However, the data have suggested an inappropriately high utilization of catheters as a means of vascular access. Observational studies such as those using data from DOPPS have suggested that the high prevalence of catheter use in the United States may explain a large fraction of the higher US mortality in the HD population compared with similar patients in Europe.[136] However, other recent, well-done studies have suggested that much of the mortality risk is related to noncatheter patient factors.[137] Regardless, the elevated mortality risk and frequent complications with catheters create major challenges for nephrologists.[138]

Thrombosis with loss of catheter function is a frequent complication leading to the need for intervention. No uniform solution to prevent these events has been established. Catheters are routinely locked, with the installation of high-dose heparin solutions injected into both lumens, but this procedure does not completely prevent the problem. The use of heparin for locking may occasionally result in bleeding through an error in dose, failure or inability to remove the heparin solution before use, and/or leakage of heparin between dialysis treatments. Even the addition of systemic therapy with low-dose warfarin did not prevent thrombosis in a clinical trial setting and led to an increase in complications.[139] Small studies have explored alternatives to heparin, such as locking catheters with citrate-containing solutions.[140] One such study has demonstrated that locking the catheter with recombinant tissue plasminogen activating factor once a week and heparin the other 2 days compared with heparin 3 days a week reduces both thrombosis and the incidence of bacteremia.[141,142] However, the cost of implementing this strategy would be prohibitive in the United States.

Infection isolated to the catheter itself or systemic infection with the catheter as a source is a substantial complication of catheter use resulting in morbidity, costly hospitalizations, and mortality. The frequency of bacteremia associated with HD catheters has been estimated to be two to four episodes/1000 patient-catheter days,[143] a frequency 10- to 20-fold higher than rates of infection estimated in patients with AV fistulas. These catheter-related infections can result in more complex infections such as osteomyelitis, endocarditis, or septic arthritis, despite antibiotic therapy.

Careful management of catheter-related infection is critical, including an adequate duration of antibiotic treatment, an aggressive approach to catheter removal and replacement, and delayed replacement until the patient is symptom free (so-called "line holiday").[144–146] When the infection is isolated to the exit site or does not involve the tunnel, studies have suggested that catheter exchange can be successfully undertaken. Attempting to treat infected catheters with antibiotics alone usually results in failure, although some organisms (e.g., *Staphylococcus epidermidis*) may clear. The combined use of antibiotic installation into catheter lock solutions, in addition to systemic antibiotics, may increase success in preserving the existing catheter.

Several approaches to the prevention of infection with catheters have demonstrated potential promise. The use of mupirocin ointment at the exit site may reduce the incidence of infection.[147,148] Locking catheters with antibiotic-containing solutions as a routine after all HD treatments may also be helpful in reducing the incidence of catheter-related infections.[146,148–150] With the prophylactic use of antibiotics to minimize bacterial catheter infections, the issue of antibiotic-resistant bacterial selection is always a concern. The use of catheters impregnated with antibacterial agents so far has not been as promising in preventing infection.[151]

Even if the thrombotic and infectious complications could be solved, catheters may not support the blood flow rates needed to achieve an adequate HD dose in some patients. Additionally, long-term catheter use results in vascular damage that often leads to central vein stenosis and in situ thrombosis.[152] In the end, catheters must remain an interim solution or the access of last resort in the patient on maintenance HD.

MAINTENANCE OF VASCULAR ACCESS FUNCTION

After some form of long-term vascular access is achieved, maintaining patency and adequate function is paramount. Numerous strategies have been explored to accomplish this goal. Cannulation methods have received some attention, with speculation that methods such as the buttonhole technique might provide advantages over other methods. One nonrandomized study has compared the rope ladder technique (70 patients), in which cannulation occurs along the entire length of the fistula, with the buttonhole procedure (75 patients), where repeated cannulation occurs in the exact spot over 9 months.[153] Unsuccessful cannulation occurred more frequently in the buttonhole group, but there were fewer complications, including hematoma and aneurysm formation, compared with the rope ladder group. More recent data have suggested that the buttonhole technique is associated with a higher rate of infection, without evidence of improved fistula survival.[154] At present, the buttonhole technique has fallen out of favor due to its higher rate of bacteremia, but is still an important option for patients doing their own cannulation for home HD. Guidelines for needle insertion were created through the National Kidney Foundation's effort to develop management standards, but these guidelines did not include definitive recommendations regarding this question of cannulation methodology.[82] However, clinical experience has suggested that good cannulation technique, such as using the entire length of the AV fistula or graft, and careful post-HD needle site care are both important for access longevity.

Other strategies to extend access life span include methods of AV graft and fistula monitoring for early signs of potential failure, treatments to prevent access dysfunction or loss, and interventions to fix problems found during monitoring or when shunt thrombosis occurs. There is an accepted nomenclature for access failure that categorizes access as having primary or secondary patency. The definition of primary access patency is an access capable of supporting adequate HD without any intervention since the time of original placement. After an intervention has occurred for any reason (e.g., reduced flow, total thrombosis), secondary patency is then invoked to characterize the status of the vascular access.

MONITORING AND SURVEILLANCE

The natural history of vascular access loss has suggested that observable and physiologic changes in access venous pressure or total blood flow occur before and predict impending loss of function. If detectable change in either pressure or flow were observed reliably, an intervention to prevent emergent loss of access function would be possible theoretically. The question is whether salvaging the access before total loss of function occurs offers any advantage.

A consistently applied program of AV shunt monitoring is recommended by most dialysis guidelines. Although there are many techniques and devices available to accomplish this, a good physical examination of the AV shunt remains the most important monitoring technique.[155] This can be taught to dialysis technicians, nurses, and nephrologists. For example, if an AV fistula has a bounding pulsation with increasing aneurysm size or does not flatten when the fistula arm is raised above the head, then a venous outflow or central venous stenosis should be suspected.

Other hints to access dysfunction may come from information gathered from the monthly blood tests, as well as through feedback from the dialysis staff. For example, if a decline in HD urea clearance cannot be readily explained, a poor access flow rate may be the root cause. Also, if the dialysis nurse or technician is experiencing difficulty in cannulation, or the staff reports prolonged hemostasis time (or even bleeding from the AV shunt at home reported by the patient), arterial inflow or venous outflow stenosis may be the respective culprits.

Currently, vascular access shunt pressure can be monitored by measuring static or dynamic venous pressures using several standardized methods. To detect stenosis adequately, these procedures must be done serially using consistent techniques; the change or trend in measurements over time is more important in predicting a stenosis than an absolute threshold value. Shunt blood flow measurement that uses indictor flow detection by ultrasound (sometimes termed the "ultrasound dilution method using the Transonic device" [Transonic, Ithaca, NY]) can be done routinely and easily. Regardless of method, monitoring of HD access should be championed by a member of the dialysis clinic and be part of continuous quality improvement.

Despite the theoretic advantage of access monitoring, the impact of aggressive surveillance programs on maintenance of vascular access has been controversial.[156] In lending support to regular monitoring, a nonrandomized report comparing three monitoring schemes in a single dialysis program has demonstrated significant improvement in short-term AV graft and fistula survival using blood flow monitoring versus dynamic pressure monitoring or no monitoring.[157] In addition to access survival, there were fewer missed HD sessions and reduced overall patient costs when using the blood flow monitoring protocol. The overall goal of surveillance protocols, however, is to prolong the function of an AV access. The unfortunate natural history of access failure, especially with AV grafts, is the occurrence of restenosis after intervention procedures such as angioplasty (venoplasty). There may be no difference in eventual access survival whether preemptive intervention is practiced versus simply responding to a thrombosis event when it occurs. A review of surveillance trials did not definitively support routine monitoring of HD access.[158] Therefore, access surveillance is a topic of ongoing research.

Whether the additional intervention of stent placement improves access survival has received a great deal of attention. A randomized trial of 190 HD patients has provided evidence that use of a stent is superior to balloon dilation alone with stenosis at the venous end of an AV graft.[159] New materials for manufacturing stents are under investigation, and the place for drug-eluting stents, although well studied in other vascular systems, needs to be addressed in regard to AV access.[156] The subspecialty of interventional nephrology has emerged in the field of nephrology as a result of the need for timely treatment of vascular access dysfunction and prevention of associated complications. Supported by the creation of the American Society of Diagnostic and Interventional Nephrology, reports have documented the quality and safety of the care delivered by these physicians.[160,161] Interventional nephrologists have been reporting experience with AV access, as well as additional experience with related arterial disease, such as subclavian and renal artery stenosis and peripheral vascular disease.[162] No longer just responsible for monitoring the vascular access, suitably trained nephrologists have tools at their disposal to intervene on behalf of their patients. See Chapter 68 for a further discussion of interventional nephrology.

PROPHYLACTIC THERAPY FOR ACCESS MAINTENANCE

Access loss occurs from the combined pathologic events of thrombosis and hyperplasia leading to stenosis. Strategies to prevent these pathologic processes have been used with the goal of prolonging access survival, so far with no clear prevention regimen established. Thrombosis may be the final pathway to access failure but, so far, agents well known to prevent thrombosis in other vascular diseases have not consistently enhanced vascular access survival. A multicenter trial in the Veterans' Affairs Heath System with clopidogrel plus aspirin had to be discontinued because of bleeding problems in the treatment group.[163] The Dialysis Access Consortium (DAC) Study Group has reported that although clopidogrel reduces the early thrombosis rate of new AV fistulas, it does not increase the number of usable fistulas.[164] Some small randomized trials have suggested that agents such as fish oil may be effective in reducing thrombosis in grafts, but this approach has yet to receive further study or widespread use.[165,166] The DAC graft study has provided support for the use of dipyridamole and low-dose aspirin started immediately after placement of an AV graft,[167] although the practice has not been widely adopted.

The myointimal proliferation and resulting stenosis that occur, particularly with AV grafts, have also been a focus of prevention, especially because this process may be the ultimate final pathway of access failure.[168] Strategies to prevent cell proliferation in AV grafts have been adapted from studies of other vascular diseases, such as cardiac disease. For example, a retrospective study using angiotensin-converting enzyme (ACE) inhibitors in patients with AV grafts has suggested a reduced risk for graft loss compared with patients not receiving this class of medication.[169] Interventions that have been shown to reduce proliferation in other vascular diseases are being studied. Brachytherapy has been used for some time and has been shown to be safe and have promise in reducing stenosis in AV grafts.[170] Far-infrared light therapy, which improves endothelial function, was shown in a randomized trial to improve patency rates of new AV fistulas when administered repeatedly[171] and may become a future tool to improve AV graft survival.[172] Experimental grafts created by biologic autologous materials, as already described, may eventually provide one solution for access maintenance.

Finally, personalized medicine may eventually be integrated into vascular access protocols as genetic studies uncover possible predictors of risk for access loss. One such study has suggested that polymorphism in the methylenetetrahydrofolate reductase gene *(C677T)* may be a predictor of access thrombosis, presumably through its effect on homocysteine.[173] Another study has suggested that the complex relationship between transforming growth factor beta 1 (TGF-β1) and polymorphisms in the plasminogen activator inhibitor type 1 gene may predict patients who are at risk for access thrombosis.[174] Preknowledge of genetic determinants may allow tailored prophylactic therapy to be administered in the future to patients at risk.

VASCULAR ACCESS TIMING AND DECISION MAKING

In an ideal world, discussion, planning, and placement of HD vascular access would be done with a nephrologist in a structured and orderly manner. Unfortunately, the reality is that 36% of patients starting HD in the United States have little or no pre-ESKD nephrology care.[10] Even in those individuals whose care is managed by a nephrologist, the decline in kidney function in CKD is rarely a linear predictable process and is often characterized by abrupt changes.[175] Therefore, even the best laid-out plans for timely HD access may be foiled, with the patient starting dialysis with a catheter. On the other hand, an unwavering adherence to dialysis guidelines, which suggest consideration of HD vascular access placement when the eGFR falls below 20 mL/min/1.73 m^2, would certainly result in unnecessary surgery and procedures in some patients who will never need or want dialysis.

Although registry data and retrospective studies using large databases have consistently found AV fistulas associated with best outcomes, the right access for the individual patient is much more than simply placing an AV fistula in all patients. The decision is complex, because clinical outcomes and cost analysis must take into account the morbidity and mortality of each access type, success rate of the access, and cost of maintaining patency, as well as the cost of alternative access types. A detailed model incorporating many of these variables was recently published and provides insight into the decision-making process that must also take into account patient factors such as age, gender, comorbidities, and competing risk of death.[176] Recent studies examining older adults who have started dialysis also found that a single-minded, fistula-first approach to HD vascular access may not be optimal.[177]

GENERAL PRINCIPLES OF HEMODIALYSIS: PHYSIOLOGY AND BIOMECHANICS

Therapeutic HD removes solutes principally by diffusion and, to a lesser extent, by convection across a semipermeable membrane. The driving force for solute diffusion is the transmembrane concentration gradient (Fig. 63.11A), and the driving force for convection, commonly called "ultrafiltration," is the transmembrane hydrostatic pressure (see Fig. 63.11B). Selective removal of solutes is achieved by restricting the pore size in the membrane to admit small molecules and reject large molecules and by including desirable solutes in the dialysate, effectively preventing their removal (and often prompting their influx).

NATIVE KIDNEY VERSUS ARTIFICIAL KIDNEY

The native kidney's glomerular filter separates solutes based on molecular size, a function that the dialyzer attempts to mimic. Most of the soluble large molecules found in the blood are the products of complex intracellular synthetic processes that require energy. Most continue to be active, serving to signal and regulate processes in distant organs. Their loss by filtration through the kidneys would be a liability best avoided. Loss is

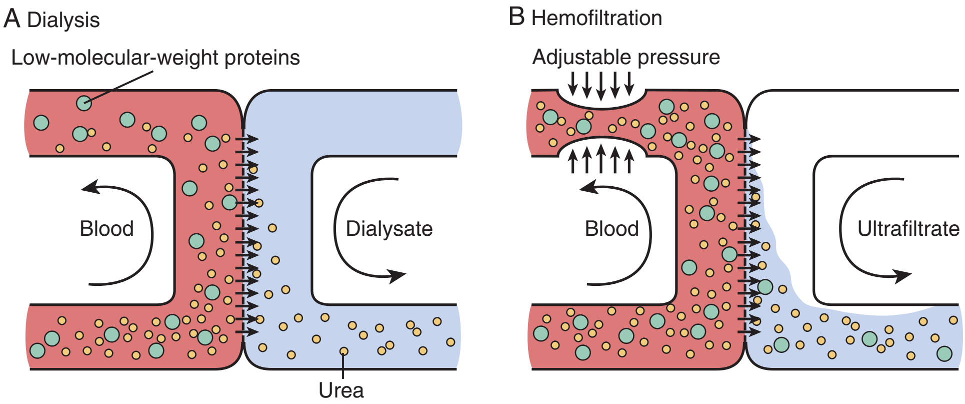


Fig. 63.11 Diffusion versus convection. (A) Diffusion across a semipermeable membrane. The driving force for solute diffusion is the transmembrane concentration gradient. Small solutes with higher concentrations in the blood compartment, such as potassium, urea, and small uremic toxins, diffuse through the membrane into the dialysate compartment. Dialysis dissipates this concentration gradient (i.e., the molecular concentration gradient decreases with dialysis). Larger solutes and low-molecular-weight proteins such as albumin diffuse poorly across the semipermeable membrane. (B) Convection across a semipermeable membrane. The driving force for convection, commonly called ultrafiltration, is the transmembrane hydrostatic pressure. When applied to the blood compartment, solvent flows across the membrane into the dialysate compartment, bringing along solutes. For solutes with a sieving coefficient close to 1, there is no change in concentrations in the blood compartment with time. (From Meyer TW, Hostetter TH. Uremia. *N Engl J Med.* 2007;357(13):1316–1325.)

first prevented by the bilipid cell membrane barrier that keeps precious molecules sequestered intracellularly. Those that are secreted or leak out of the cell are often bound to serum macromolecules, most notably serum albumin, a well-known transport protein. Although the glomerular membrane is highly permeable when compared with cell membranes in general, albumin and its bound ligands, as well as other macromolecules, are poorly filtered and thus protected from loss. Small proteins and peptides that leak through the filter are efficiently reabsorbed by the proximal tubule where they are broken down and their subunits reused.

Smaller molecules are often end products of metabolism or ingested intruders that the kidney effectively eliminates by filtration without reabsorption. Precious small molecules are reclaimed after filtration by selective reabsorptive mechanisms in the renal tubules. Dialyzers lack the latter vital functions, so losses are prevented by including the measurable, desirable small solutes in the dialysate, effectively eliminating the gradient for diffusive loss. Fortunately, most of these small solutes are abundant and relatively inexpensively added to the dialysate.

Although both the native and artificial kidney are excretory organs, and both use semipermeable membranes to separate small from large particles, each operates on a different principle. The natural kidney is a selective filtration or convection device driven by blood pressure generated by the heart, with highly selective reabsorption and secretion downstream. In contrast, the dialyzer separates molecular species primarily by simple diffusion, without the need for pressure generation or reabsorption. Urea, for example, is highly reabsorbed after filtration by the glomerulus, but selective reabsorption plays no role in hemodialyzers. The rapid equilibration of urea across red blood cell (RBC) membranes facilitates urea removal during dialysis but plays no role during glomerular filtration. In contrast, creatinine and most other water-soluble compounds are poorly reabsorbed by the native kidney and exhibit slow or no transport across RBC membranes within the short time frame that blood transits through the dialyzer. Consequently, clearance rates of creatinine by the native kidneys are higher, and by the dialyzer lower, than their respective urea clearances. The artificial hollow fiber kidney contains 8000 to 10,000 fibers, each with an internal diameter of approximately 200 μm and length of 250 mm, providing a surface area for exchange of about 1.5 m^2. Each of the 1 to 2 million functioning nephrons in the two native kidneys has a proximal tubule diameter of about 40 μm and a length of about 14 mm, providing a minimum surface area for proximal reabsorption of approximately 3 m^2 (ignoring microvilli).

Another major difference between the native kidney and artificial filter is active tubular excretion. Just as the native kidneys reabsorb desired substances, they also actively secrete waste products that are filtered poorly by the glomerulus because of significant protein binding. For example, the native kidney clearance of hippurate, which is 56% protein-bound, in HD patients with residual kidney function is 6.6 times that of native urea clearance, suggesting active tubular secretion of the substance.[178] Other metabolic functions of the native kidney, many of which are performed by the tubular and interstitial cells, are also not found in the artificial kidney. Known native kidney synthetic functions include the synthesis of erythropoietin and the 1-hydroxylation of 25-hydroxy vitamin D.

For endogenous solutes, the first pathway on the route to elimination by native or artificial kidneys is diffusion through intracellular and extracellular pathways, including passive or facilitated diffusion across membranes. Thus, diffusion is a vital transport mechanism for native kidney and artificial kidneys. Because removal of small solutes appears to be the major function of both excretory methods, dialyzer clearance of small solutes can be compared with similar clearances by the native kidney as a reasonable first step toward assessing hemodialyzer adequacy.

CLEARANCE

The goals of dialysis are straightforward—removal of accumulated fluid and retained solutes, some of which are toxic. With respect to toxins, the ultimate goal is to maintain concentrations below the threshold at which uremic symptoms and signs begin to appear. However, the levels of retained toxic solutes are not used as performance measures for dialysis because their identities are unknown, and their generation rates probably vary from patient to patient and from time to time (see later). Instead, performance of dialysis is judged by the clearance of representative solutes.

As already noted, most targeted solutes are small, so the clearance of a small representative solute can be used to measure the most important dialyzer function, which is to lower the concentration of small toxic solutes in the patient. This is an inescapable conclusion based on the observation that dialysis works extremely well to reverse life-threatening uremia rapidly. The mechanism for this lifesaving property of dialysis is not mysterious; it is simply the result of solute removal by diffusion across the semipermeable dialysis membrane. Earlier cellulose-based dialyzer membranes removed solutes with molecular weights above 3000 Da very poorly, yet were effective in reversing uremia, so small solutes are the obvious primary culprits accounting for the life-threatening aspects of the uremic syndrome. It is therefore reasonable to use the clearance of a representative small solute as a measure of this fundamental dialysis function.

Clearance is recognized as the best measure of first-order processes such as diffusion and filtration. A zero-order process such as urea generation by the liver is not influenced by the solute concentration, but first-order removal processes, such as diffusion, are driven by the concentration, rendering the removal rate directly proportional to the concentration. Clearance (K) is the proportionality constant, as shown in this equation:

$$K = \text{removal rate}/\text{concentration} \qquad [1]$$

K has value as an expression of a first-order process that is independent of the solute removal rate or its concentration. For intermittent dialysis, the main advantage of the clearance expression is that it tends to remain constant, despite rapid changes in both the solute concentration and removal rate during the procedure.

In a simple flowing system, the removal rate is the difference between the inflow concentration (C_{in}) and the outflow concentration (C_{out}) multiplied by flow (Q). From the first equation, clearance can also be expressed as the extraction ratio (E) multiplied by flow:

$$E = (C_{in} - C_{out})/C_{in} \qquad [2]$$

$$K = Q \times E \qquad [3]$$

For a constant-flow system, the extraction ratio is also constant over time, despite marked changes in concentration. E is the fraction of total inflow (Q), and K is the absolute flow that is completely cleared of the solute; both tend to remain constant during dialysis. Clearance is affected by the flow of both blood and dialysate, as well as other variables, such as the convective filtration rate (see later), but is independent of concentration.

Although clearance is independent of solute concentration, the converse is not true (i.e., concentrations depend on clearance, and solute concentrations are used to measure clearance). During a period of steady-state kinetics, in which generation equals removal, the concentration of a solute is inversely proportional to its clearance if its generation rate is fixed. Because dialysis is simpler than native kidney function and removes solutes primarily by diffusion, the calculation of clearance is nearly the same for all easily dialyzed substances if one assumes that these solutes are distributed in a single mixed pool in the patient. Application of this principle allows selection of an easily measured solute (e.g., urea) to assess dialyzer performance, with the expectation that the measured clearance will correlate inversely with patient levels of similar, easily dialyzed solutes. The measured solute need not be toxic, but its removal must occur in parallel with removal of the toxic solutes. Generation rates of various solutes differ, but if each is relatively constant from week to week (e.g., creatinine generation), then the measured clearance of a representative solute can be used to reflect the effectiveness of the dialysis for clearing all easily dialyzed solutes. This principle, which forms the basis of established standards for measuring and prescribing HD, has logical merit, but its applicability has been challenged and may require modification for solutes that are strongly sequestered in remote body compartments (see later).[82,179–181] In addition, after adjustment for body size and possibly sex (see later), all patients appear to require the same weekly clearance. Furthermore, the dose requirement or need for dialysis does not seem to vary from time to time in the same (anuric) patient, provided a minimum threshold clearance is delivered during each treatment.

Native kidneys appear to clear small solutes at a rate far above the minimum required to sustain life. For example, removal of a kidney for transplantation can be done without major adverse consequences in the donor. Although the reason for this surplus function is unknown, it helps explain how modern intermittent HD, which achieves a continuous equivalent small solute clearance that is only about 10% to 15% of the normal GFR of normal kidneys, is able to sustain life (see discussion of continuous equivalent clearance).

CLEARANCE VERSUS REMOVAL RATE

The clearance of a solute must be distinguished from its absolute removal rate. Clearance is best envisioned as a measure of removal expressed as a fraction of the remaining solute and is therefore independent of the concentration. Two substances may have the same clearance, but if one is present at half the concentration of the other, the removal rate will also be half of the other's. In practical terms, it is impossible to compare dialyzers by measuring removal rates alone because removal depends on the solute concentration. Measurement of clearance eliminates this requirement, allowing use of a single term to make valid comparisons among purgative instruments.

Similarly, finding a lower concentration of a solute in a patient does not indicate that the clearance is higher; it may simply reflect a lower generation rate. If a steady state exists where input equals output, the removal rate of a substance is simply a measure of its generation rate, revealing little about the effectiveness of the dialyzer. If the clearance decreases, and the generation rate does not change, the patient's solute concentration will increase until a new steady state is reached, when the removal rate again matches the generation rate.

SERUM UREA CONCENTRATION VERSUS UREA CLEARANCE

The serum urea concentration has proven to be a poor surrogate for uremic toxicity. The determinants of urea

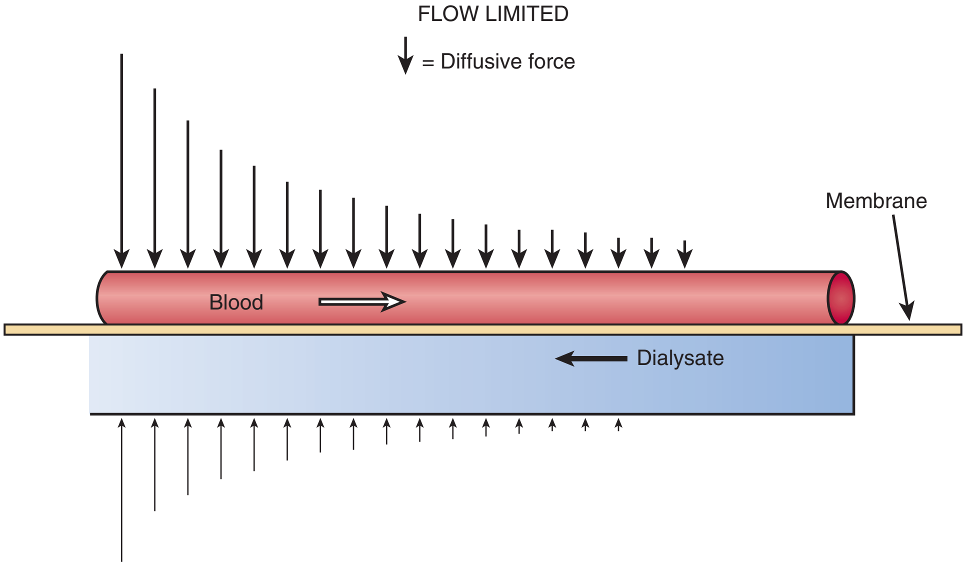


Fig. 63.12 Flow-limited clearance. A logarithmic decline in solute concentrations, indicated by the *arrows* on either side of the membrane, is depicted from the dialyzer inlet at the *left* to the blood outlet on the *right*. This predictable decline is attributable to relatively rapid diffusion of solute across the membrane and forms the basis for Eq. 5. Flux is equal to the product of membrane-solute K_0A and the log mean gradient. Solute flux and removal are maximized by countercurrent flow of blood and dialysate.

concentration are generation and clearance. Whereas urea clearance by the dialyzer should correlate with the clearance of other small (dialyzable) solutes that are presumably responsible for uremic toxicity, the generation of urea as an end product of protein catabolism correlates poorly with uremic toxicity, despite cell culture and animal studies that have suggested that urea disrupts cellular function and promotes metabolic disturbances.[182] In fact, patients with higher urea generation rates have better outcomes, probably as a reflection of better appetite and higher protein intake.[183] It is difficult to dissect the clearance factor from the generation factor in a single blood urea nitrogen (BUN) measurement but, as explained later, this can be differentiated by modeling the change in BUN during a dialysis treatment. For purposes of measuring the dose of dialysis and dialysis adequacy, only the relative change in urea concentration during dialysis is used to model clearance; the absolute concentrations are ignored. Thus, despite its lack of intrinsic toxicity and the poor correlation with overall uremic toxicity, urea measurements during dialysis can be used to assess dialysis effectiveness and adequacy. The change in urea concentration during dialysis, which reflects its clearance, is used as a surrogate for the clearance of other small, easily dialyzed solutes, some of which must be toxic, or conventional diffusive dialysis would not reverse the life-threatening component of uremia. This logic justifies the use of urea clearance as an index of dialysis adequacy while acknowledging that isolated urea concentrations cannot be used for this purpose.

FACTORS THAT AFFECT CLEARANCE IN A FLOWING SYSTEM

Diffusive clearance in a flowing system depends on the rates of blood and dialysate flow, as well as the targeted solute membrane permeability. The biomaterials used to make the hollow fiber dialyzer, together with the pore size and thickness of the membrane, determine its clearance, or the membrane permeability constant (K_0), for a given solute. Multiplying K_0 by the surface area for diffusion (A) yields the permeability, or mass transfer area coefficient (K_0A), of an entire dialyzer. K_0A is expressed in milliliters per minute and is independent of solute concentration, similar to clearance. The predictable exponential decline in concentration gradient along the dialyzer membrane from blood inflow to outflow (Fig. 63.12) is the basis for the calculation of K_0A from the blood and dialysate flow rates and is widely used in mathematical models of solute kinetics. For countercurrent dialysate and blood flow,[184] where K_d is dialyzer clearance, the following equation is applicable:

$$K_0A = (Q_bQ_d/[Q_b - Q_d])\ln(Q_d[Q_b - K_d]/Q_b[Q_d - K_d]) \quad [4]$$

Analogous to clearance, which expresses the dialyzer removal rate normalized to the inflow solute concentration, K_0A is an expression of dialyzer performance normalized to blood (Q_b) and dialysate flow rates (Q_d). K_0A is sometimes called the "intrinsic clearance" of a dialyzer and can be viewed as the maximum clearance possible for a particular solute and dialyzer at infinite Q_b and Q_d. Note that K_0A is both dialyzer- and solute-specific. It is the best parameter for comparing dialyzers, with higher values indicating more efficient solute removal.

A useful rearrangement of this equation provides a measure of clearance at any blood and dialysate flow rate, as shown in the following:

$$K_d = Q_b\left[\frac{e^{K_0A\left(\frac{Q_d - Q_b}{Q_dQ_b}\right)} - 1}{e^{K_0A\left(\frac{Q_d - Q_b}{Q_dQ_b}\right)} - \frac{Q_b}{Q_d}}\right] \quad [5]$$

The preceding expression of clearance does not include the contribution of convective solute removal by ultrafiltration (Q_f) during therapeutic HD. Convective clearance results from bulk movement of solute across the membrane driven by hydrostatic pressure. Simultaneous filtration across the same membrane used for dialysis removes additional solute, but the amount removed is inversely related to the efficiency of the dialysis. For example, if the dialyzer removes solute very efficiently by diffusion, with an extraction ratio approaching 100%, addition of ultrafiltration adds very little or nothing to the removal rate, which cannot exceed 100% of the inflow. The effect of ultrafiltration on clearance is expressed as follows[185]:

$$K_d = Q_b([C_{in} + C_{out}]/C_{in}) + Q_f(C_{out}/C_{in}) \quad [6]$$

where Q_b is the blood inflow rate, C_{in} is the inflow concentration, C_{out} is the outflow concentration, and Q_f is the ultrafiltration rate, in millimeters per minute. As C_{out} approaches zero, the dialysis component of clearance maximizes, and the Q_f component extinguishes.

DIALYSANCE

For peritoneal dialysis (PD), after the infusion of fresh dialysate, solute begins to accumulate in the peritoneal fluid. As the level increases, the concentration gradient from blood to dialysate decreases, which causes the removal rate and clearance to decrease, both eventually falling to zero as equilibrium is reached. However, the flux of solute per unit of concentration gradient, also known as "dialysance" (D), remains constant:

$$D = (\text{removal rate})/(\text{concentration gradient}) \quad [7]$$

Dialysance can be formulated as the initial clearance when the dialysate concentration is zero or, in a flowing HD system, as the solute removal rate per mean solute gradient along the membrane (blood minus dialysate concentration), and is equivalent to the dialyzer's K_0A for the measured solute.

DETERMINANTS OF CLEARANCE

A number of variables affect clearance during dialysis (Table 63.4). Solute-related variables include the physical and chemical properties of the substance to be removed (e.g., size, charge, protein binding) and its distribution in the body (e.g., intracellular, extracellular, interstitial). Treatment-related variables include the permeability of the membrane to solutes of various sizes, dialysis treatment time, membrane surface area, and flow rates of blood and dialysate (see preceding equations).

Molecular size and membrane permeability together limit the rate of movement for individual molecular species. In flowing systems, the concentrations of larger molecules tend to remain constant along the length of the dialyzer, uninfluenced by blood and dialysate flow. Their clearances, which are low because of their large size, are limited by the size and permeability of the membrane alone and independent of flow rates (Fig. 63.13). Smaller molecules tend to be cleared at the proximal end of the dialyzer, leaving the more distal end for further enhancement of clearance as flow is increased (see Fig. 63.12). In this situation, clearance is said to be flow-limited. Note that flow-limited clearance for small solutes and membrane-limited clearance for larger solutes may occur at the same time in a dialyzer. The relation between flow and clearance is shown graphically in Fig. 63.14 for both limiting scenarios.

The molecular activity of a solute determines its capacity for movement across the dialysis membrane. Because water-soluble solutes are active only in the water phase of the blood, only the water component (~90% of normal blood

Table 63.4 Factors Influencing Effective Clearance

	Treatment-Related (in Order of Importance)	
Solute-Related	**Small Molecules**	**Large Molecules**
Molecular size	Blood and dialysate flow	Membrane permeability
Molecular charge	Membrane surface area	Treatment time
Macromolecular binding	Treatment time	Membrane surface area
Body distribution and sequestration	Membrane permeability	Blood and dialysate flow

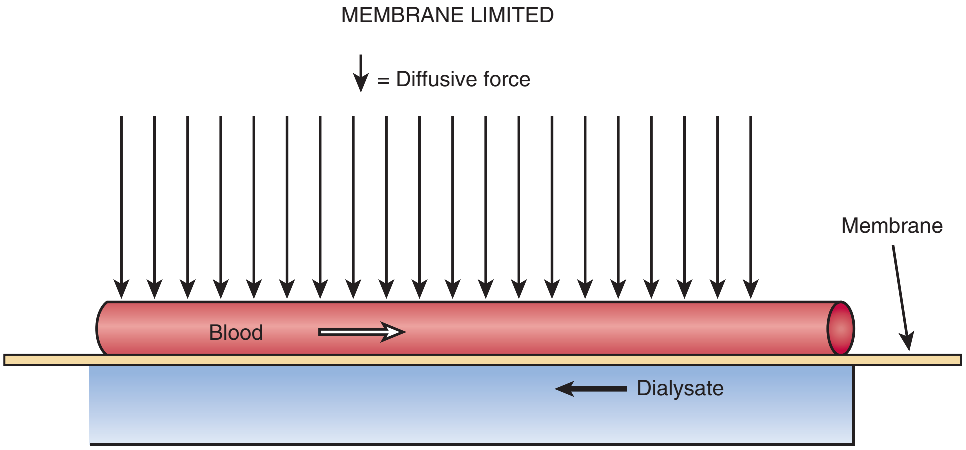


Fig. 63.13 Membrane-limited clearance. The diffusive force is the solute concentration gradient. Solute concentrations along the membrane from dialyzer inflow to outflow for both blood and dialysate are relatively constant because transport across the membrane is limiting and relatively low.

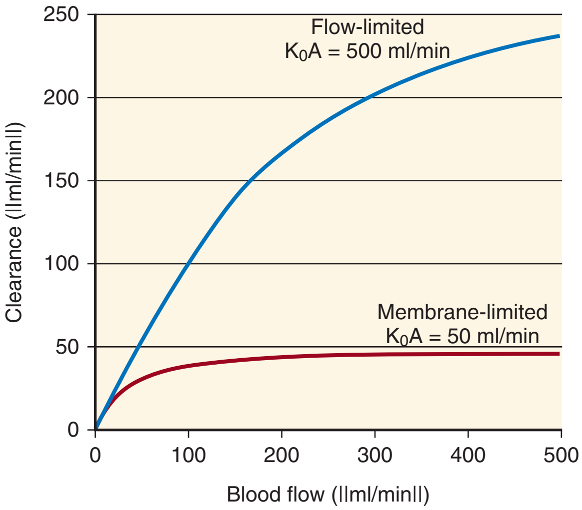


Fig. 63.14 Flow-limited versus membrane-limited clearances. Flow limits clearance when the membrane is not fully exposed to inflow solute concentrations (see Fig. 63.12). This typically occurs for small, easily dialyzed solutes. In contrast, larger solutes tend to saturate the membrane along its entire length at lower blood (or dialysate) flow rates (see Fig. 63.13). For these less easily dialyzed solutes, further increases in flow have no influence on clearance. *K_0A,* Mass transfer area coefficient.

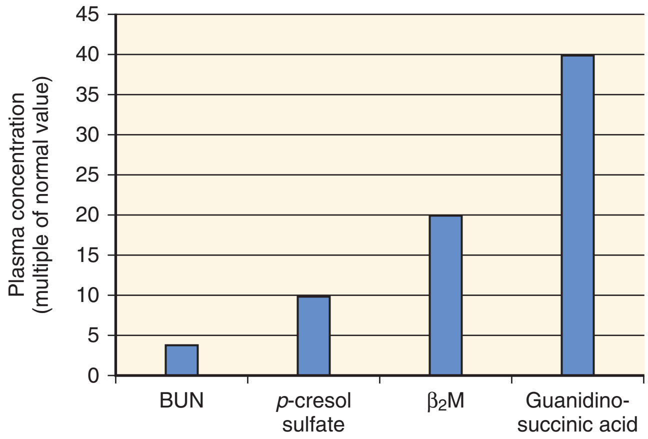


Fig. 63.15 Effect of protein binding, molecular weight, and sequestration on solute concentrations in hemodialysis (HD) patients. Conventional thrice-weekly dialysis is very effective at removing blood urea nitrogen *(BUN)*, resulting in an average urea level of about four times the normal value in a patient undergoing HD. Binding of *p*-cresol to albumin and the large molecular size of beta-2 microglobulin limit their removal with conventional dialysis, resulting in levels about 10 to 20 times that of normal, respectively. Plasma guanidinosuccinic acid levels are even higher (≈40 times normal) because of increased production in kidney failure and intracellular sequestration, making it difficult to remove during dialysis. Although the plasma levels of these other solutes are several orders of magnitude higher than normal, their absolute levels are much lower than that of urea, and it is unclear if they exert any toxicity. (Modified from Meyer TW, Hostetter TH. Uremia. *N Engl J Med.* 2007;357(13):1316–1325.)

volume) participates in the dialysis process. Blood flow through the dialyzer for water-soluble solutes (nearly all) should be expressed as blood water flow, or about 90% of whole blood flow. Similarly, blood concentrations should be expressed as blood water concentrations, which are about 7% higher than whole serum concentrations. Note that BUN is a misnomer; serum urea nitrogen is actually measured. For charged molecules, the Donnan effect acts in the opposite direction, reducing the effective blood activity.[186] Correcting for this reduced activity, the effective sodium concentration on the blood side of the membrane is about 3 mEq/L lower than its actual concentration. The Donnan effect for blood equilibrated with dialysate is attributable to nondialyzable plasma proteins, mostly albumin, which has a net negative charge (≈17 mEq/mmol albumin). The asymmetric charge distribution across the membrane effectively "captures" a small fraction of the positively charged sodium ions on the plasma side, reducing their potential for diffusion.

The size of the molecule is the most important intrinsic physical feature governing its removal (Fig. 63.15). In general, the rate of movement, or flux (J), of smaller molecules is higher than the flux of larger molecules. Other factors such as binding to plasma proteins, shape, charge, and sequestration in the intracellular compartment must be considered when predicting clearance (see Fig. 63.15).

DIALYZER CLEARANCE VERSUS WHOLE-BODY CLEARANCE

A distinction must be made between clearance across the dialyzer and clearance across the patient. For each of these, the removal rate (first equation) is the same, but the denominator differs. Dialyzer clearance is an expression of solute removal as a fraction of the blood concentration (adjusted for blood water content) at the dialyzer inflow port. In contrast, whole-body clearance is an expression of removal as a fraction of average concentrations throughout the body. The average whole-body concentration is substituted for dialyzer inflow concentration in the denominator of the standard clearance formula (first equation). Whole-body concentrations are higher than serum concentrations during dialysis because of solute disequilibrium, or nonuniform distribution of solutes, so whole-body clearances are always lower than dialyzer clearances. Higher concentrations throughout the body result from solute sequestration or a delay in the diffusive movement of solute from remote body compartments to the patient's blood, which is the immediately dialyzed compartment.

Typical solutes that exhibit disequilibrium distribute preferentially in the intracellular compartment and diffuse slowly across the cell membrane to the extracellular compartment. Such solutes are sometimes labeled as "difficult to dialyze." For example, therapeutic dialysis is not recommended for removal of digoxin in patients with digoxin intoxication,[187] even though it is water-soluble and easily removed in vitro, with a high clearance across the dialyzer membrane. Removal in vivo, however, is limited by sequestration in remote tissue compartments. Solutes such as digoxin have an apparent large distribution, often larger than total body water volume.

Even urea, one of the most diffusible solutes, in normal physiologic conditions is often termed an "ineffective osmole" and exhibits disequilibrium during HD. Whole-body urea clearance can be calculated by substituting the equilibrated postdialysis BUN (Fig. 63.16) for the immediate postdialysis BUN in the modeling equations. Experience has shown that urea sequestration can be predictable from the rapidity or rate of solute removal. This observation has led to the

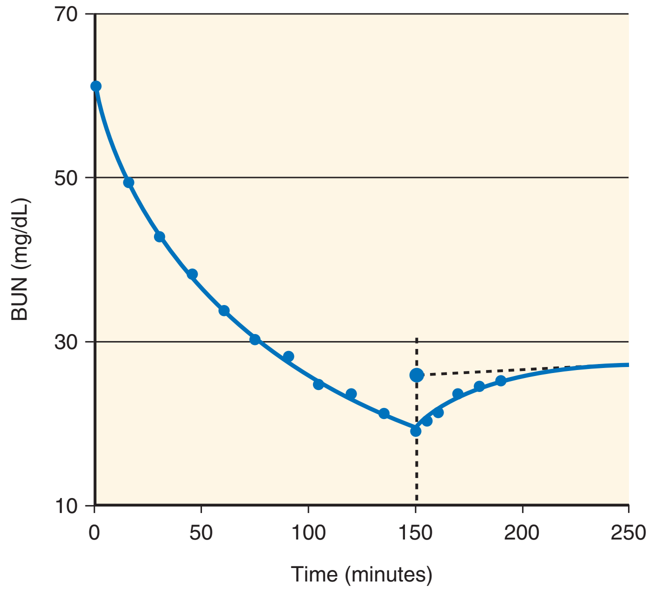


Fig. 63.16 Equilibrated postdialysis blood urea nitrogen *(BUN)*, the basis for eKt/V. Precise measurements of BUN every 15 minutes during a typical patient's 2.5-hour hemodialysis shows a logarithmic decrease in concentration during the treatment and a rapid rebound that is complete approximately 1 hour later. The double-pool model shown in Fig. 63.21 and the *solid line* in the graph predict the concentrations accurately. The equilibrated postdialysis BUN is an extrapolated value shown as the *large solid circle*.

development of simplified equations to estimate the equilibrated Kt/V (eKt/V) from single-pool Kt/V (spKt/V),[188,189] where Kt/V is dialyzer clearance × time over volume:

$$eKt/V = spKt/V - 0.6(spK/V) + 0.03 \quad [8]$$

$$eKt/V = spKt/V[t/(t+35)] \quad [9]$$

When eKt/V is the chosen standard of care measure of hemodialysis adequacy, these equations allow its calculation without requiring the patient to wait 30 to 60 minutes after dialysis has stopped to measure the equilibrated BUN.

EFFECT OF RED BLOOD CELLS PASSING THROUGH THE DIALYZER

In the blood compartment, solutes may diffuse slowly or not at all out of RBCs during transit through the dialyzer.[190,191] Water content of red blood cells is about 64% by volume, so this may account for the additional transport of water-soluble toxins during dialysis.[192] Clearances of solutes depend on how readily they move out of RBCs. For example, creatinine and uric acid are small molecules with clearances similar to that of urea when measured in a saline solution in vitro. Measured in vivo, however, their clearances are lower than urea clearance because movement out of the RBCs as it passes through the dialyzer is much slower than that of urea, limited by lack of specific urea-like transporters in the RBC membrane.[193–195]

SEQUESTRATION IN REMOTE COMPARTMENTS

Other solutes such as potassium and phosphorus are cleared easily in vitro, but their removal is limited by cellular and bone sequestration in patients, explaining the need for dietary restriction and for the use of intestinal binders to diminish absorption and, ultimately, the serum concentration of these solutes. Phosphorus is rapidly removed from the intravascular compartment, causing hypophosphatemia during HD in most conventionally treated patients. However, for 2 to 4 hours after dialysis, the serum phosphorus concentration increases relatively rapidly, returning to near-predialysis levels (Fig. 63.17). Thus, sequestration and its consequent postdialysis rebound account for the failure of conventional HD alone to normalize interdialytic concentrations. Low blood phosphate generated during dialysis may also account for at least part of the postdialysis disequilibrium syndrome, discussed elsewhere in this chapter. A pseudo–one-compartment model has been proposed to explain the behavior of phosphorus with HD.[196] Extrapolating further, one can speculate that the magnitude of sequestration for the toxic solutes responsible for the reversible life-threatening aspect of uremia cannot be as pronounced as for phosphorus; otherwise, HD would not be successful.

COMPONENTS OF THE EXTRACORPOREAL CIRCUIT

The HD system for a single patient in the 1940s was about the size of a twin bed. Modern HD machines are about the size of a three- to four-drawer filing cabinet. Central to the dialysis delivery system is the artificial kidney or dialyzer, which acts as the point of exchange between blood and the dialysate. The system is designed to deliver blood and properly constituted dialysate to the dialyzer, where diffusion and convection occur. Technologic advances have allowed the development of online monitors that accurately monitor and regulate the blood flow and dialysate flow rates, circuit pressures, and dialysate composition and temperature. Additional advances include automated safety mechanisms designed to detect blood leaks and air in the circuit and online devices that monitor vascular access, hematocrit, and dialysis adequacy during each treatment.

BLOOD CIRCUIT

During dialysis, the steady flow of blood required may be obtained from a central venous catheter or from an AV fistula or graft. If a catheter is used, blood enters the extracorporeal circuit from the ports along the sides of the double-lumen catheter (arterial lumen) and returns through the port at the distal tip (venous lumen). Alternatively, the graft or fistula is cannulated with two needles, with blood flowing from the arterial needle into the blood tubing and dialyzer and returning to the patient through the venous needle. The driving force for the blood circuit is a peristaltic roller pump, which sequentially compresses the pump segment of the tubing against a curved rigid track, forcing blood from the tubing. Elastic recoil refills the pump tubing after the roller has passed, readying it for the next roller. Because of the elastic recoil, and because most pumps have only two or three rollers, blood flow through the dialyzer is pulsatile. Increasing the number of rollers makes the flow less pulsatile but increases the risk of hemolysis and damage to the pump segment.

An alternative configuration of the dialysate delivery system allows the use of a single needle in the vascular access or a

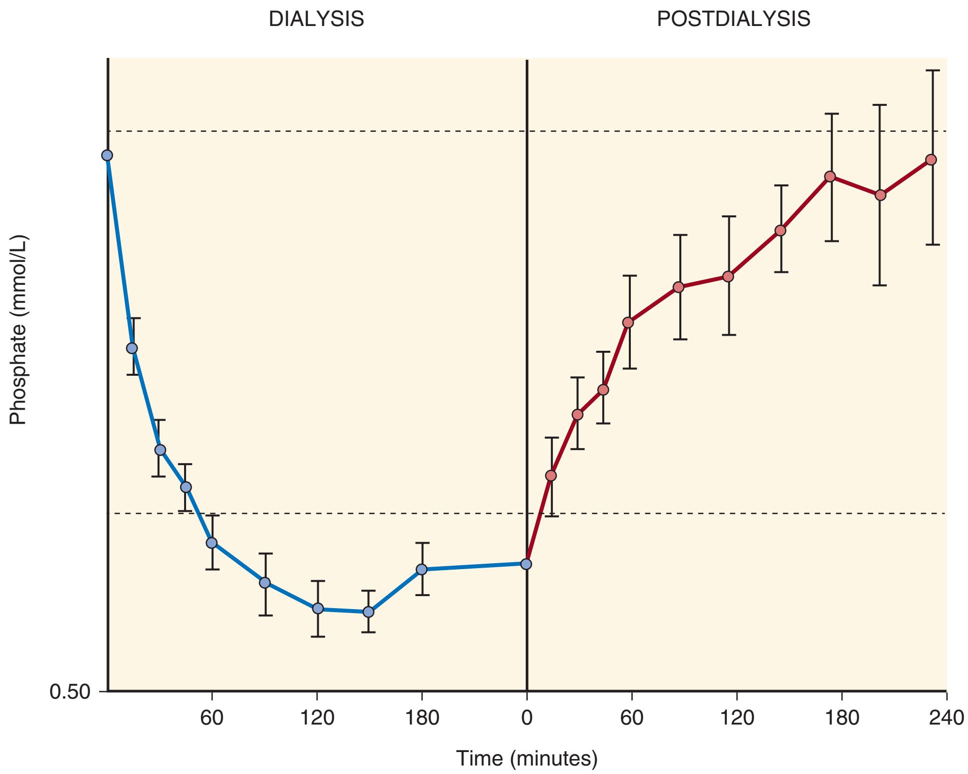


Fig. 63.17 Effect of sequestration on serum phosphate levels and removal during dialysis. Measurements of serum inorganic phosphorus concentrations were taken at 15-minute intervals during both high flux and standard hemodialysis. The rapid flux of phosphorus due to its relatively high mass transfer area coefficient caused levels to fall into the hypophosphatemic range (below the lower *dotted line*) during most of the 4-hour dialysis. A marked rebound continued for 4 hours after dialysis ended. (Modified from DeSoi CA, Umans JG. Phosphate kinetics during high-flux hemodialysis. *J Am Soc Nephrol.* 1993;4:1214–1218.)

single-lumen catheter for dialysis.[197,198] This arrangement uses one blood pump and two pressure-controlled blood line clamps or two pressure-controlled blood pumps. The advantage is reduced trauma to the vascular access, especially during initial cannulation of a new fistula, and potentially reduced dialysis catheter use after surgical revision of a vascular access.[198] However, recirculation and hemolysis may be increased, and efficiency (adequacy) may be compromised.[197,199,200] Increasing the effective blood flow rate to 250 mL/min, increasing the length of the dialysis session, or using a larger dialyzer may improve the efficiency of solute clearance, but careful monitoring of adequacy and for complications is essential.[199]

Pressure monitors are located proximally to the blood pump and immediately distal to the dialyzer. The proximal or arterial pressure monitor guards against excessive suction on the vascular access site by the blood pump. Accepted ranges for arterial inflow pressures are −20 to 80 mm Hg, but may be as low as −200 mm Hg when the blood flow rate (Q_b) is high. The distal or venous pressure monitor gauges the resistance to blood return in the vascular access; acceptable values range from +50 to +200 mm Hg. When the upper or lower limits of arterial or venous pressures are exceeded, an alarm sounds, and the blood pump turns off. Excessively low arterial pressures may be caused by kinks in the tubing, improper arterial needle position, hypotension, or arterial inflow stenosis. Blood clotting in the dialyzer, kinking or clotting in the venous blood lines, improperly positioned venous needles, infiltration of a venous needle, or venous outflow stenosis can cause high venous pressures. Accurate measurements of the arterial and venous pressures are essential to determining the transmembrane pressure (TMP), which partly determines the ultrafiltration rate. Excessively high pressures anywhere in the blood compartment may rupture the dialyzer membrane or disconnect the blood circuit, leading to an abrupt decrease in pressure in the blood circuit. The automatic shutoff of the blood pump in this circumstance is potentially lifesaving.

Two additional safety devices, the venous air trap and air detector, are located in the blood line distal to the dialyzer. Air may enter the blood circuit through loose connections, improper arterial needle position, or the saline infusion line. The venous air trap prevents any air that may have entered the blood circuit from returning to the patient. If air is detected in the venous line after the air trap, the machine sounds an alarm, and a relay switch turns off the blood pump. Excessive foaming of blood also triggers the air detector. These safety features prevent air embolism, which can lead to stroke, cardiac and/or respiratory failure, and death if not immediately recognized.[201] However, microbubbles formed during dialysis may escape detection and lodge in organs such as the brain and lungs, possibly contributing to the higher incidence of pulmonary hypertension and cognitive decline observed in dialysis patients.[202,203] Ensuring a high blood level in the venous air trap, avoiding extremely high blood flow rates, and adequately priming a dry-packed dialyzer may reduce the incidence of such microemboli.[201,204]

HEMODIALYZERS

A hemodialyzer, or dialyzer, is often called an "artificial kidney." Its configuration allows blood and dialysate to flow, preferably in opposite directions, through individual compartments separated by a semipermeable membrane. By convention, blood entering the hemodialyzer is designated arterial, and blood leaving the hemodialyzer is venous. The many available hemodialyzers differ mainly in the composition, configuration, and surface area of the membrane. Hemodialyzers influence the efficiency and quality of dialysis through their membranes, which determine their K_0A value, and through the blood and dialysate flow rates, which determine the clearance values (see also K_0A; Table 63.5).

Virtually all commercial dialyzers available in the United States are hollow fiber dialyzers. These hemodialyzers are constructed with a cylindric plastic casing (usually polycarbonate) that encloses several thousand hollow fiber semipermeable membranes stretched from one end to the other and anchored at each end by a plastic potting compound, usually polyurethane. The blood compartment or fiber bundle volume of the hollow fiber dialyzer ranges from 60 to 150 mL and, in contrast to older dialyzer designs, does not expand during dialysis. Each fiber has an inner diameter of approximately 200 μm. Along with the semipermeable membrane, the potting material separates the blood compartment from the dialysate compartment, where dialysate flows between and around each fiber. Blood flows to or from the open end of each fiber through a removable header attached to the blood tubing. Apart from lowering blood priming volume, the hollow fiber design also improves the efficiency of solute exchange by increasing the contact area between blood and dialysate. Additional maneuvers to maximize surface area include insertion of spacer yarns between fibers and a wavy moiré configuration[205] of the fibers to prevent loss of surface area through fiber to fiber contact. The arterial port design also influences the distribution of blood flow through the hollow fibers and can reduce dialysis efficiency.[206] Thrombosis and the need for potting compound are major disadvantages of the hollow fiber design. The potting compound absorbs chemicals used to disinfect newly manufactured dialyzers (e.g., ethylene oxide) or reprocessed dialyzers (e.g., formaldehyde, peracetic acid, glutaraldehyde) and acts as a reservoir for these chemicals, allowing them to leach out slowly during dialysis into the patient's blood.[207]

Table 63.5 Key Factors That Affect the Solute Clearance of a Hemodialyzer

Parameter	Key Factors	Clearance
Properties of the membrane	Membrane porosity	↑
	Membrane thickness	↓
	Membrane surface area	↑
	Membrane charge	Varies
	Membrane hydrophilicity	↑
Properties of the solute	Molecular weight and size	↓
	Charge	Varies
	Lipid solubility	↓
	Protein binding	↓
Blood side	Unstirred blood layer	↓
	Blood flow	↑
Dialysate side	Dialysate channeling and unstirred layer	↓
	Dialysate flow	↑
	Countercurrent direction of flow	↑

MEMBRANE COMPOSITION

The biomaterials used to make the hollow fiber dialyzer dictate its clearance and ultrafiltration characteristics, as well as its biocompatibility. Two major classes of membrane material are available commercially: (1) cotton fiber, or cellulose-based membranes; and (2) synthetic membranes. Unmodified cellulose-based membranes contain many free hydroxyl groups, which activate white blood cells, platelets, and serum complement via the alternate pathway (see later).[208] Treating the cellulose polymer with acetate and tertiary amino compounds improves membrane biocompatibility, presumably through the covalent binding of the hydroxyl groups to form acetylated cellulose and laminated cellulose, such as Cellosyn or Hemophan.[209,210]

The major polymers in synthetic membranes are polyacrylonitrile, polysulfone, polycarbonate, polyamide, polyether sulfone, and polymethylmethacrylate. Although these membranes are thicker, they can be rendered more permeable than the cellulose membranes, yielding greater fluid and solute removal. The larger pore sizes in the synthetic membranes remove higher molecular weight substances, such as beta-2 microglobulin, more efficiently.[211–213] Some membranes, such as polyacrylonitrile, polyamide, and polymethylmethacrylate membranes, have low hydrophilicity, providing significant protein adsorption and enhancing their removal.[214]

MEMBRANE BIOCOMPATIBILITY

Dialyzer membranes may interact with blood components during dialysis to activate white blood cells, platelets, and the complement cascade via the alternative pathway with generation of the anaphylatoxins C3a and C5a.[208,213–216] The degree to which the membrane activates blood components determines what is called its "biocompatibility." Through activation of white blood cells, platelets, and the complement cascade, bioincompatible membranes may cause allergic reactions, hypoxemia, transient neutropenia (due to leukosequestration), altered immunity, tissue damage, anorexia, protein catabolism, or an inflammatory state. Because dialysis is repetitive, the effects of low-grade subclinical membrane interactions during each treatment may be cumulative, eventually resulting in adverse clinical outcomes such as infection, accelerated atherosclerosis, frequent hospitalization, and death.

In addition to the capability of the membrane to activate blood elements, its absorptive capacity also can influence its biocompatibility. Some synthetic membranes, such as polyacrylonitrile, are more hydrophobic, bind proteins to a greater extent, and may ameliorate bioincompatible inflammatory reactions through their ability to bind anaphylatoxins such as C3a and C5a and cytokines.[213,214] Therefore, measurements of these elements in blood may not reflect accurately the true capability of a membrane for inciting the complement cascade, producing cytokines, and inducing an inflamed state.[213] In general, however, synthetic membranes and modified or substituted cellulose membranes are more biocompatible than unmodified cellulose membranes,[208,213,214] but issues may still arise, as in a recent report of increased blood levels of bisphenol A in patients treated with a polysulfone membrane.[217]

Because bacterial contaminants in product water (see "Water Treatment") also can activate complement and leukocytes if they are in contact with blood, it is difficult to determine the relative contribution from bioincompatible membrane versus nonsterile or inadequately purified water to the inflamed state seen in dialysis patients. With increasing use of modified cellulose and synthetic membranes and closer attention to water quality, the distinction has become even more difficult. Studies evaluating the relative biocompatibility of substituted cellulose versus synthetic membranes have reported no difference.[210,218,219] Ongoing efforts to improve biocompatibility include coating the membrane with heparin or vitamin E.[220,221]

MEMBRANE PERMEABILITY AND SURFACE AREA

Dialyzers perform two important functions, elimination of unwanted solutes and removal of excess fluid. The thickness, porosity, composition, and surface area of a membrane determine its ability to clear solutes and remove water. In general, the thinner the membrane and the more porous the membrane, the more efficient is the transport of solutes and fluid across the membrane. The urea K_0A (or clearance) of the dialyzer describes its ability to eliminate low-molecular-weight substances, the vitamin B_{12} and beta-2 microglobulin (B_2M) K_0A (or clearance) capacity to remove higher molecular weight substances, and the ultrafiltration coefficient (K_{Uf}) ability to remove water (Table 63.6).

Most hemodialyzers have a membrane surface area of 0.8 to 2.1 m^2. The desirable increase in solute transport associated with larger membranes can be achieved by increasing the length, increasing the number, or decreasing the diameter of the hollow fiber,[222] but each maneuver has undesirable effects when carried too far. Lengthening the fiber increases the shear rate and resistance to blood flow, magnifying the pressure decrease between blood entering and exiting the dialyzer. Increasing the shear rate can damage RBCs, and the increased pressure can increase the ultrafiltration rate. However, the higher filtration rate at the arterial inflow end of the dialyzer is partially offset by the dissipation of pressure at the venous end, reducing the contribution of ultrafiltration to solute clearance and offsetting this potential advantage of greater surface area.[205] Increasing the number of hollow fibers increases surface area but expands the extracorporeal blood volume, which can compromise the patient's hemodynamic stability. Smaller diameter fibers can offset this disadvantage but, as the fiber diameter decreases, resistance to blood flow increases, enhancing not only filtration but also backfiltration and clotting.[223] As fibers thrombose, the effective surface area for diffusion decreases, and solute clearances decrease. Because of these adverse effects, the minimal acceptable internal fiber diameter is 180 μm.[223] The design and geometry of the hollow fiber dialyzer represent a delicate balance among these factors.

Table 63.6 Characteristic Values for Standard, High-Efficiency, and High-Flux Dialyzers[a]

	Standard	High Efficiency	High Flux
Blood flow rate (mL/min)	250	≥350	≥350
Dialysate flow rate (mL/min)	500	≥500	≥500
K_0A urea	300–500	≥600	Variable
Urea clearance (mL/min)	<200	>210	Variable
Urea clearance/body weight (mL/min/kg)	< 3	>3	Variable
Vitamin B_{12} clearance (mL/min)	30–60	Variable	>100
Beta-2 microglobulin clearance (mL/min)	<10	Variable	>20
Ultrafiltration coefficient (mL/hr/mm Hg)	3.5–5.0	Variable	>20
Membrane	Cellulose	Variable	Variable

[a]See text.

HIGH-EFFICIENCY AND HIGH-FLUX DIALYZERS

Historically, low dialyzer membrane permeability has limited the efficiency of HD, requiring more than 6 hours for each treatment. As dialyzer design improved, treatment times were shortened progressively. In the late 1980s, with the advent of more permeable dialyzer membranes, improved techniques to reduce bacteriologic contamination of bicarbonate dialysate (see later), more precise ultrafiltration control, and more reliable vascular access to achieve adequate blood flow, treatment times decreased to 2 to 3 hours three times weekly in the United States. These substituted cellulose and synthetic membranes ushered in the era of high-efficiency and high-flux dialysis.

The distinction between high-efficiency and high-flux dialyzers is imprecise, and sometimes these terms are used interchangeably. Both types of dialyzers have improved solute and fluid clearance compared with standard hemodialyzers and take advantage of higher blood and dialysate flow rates to reduce dialysis time while maintaining an adequate dose. High-efficiency dialyzers have a high K_0A and a high clearance of small molecules, such as urea, compared with standard dialyzers (see Table 63.6). High-flux dialyzers have a highly permeable membrane for larger molecules, such as vitamin B_{12} and B_2M, and have a higher K_{Uf} than high-efficiency dialyzers, but not necessarily high urea clearances (see Table 63.6). As evident from the foregoing discussion, the two dialyzer designs frequently overlap—hence, the imprecision (see Table 63.6).

High-efficiency and high-flux dialyzers contain a substituted cellulose or synthetic membrane. Both membranes improve dialyzer permeability because substituted cellulose membranes can be made thinner to increase porosity and surface area, and synthetic membranes can be manufactured with more and larger pores. Both high-efficiency and high-flux dialysis require the use of bicarbonate dialysate and volume-controlled filtration. When acetate was used as the prominent base in dialysate, its rate of diffusion into blood exceeded the metabolic capacity of the body, leading to acidosis, vasodilation, and intradialytic hypotension (see "Dialysate Composition"). The high K_{Uf} of these dialyzers creates the potential for hemodynamic instability with pressure-controlled filtration;

hence, the need for volume-controlled filtration (see "Dialysate Circuit").

Because of their relative porosity, high-flux dialyzers can remove larger molecules such as B_2M, which are not removed at all by standard cellulosic dialyzers.[211–213] B_2M removal reduces the risk of carpal tunnel syndrome and other complications in long-term patients.[211,224,225] Initial results have suggested that removal of other large molecules may offer additional benefits, such as an enhanced response to erythropoietin,[226] higher leptin removal, possibly leading to a better appetite,[227] and perhaps lower mortality and hospitalization rates.[225,228] However, potential adverse consequences include more efficient removal of amino acids,[229] albumin,[230] and drugs such as vancomycin.[231] Theoretically, the backfiltration that occurs during high-flux dialysis can increase the exposure of patients to endotoxin from dialysate, but this theoretic concern has not been verified clinically.[232,233]

Despite early promise in patients undergoing maintenance hemodialysis, randomized controlled or crossover trials comparing high-flux dialysis with standard dialysis have found no difference in the incidence of hypotension and intradialysis symptoms,[211,234] control of blood pressure,[235] neuropsychological function,[236] hemoglobin concentration, and use of erythropoiesis-stimulating agents (ESA)s[237–239] or markers of inflammation, oxidative stress, and nutritional status.[238] Three large randomized controlled trials, the Hemodialysis (HEMO) Study,[240,241] the Membrane Permeability Outcome (MPO) Study,[242] and the EGE Study (Multiple Interventions Related to Dialysis Procedures in Order to Reduce Cardiovascular Morbidity and Mortality in Hemodialysis Patients)[243] detected no significant difference in mortality or morbidity between patients treated with standard versus high-flux membranes. Post hoc subgroup analyses have demonstrated that high-flux dialysis reduces cardiovascular events in patients with longer dialysis vintage in the HEMO Study,[240,241] reduces mortality in patients with low serum albumin concentrations (<4.0 g/dL) and diabetes in the MPO Study,[242] and reduces cardiovascular mortality in patients with an AVF or diabetes in the EGE study.[243] A meta-analysis of available data has concluded that high-flux hemodialysis may reduce cardiovascular mortality by 15% but does not alter infection-related or all-cause mortality.[244]

Convection Versus Diffusion

Further increases in dialyzer flux may not lead to improved middle molecule clearance or outcomes because some solutes that accumulate in kidney failure are protein-bound or sequestered inside the cell.[245] Because of slow diffusion caused by low free concentrations of protein-bound solutes and delayed diffusion across cell membranes, the removal of such solutes remains time-dependent. Initial experience has suggested that hemofiltration (HF) and hemodiafiltration (HDF) may augment the removal of larger molecules and protein-bound solutes through increased convective clearance.[245–247] However, randomized controlled studies comparing HF and HDF with HD have found no difference in hemoglobin or ESA resistance,[248,249] serum phosphate levels,[250] health-related quality of life,[251] cardiovascular parameters such as left ventricular mass and pulse wave velocity,[252] and intradialytic hypotension,[253,254] although some studies have reported less intradialytic hypotension with HDF.[255,256] Two of the three largest randomized controlled studies, the Convective Transport Study (CONTRAST)[257] and Turkish Online Haemodiafiltration Study (OL-HDF),[254] have reported no difference in cardiovascular and all-cause mortality, but subgroup analyses have suggested a benefit in patients treated with greater convective and replacement volumes (>22 L and 17.4 L, respectively, so-called "high efficiency" HDF). The ESHOL (Estudio de Supervivencia de Hemodiafiltracion On-Line) Study,[256] which achieved about 23 L of convective volume per session, reported lower cardiovascular mortality, all-cause mortality, and hospitalization in patients randomized to HDF. Recent meta-analyses have concluded that convective therapies may reduce intradialytic hypotension[247,258,259] and cardiovascular mortality,[247,259] but additional high-quality randomized and controlled studies addressing the effect of convective volume on outcomes are needed.[260]

Further technologic advances have allowed the development of high-retention onset or mid- or high-cutoff hemodialyzer membranes, which have larger but more homogenous pore sizes that will clear molecules up to 45 kDa in size yet prevent the excessive loss of albumin.[261–263] Preliminary clinical data have suggested that HD with these membranes allows the clearance of larger middle molecules, such as advanced glycation end products and inflammatory mediators, comparable to that achieved with HDF, and may improve the residual syndrome (symptoms that remain in patients despite adequate dialysis) and reduce cardiovascular risk.[261–265]

DIALYSATE CIRCUIT

Another major function of the HD system is the preparation and delivery of dialysate to the dialyzer. Most dialysis clinics use a single-pass delivery system, which discards the dialysate after a single passage through the dialyzer, and a single-patient delivery system, which prepares dialysates individually and continuously at each patient station by mixing liquid concentrates with a proportionate volume of purified water. To ensure that the dialysate concentrates are diluted safely and accurately, the delivery system has many built-in safety monitors. Some clinics use a central multipatient delivery system. The dialysate is mixed in an area separate from patient care and then piped into each patient station, or the concentrate is piped to each station before mixing. These centralized systems lower patient care costs and reduce staff back injuries from carrying the individual concentrate jugs. However, a disadvantage is the additional effort and cost required to modify the dialysate concentration of electrolytes such as calcium and potassium for individual patients.

The dialysis machine warms purified water to physiologic temperatures and then deaerates it under vacuum. Because the patient is exposed to 100 to 200 L of dialysate during each treatment, the dialysate must be warmed to avoid hypothermia. In practice, the dialysate temperature is maintained at 35° to 37°C. If the dialysate is too hot, protein denaturation (>42°C) and hemolysis (>45°C) occur. To ensure safety, the temperature monitor in the dialysate circuit sounds an alarm, and a bypass valve diverts the dialysate directly to the drain, automatically bypassing the dialyzer if the dialysate temperature is outside the limits of 35° to 42°C. Without deaeration, dissolved air will come out of solution as negative pressure is applied during dialysis, creating air bubbles in the dialysate, which leads to malfunction of the blood leak detector and conductivity detector, increased channeling, masking parts of the membrane, and reduced surface area.

The heated and deaerated product water is then mixed proportionately with the concentrate to produce dialysate. Improperly proportioned dialysate may cause severe electrolyte disturbances in the patient and lead to death. Because the primary solutes in the dialysate are electrolytes, the electrical conductivity of the dialysate varies directly with the concentration of solutes. Based on this principle, the conductivity monitor downstream from the proportioning pump continuously measures the electrical conductivity of the product solution to ensure proper proportioning. It has a narrow range of tolerance, is usually redundant, and must be calibrated periodically using standardized solutions or by laboratory measurements of electrolytes in the dialysate. Changes in temperature, the presence of air bubbles, or malfunction of the sensor (usually an electrode) can alter the dialysate conductivity.

The dialysate pump, located downstream from the dialyzer, controls dialysate flow and dialysate pressure. Although many dialyzers require a negative dialysate pressure for filtration, the circuit also must be able to generate positive dialysate pressures within the dialyzer because positive pressure is required to limit filtration when using dialyzers with a high K_{Uf} or under conditions that increase pressure in the blood compartment. The dialysate circuit regulates the pressure by controlled constriction of the dialysate outflow tubing while maintaining a constant flow rate. In addition, the dialysate delivery system controls the filtration rate, either indirectly by altering the TMP (pressure-controlled ultrafiltration) or directly by modifying the actual filtration rate (volume-controlled ultrafiltration). Earlier systems used manual pressure-controlled filtration, requiring dialysis personnel to calculate and enter the TMP, closely monitor the filtration rate, and recalculate and adjust the TMP as needed. For dialyzers with a K_{Uf} greater than 6 mL/hr/mm Hg, dialysate delivery systems with built-in balance chambers and servomechanisms that accurately control the volume of the fluid removed during dialysis (volume-controlled filtration)[266] are mandatory to prevent excessive fluid gain or removal.

When blood is detected in the dialysate, the blood leak monitor located in the dialysate outflow tubing sounds an alarm and shuts off the blood pump. Blood in the dialysate usually indicates membrane rupture and may be caused by a TMP exceeding 500 mm Hg or by a damaged dialyzer membrane from the bleach and heat disinfection used to reprocess dialyzers for repeated use. Although a rare complication, membrane rupture can be life-threatening because it allows blood to come into contact with nonsterile dialysate.

ONLINE MONITORING

In addition to delivering dialysate to the dialyzer and the many built-in safety features described previously, modern dialysis machines also record and store such varied, real-time data as patient vital signs, blood and dialysate flows, arterial and venous pressures, delivered dialysis dose, plasma volume, thermal energy loss, and access recirculation. Linking computerized medical information systems with dialysis delivery systems can facilitate and improve patient care by allowing the integration of patient data while maintaining treatment records. These capabilities may be most important among patients receiving in-center self-care or home-based HD, where there are fewer personnel present at the point of care.

MONITORING CLEARANCE

Online monitoring of clearance may provide the best assessment of dialysis adequacy.[267–271] Online monitors record urea clearances by measuring the urea concentration in the dialysate, either continuously or periodically,[267,271,272] determining dialyzer sodium clearance by pulsing the dialysate with sodium and measuring dialysate conductivity at the dialyzer inlet and outlet (ionic dialysance),[267,268,273,274] or determining clearance of uremic solutes by measuring ultraviolet light absorbance of spent dialysate.[267,275,276] Most online methods for monitoring urea or sodium kinetics provide Kt/V based on whole-body clearance in addition to dialyzer clearance.[268,273] Online urea monitoring has not gained popularity, possibly because of the need for repeated calibration and the added expense for additional disposable supplies. Online clearance monitoring removes the expense and risks of blood sampling, reduces dialysis personnel time, allows more frequent determination of delivered dose, and provides real-time measurements for instant feedback.[268–270] However, reported clearance values may differ, depending on the online equipment used and adjustments applied by the instrument's software to match the urea Kt/V more closely.[274,275] Drawbacks of online clearance monitoring include the need for multiple measurements of K_d to obtain an average for the entire dialysis, accurate monitoring of treatment time, the need for blood urea sampling to allow determination of protein catabolic rate (a marker of nutrition), and the need to measure or estimate V to allow the calculation of Kt/V from the online K_d measurements.[277]

MONITORING HEMATOCRIT AND RELATIVE BLOOD VOLUME

The hematocrit can be measured online during dialysis using an ultrasound determination of plasma protein concentration[278] or an optical measure of hemoglobin concentration or hematocrit.[267,269,270,279] Of these, the optical technique is most widely available. In theory, patients treated with dialysis who are prone to hypotension and cramping may benefit from online monitoring of hematocrit because their symptoms are usually caused by a decrease in circulating blood volume when the ultrafiltration rate exceeds intravascular refilling from the interstitium and intracellular spaces.[267,269,270,279,280] The degree of hemoconcentration reflects the immediate magnitude of intravascular volume depletion and allows determination of changes in the blood volume or relative blood volume. Theoretically, monitoring the hematocrit online and altering the filtration rate during dialysis to minimize excessive hemoconcentration may reduce the occurrence of symptoms during dialysis and optimize the dry weight.[267,270,281–283] In practice, using just the relative blood volume to guide the filtration rate has not been very successful in ameliorating symptoms, likely because of inaccuracies in the measurements, the varied compensatory cardiovascular responses to volume depletion within and among individual patients, and a dialysis-induced reduction in arteriolar tone and left ventricular function (myocardial stunning).[267,270,279,280,284–289]

Online determination of relative blood volume varies significantly among devices and underestimates the true decline in blood volume because of intravascular translocation of blood from the microcirculation (capillaries and venules), which has a lower hematocrit, to the larger vessels.[279,280,282] Because of the previous limitations, identifying the pattern of the relative blood volume decline and using a computer-controlled biofeedback system to modify the filtration rate continuously during dialysis (see later), in combination with clinical assessment such as symptoms and bioimpedance spectrometry, have shown more promise.[267,269,270,290–292] The absolute or total blood volume may be more predictive of intradialytic symptoms; automated online monitoring of absolute blood volume is not currently available, although mathematical models may allow derivation of the total blood volume from relative blood volume.[280,293–296]

COMPUTER CONTROLS

Solute removal during HD reduces plasma osmolarity, favoring fluid shift into the cells and thwarting efforts to achieve net fluid removal.[297–299] Raising the dialysate sodium concentration (sodium ramping) helps preserve plasma osmolarity and may allow continued fluid removal but leads to increased thirst, excessive interdialytic weight gain (IDWG), and hypertension, although the latter is not a consistent finding.[269,297–303] Computer-controlled sodium modeling changes the dialysate sodium concentration automatically during dialysis, usually starting at 150 to 155 mEq/L and stepping down to 135 to 140 mEq/L near or at the end of dialysis, and offers the theoretic benefit of reducing intradialytic symptoms (hypotension and cramps) while minimizing thirst, IDWG, and hypertension. In practice, however, thirst, increased IDWG, and hypertension persist because of predialysis hyponatremia (on average, 134–136 mEq/L) and an overall positive sodium balance during dialysis, despite the step-down.[269,300,304,305] Instead, individualizing dialysate sodium to maintain a sodium gradient of 0 to −2 mEq/L with respect to the predialysis plasma sodium concentration results in less thirst, lower IDWG, improved blood pressure control, and perhaps diminished intradialytic symptoms because of lower ultrafiltration requirements.[297–300,306] Such individualization may be accomplished through the use of online conductivity monitoring or by estimating each patient's inherent plasma sodium concentration (sodium set point) using an average predialysis sodium concentration. This can be measured with direct potentiometry or mathematically corrected for the Gibbs-Donnan effect—sodium that is not available for diffusion because of trapping by negatively charged proteins.[297–300,305] To complicate matters further, data from DOPPS have suggested a differential effect of dialysate sodium on outcome; HD patients with the lowest predialysis plasma sodium levels have a lower mortality when dialyzed against a high-dialysate sodium, despite an increase in IDWG.[302] In addition, measured dialysate sodium levels deviate significantly from prescribed levels in over 40% of assays, ranging from 13 mEq/L higher to 6 mEq/L lower than prescribed,[307–309] making it difficult to interpret studies evaluating the clinical effects of varying dialysate sodium concentrations.[297–299,310] Given our incomplete understanding of the effects of altering dialysate sodium levels on morbidity and mortality, the indiscriminate use of dialysate sodium ramping and sodium modeling in its current form should be abandoned, and individualization of dialysate sodium should be done cautiously.[289,297–299,311,312]

Ultrafiltration modeling provides a variable rate of fluid removal during dialysis according to a preprogrammed profile (e.g., linear decline, stepwise changes, exponential decline of filtration rate with time). Theoretically, altering the filtration rate during dialysis allows time for the blood compartment to refill from the interstitial compartment, leading to less hypotension and cramping. As with sodium modeling, standalone ultrafiltration modeling is relatively crude, and altering the ultrafiltration rate in response to blood volume monitoring may be of more benefit (see earlier, "Monitoring Hematocrit and Relative Blood Volume"). The effects of sodium and ultrafiltration modeling may be difficult to separate because they are often used together.[267,269,270,304]

Recent technologic advances include the development of dialysis machines with biofeedback systems, allowing for computer-controlled adjustments of treatment parameters based on real-time input from the online monitors. The most common system in use monitors the blood volume (see earlier) and adjusts the ultrafiltration rate and dialysate conductivity to prevent it from decreasing below a preset value during dialysis. Small studies have demonstrated that this device ameliorates symptoms in hypotension- and nonhypotension-prone patients. [267,269,270,282,283,290,313,314]

Although the ability to monitor plasma conductivity throughout dialysis may ensure sodium balance, despite constant modifications to the dialysate conductivity, and may reduce the problem of thirst, IDWG, and hypertension,[267] most authorities have advised against sodium modeling, even with a feedback control system (see previous discussion).[181,269,270,313,315] Instead, automated control of the dialysate temperature to maintain isothermic dialysis (constant body temperature) has shown promise and is superior to thermoneutral dialysis (using lower but constant dialysate temperature) in reducing intradialytic hypotension without incurring a sodium load.[267,270,316–319] Other recent studies have suggested that individualized dialysate cooling (0.5°C lower than body temperature, averaged over six preceding treatments) reduces episodes of intradialytic hypotension by 70%, raises mean arterial blood pressure during HD by 12 mm Hg, and abrogates both HD-associated cardiomyopathy and brain white matter changes.[320,321] Although these online monitors and automated biofeedback systems are expensive, they have the potential to reduce hypotension, detect vascular access dysfunction, and increase dialysis efficiency while minimizing blood sampling.[271,282,283,322,323] By improving patient care, they may prove not to be cost-prohibitive in the long run.

DIALYSATE

During HD, blood flows in the blood compartment in one direction, and iso-osmotic dialysate flows in the opposite direction in the dialysate compartment (see Fig. 63.12). This countercurrent flow optimizes the concentration gradient for solute removal. Preparation of the dialysate and its composition is critical to the success of dialysis. The solution must be prepared from properly treated water (see later) that includes reducing the concentration of endotoxins to prevent pyrogenic reactions in the patient. Sterility is not required because the semipermeable membrane excludes large particles such as bacteria and viruses. The concentrations of vital solutes added to the dialysate reflect those normally maintained in the body by the native kidneys (Table 63.7).

The dialysate is essentially a physiologic salt solution that creates a gradient for removal of unwanted solutes and maintains a constant physiologic concentration of extracellular electrolytes (see later).

WATER TREATMENT

Because HD patients are exposed to as much as 600 L of dialysate water/week, treating the water used to generate dialysate is essential to avoid exposure to harmful substances such as aluminum, chloramines, fluoride, endotoxins, and bacteria.[324–328] Technical advances such as high-flux dialyzers, reuse or reprocessing of dialyzers, and bicarbonate-based dialysate have made high water quality even more imperative. To avoid these complications, tap water is softened, exposed to charcoal to remove contaminants such as chloramine, filtered to remove particulate matter, and then filtered again under high pressure (reverse osmosis) to remove other dissolved contaminants (Fig. 63.18). A complete review of this topic is beyond the scope of this chapter, and readers are referred to reviews on the topic.[324–329] Highlights are discussed later.

Hazards Associated With Dialysis Water

Improperly treated water contains potentially harmful substances and can cause patient injury or death.[324–330] Accumulation of aluminum in the body may cause osteomalacia, microcytic anemia, and dialysis-associated encephalopathy (dialysis dementia and movement disorders).[331–333] Treating water to keep aluminum levels below 10 mg/L has markedly reduced aluminum-associated diseases.[334,335] Chlorine is added to municipal water as a bactericidal agent and interacts with organic material in the water to form chloramines. Alternatively, chloramine may occur naturally or may be added directly to municipal water as a bactericidal agent. Unfortunately, in contrast to chlorine, direct exposure of the blood to chloramine causes acute hemolysis and methemoglobinemia.[327,328,336–338] Fluoride can cause cardiac arrhythmias and death acutely[339,340] and osteomalacia chronically.[341] Excess calcium and magnesium have been linked to the hard water syndrome with a constellation of symptoms, including nausea, vomiting, weakness, flushing, and labile blood pressures.[342] Close communication with water suppliers is critical to anticipate changes in feed water quality from added chemicals and environmental conditions such as flooding or contamination, because alterations in the water purification process may be required.[327,329] With the advent of large-pore, high-flux membranes, efforts at improving water purity have focused on further reducing bacterial endotoxins, which can cause febrile reactions, hypotension, and chronic inflammation (see later).[325,327–329]

Table 63.7 Solutes Present in the Dialysate

Solute	Concentration (mEq/L)
Sodium	135–145
Potassium	0–4.0
Chloride	102–106
Bicarbonate	30–39
Acetate	2–4
Calcium	0–3.5
Magnesium	0.5–1.0
Dextrose	11
pH	7.1–7.3

Essential Components of Water Purification

Temperature-blending valves proportion incoming hot and cold tap water to yield a water temperature of about 77°F, the optimal temperature for the carbon tank and most reverse osmosis membranes. Water temperature below 77°F reduce the flow rate and thus the efficiency of the reverse osmosis system, and temperature above 100°F may damage the membrane. Multimedia depth filters then remove particulate matter from the water (see Fig. 63.19). Using cation exchange resins that contain sodium, the water softener then removes calcium, magnesium, and other polyvalent cations from the feed water, preventing these cations from depositing on and damaging the reverse osmosis membrane. Next, granular

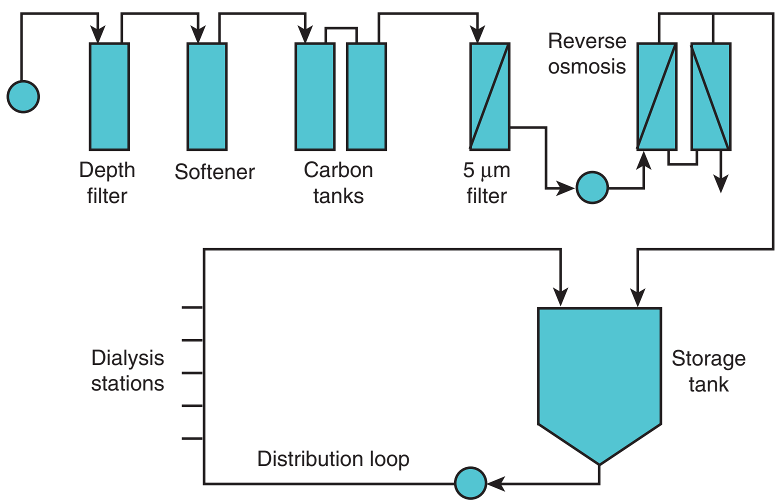


Fig. 63.18 Schematic of a typical configuration of a reverse osmosis water treatment system. Tap water undergoes filtration to remove gross particulate matter and then is softened before exposure to charcoal (carbon tanks) to remove contaminants such as chloramine. A second filtration process removes particulate matter, as well as microbiologic organisms. Finally, water is filtered under high pressure to remove dissolved contaminants such as aluminum (reverse osmosis). Product water is then either stored in a water tank or piped directly to each dialysis station.

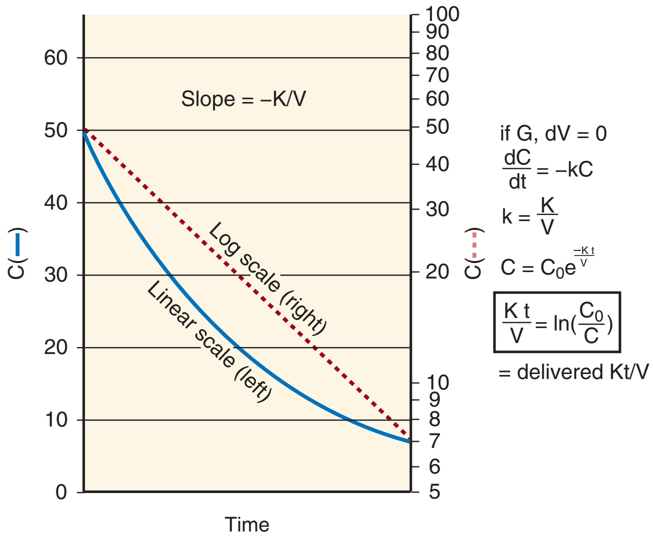


Fig. 63.19 Intradialysis urea kinetics: origin of Kt/V. *Left,* The nonlinear decrease in blood urea nitrogen during dialysis (*solid line*) becomes a straight line when plotted on a log scale (*dashed line*). The fractional rate of decrease is a constant, k = K/V, where k is the elimination constant, K is the clearance, and V is the urea distribution volume. *Right,* The solution to the equation describing first-order kinetics shows that delivered Kt/V primarily depends on the predialysis and postdialysis blood urea nitrogen values (see text and legend for Fig. 63.20 for definition of variables shown). This oversimplified equation is expanded in Fig. 63.20 to include the other important variables.

activated carbon in the carbon filtration tank absorbs chlorine, chloramines, and other organic substances from the water. Activated carbon is very porous and has a high affinity for organic material but, if not serviced properly or exchanged frequently, it can be contaminated with bacteria. Downstream, the water is then filtered through a 5-µm cartridge filter to prevent carbon particles from fouling up the reverse osmosis pump and membrane. Finally, the water is delivered to the reverse osmosis unit, which applies high hydrostatic pressure to force water through a highly selective semipermeable membrane that rejects 90% to 99% of monovalent ions, 95% to 99% of divalent ions, and microbiologic contaminants larger than 200 daltons. The water exiting the reverse osmosis unit is termed the "permeate" or "product water" and, in most clinics, can be used safely for dialysis.

When there is heavy ionic contamination of feed water, however, the product water from the reverse osmosis unit is further polished with a mixed bed ion exchange system (deionization system) and then passed through an ultrafilter to remove any bacterial contamination from the ion exchanger. The cationic resin exchanges hydrogen ions for other cations in descending order of affinity—calcium, magnesium, potassium, sodium, and then hydrogen. The anionic resin exchanges hydroxyl ions for other anions in decreasing order of affinity—nitrites, sulfates, nitrates, chloride, bicarbonate, hydroxyl, and fluoride. When the resin is exhausted, previously adsorbed ions, especially those of lower affinity, can elute into the effluent and result in levels that are more than 20 times their usual concentration in tap water, causing severe toxicity and even death.[339,343] Because of this danger, the deionization system is rarely used alone in treating water for dialysis and requires stringent monitoring of product water.

Microbiology of Hemodialysis Systems

Despite municipal treatment of tap water and the extensive water treatment system described previously, water used for dialysis still can become contaminated with bacteria and endotoxins,[324,325,327–330,344–346] mainly with water-borne, gram-negative bacteria and nontuberculous mycobacteria. Such contamination arises because the system removes the normally protective chlorine and chloramine described previously, and low flow and stagnation points in the water treatment circuit predispose to biofilm deposition. Although nontuberculous mycobacteria do not produce endotoxins, they are more resistant to germicides than gram-negative bacteria and can survive and multiply in product water that contains little organic matter.[347–350] In 1984, the Centers for Disease Control and Prevention (CDC) found nontuberculous mycobacteria in the water of 83% of surveyed dialysis centers.[348]

In addition to treating water to remove potentially harmful chemicals, routine disinfection and surveillance of the water treatment equipment, product water, and dialysate are also critically important to optimize dialysis water quality.[324,327–330,345] Because of dialyzer reprocessing and the use of high-flux dialyzers, the patient may be exposed to bacterial and endotoxin contaminants in improperly handled product water, either through direct contact of product water with the blood compartment during reprocessing or through backleak of endotoxin into the blood compartment during dialysis. Therefore, high-level disinfection to kill all microorganisms (except bacterial spores) is necessary, as well as stricter standards for water quality. In 2009, the Association for the Advancement of Medical Instrumentation (AAMI) adopted the International Organization for Standardization (ISO) guidelines for dialysis water quality, recommending a lower maximal level of 100 colony-forming units (CFU)/mL for bacteria and a maximal concentration of less than 0.25 endotoxin units (EU)/mL for endotoxin (compared with 200 CFU/mL and 2 EU/mL previously), with action levels of 25 CFU/mL and 0.125 EU/mL, respectively.[351,352] In addition to routine scheduled disinfection, the water treatment equipment and system and affected dialysis machines must be disinfected when action levels are detected on scheduled monitoring. For ultrapure dialysate, even more stringent criteria are in place, including a bacterial count of less than 0.1 CFU/mL and endotoxin level of less than 0.03 EU/mL.[353] A meta-analysis has reported that ultrapure dialysate use is associated with less inflammation and oxidative stress, higher serum albumin and hemoglobin levels, and lower ESA requirement.[354] The only randomized controlled trial to evaluate the effect of ultrapure dialysate on fatal and nonfatal cardiovascular events has found no benefit, although its power to detect a difference may have been reduced by a lower than expected endotoxin level (0.15 ± 0.22 EU/mL) in the conventional dialysate group.[243] Despite the absence of large randomized controlled trials, ultrapure dialysate use may be desirable because of these potential benefits, the potential to reduce overall cost of dialysis through decreased ESA use, and the necessity of ensuring water quality with the advent of high-cutoff dialyzer membranes.[261,263,355,356]

Water contaminated with bacteria and endotoxins can lead to pyrogenic reactions, characterized by shaking chills, fever, and hypotension in a previously afebrile and asymptomatic patient.[324,326,328,345,357] Headache, myalgia, nausea, and vomiting also may be present. Typically, the symptoms begin 30 to 60 minutes into the dialysis treatment. The source of the reaction is unlikely to be the microorganisms per se because they are too large to cross an intact dialyzer membrane. Instead, bacterial pyrogens such as lipopolysaccharide, peptidoglycans, exotoxins, and their fragments are thought to be responsible.[324,326,328,345,358] Pyrogenic reactions are typically seen in association with reprocessing of dialyzers because contaminated water gains direct access to the blood compartment during reprocessing.[207,357,359,360] In the absence of reuse, pyrogenic reactions are rare and occur only with high-level bacterial contamination of the dialysate or bicarbonate. Although the larger pore size in high-flux dialyzers may increase backfiltration and allow endotoxins to enter the blood compartment from the dialysate, synthetic membranes also adsorb endotoxins, thereby attenuating the effect of imperfectly processed dialysate.[207,324,345] However, even in the absence of pyrogenic reactions, low levels of dialysate contamination with microbes may result in chronic inflammation, manifested as higher serum C-reactive protein levels, increased oxidative stress, lower serum albumin and hemoglobin levels, and ESA resistance,[354–356] reversed by the use of ultrapure dialysate (see earlier).

Monitoring Water Quality

Because of the potential complications that can occur when improperly treated water is used for dialysis, monitoring of water quality is crucial. The source water and product water must be assayed routinely to ensure that product water meets the standards for heavy metal and other ionic contaminants. In the United States, the Association for the Advancement of Medical Instrumentation (AAMI) has adopted ISO water standards for dialysis (see previously). The frequency of scheduled testing depends on the quality of the water source, type of water treatment system used, and seasonal variation in chemicals added to municipal water to ensure its potability.

Samples of source water, water obtained from critical points in the water treatment system, product water, dialysate, and bicarbonate solution must be cultured at least monthly to ensure that bacterial contamination is below the limits set forth by AAMI standards. In addition, water is tested with the Limulus amebocyte lysate (LAL) assay to determine the degree of endotoxin contamination. Although the most common test used, the LAL assay may not detect smaller endotoxin fragments that are small enough to cross even low-flux membranes to cause pyrogen reactions. The cytokine induction assay using mononuclear cells may allow improved detection of these low-molecular-weight substances.[325,345,361]

HEMODIALYSIS ADEQUACY

HISTORICAL PERSPECTIVES

In 1973, when Medicare in the United States began to fund dialysis for any citizen with a modest record of employment, regardless of age, little attention was paid to the adequacy of dialysis. If the patient was awake and functioning at any level, the dialysis was deemed successful. As dialysis evolved and the prophylactic aspect of dialysis was better appreciated, concerns raised about the adequacy of dialysis therapy led to a meeting in Monterey, California, in 1974 that served to launch the National Cooperative Dialysis Study (NCDS).[362] Sponsored by the National Institutes of Health (NIH), this first clinical trial of HD adequacy aimed to control the average BUN level at 50 mg/dL versus 100 mg/dL, but the ultimate finding was a strong association between Kt/V_{urea} and outcome.[363] Subsequent observational studies have repeatedly confirmed the higher risk of mortality when the fractional clearance during each dialysis, expressed as Kt/V, falls below 1.2.[364–366] Another controlled trial of dialysis dose and adequacy sponsored by the NIH in the late 1990s (the HEMO Study) showed no further benefit from increasing the dialyzer single-pool Kt/V above 1.3/treatment three times weekly.[241] This study also showed that previously reported benefits from doses above 1.3 observed in uncontrolled studies were subject to bias from regression to the mean and from a newly recognized dose targeting bias.[367–369] Failure to achieve the targeted dose is apparently a risk factor in itself, independent of the actual dose. Together, these findings have led the medical community and the Medicare sponsor to issue guidelines for HD adequacy that have become standards of care in the United States and later in other countries.[94,370–372] The persistently high mortality rate in the dialysis population, although often unrelated to the dialysis itself, has spurred interest in dialysis adequacy and its methods of measurement over the years. This section reviews the rationale and methods for measuring dialysis adequacy, focusing on mathematical models of solute kinetics that have been effectively put into clinical practice in nearly all HD clinics.

In view of recent discussions about the scope of dialysis adequacy, it is important to distinguish the adequacy of the treatment itself (i.e., removal of accumulated solutes and water) from global kidney replacement therapy. The clinician must treat the whole patient, including providing treatments such as psychotherapy for depression, management of anemia, nutrition, minimizing infection risks, treatment of cardiovascular disease risk factors and, above all, maintaining a good quality of life for the patient.

However, it is important to put proper emphasis on the primary reason for the patient's attendance at the dialysis clinic, the dialysis itself. Clearly, not all compounds that accumulate in kidney failure are readily removed with dialysis, such as those that are highly protein-bound or tightly sequestered. Although it remains possible that some aspects of dialysis (e.g., high-molecular-weight solute flux) may affect conditions found in kidney failure that are not directly related to the dialysis process (e.g., anemia, cardiovascular disease) but may be considered part of dialysis adequacy, it is intuitive that erythropoietin replacement or parathyroidectomy cannot cure uremia. Although discussions focused on the role of nontraditional toxins in patient outcomes are worthwhile,[373,374] they should not distract from what is still the most critical part of maintaining health on dialysis—the treatment itself. The focus of the following discussion is on solute and water removal; standards established for other aspects of kidney replacement are discussed later (see "Management of Patients on Maintenance Hemodialysis").

UREMIA: THE SYNDROME REVERSED BY DIALYSIS THERAPY

The clinical syndrome resulting from kidney failure is a toxic state caused by the accumulation of solutes normally excreted by the kidney (see also Chapter 52). These include water-soluble, freely filtered solutes, as well as protein-bound substances, which may require active renal tubular transport for final excretion. The relationship between the syndrome and kidney disease was not obvious in antiquity and, even after the relationship was known, loss of nonexcretory functions of the kidney could be equally implicated as the cause, especially because urine volume and content, which reflect oral intake, differ little from normal as the disease progresses. When urea was discovered more than 200 years ago, investigators began to find elevated concentrations of urea and other organic solutes in patients' serum that are normally found in urine. This finding confirmed suspicions of an accumulation disease, but it was not until dialysis reversed the syndrome that this hypothesis could be considered proven. Clinicians can be confident that the immediate, life-threatening aspect of uremia is a toxic state caused by small-molecule accumulation because it is rapidly reversed by HD, a process that does little else than to remove small solutes by diffusion across a semipermeable membrane (therefore, fulfilling Koch's postulates).

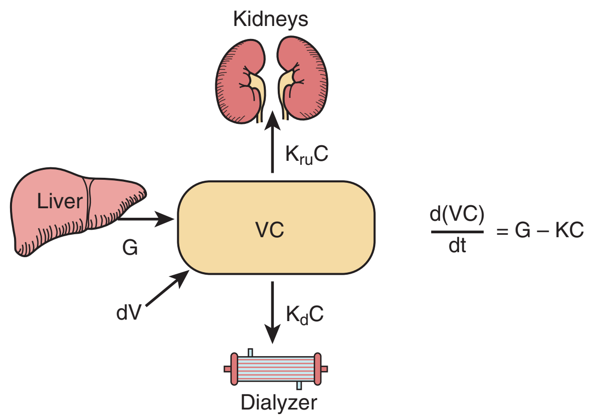


Fig. 63.20 Single-pool model of urea mass balance in a hemodialysis (HD) patient. When the patient is not undergoing dialysis (most of the time for conventional HD), K_d is 0, and removal is determined solely by K_{ru}. V is the urea distribution volume, equated to body water space; C is urea concentration; and dV is the rate of fluid gain (negative during dialysis, positive between dialyses). During HD, total clearance (K) is the sum of K_d and K_{ru}. An explicit solution is available to the differential equation that describes the rate of urea accumulation or loss (dVC/dt) as the difference between generation *(G)* and removal *(KC).*

MEASURING HEMODIALYSIS ADEQUACY

As noted in the discussion of the general principles of hemodialysis, measuring dialyzable solute levels in the blood as a method for assessing the effectiveness or adequacy of dialysis treatments has been replaced by measuring the clearance of the marker solute urea. Clearance can be measured instantaneously across the dialyzer or as an integrated parameter over time. For native kidney function, the latter is achieved by collecting timed urine specimens. For intermittent HD, collection of dialysate is impractical, so advantage is taken of the perturbations in serum urea concentrations that allow estimation of the urea clearance simply by sampling blood before and after dialysis. The magnitude of the reduction in urea concentrations during each HD session can be translated to a urea clearance much like the decrease in drug levels after a loading dose can be used to measure the drug's clearance. Application of well-established pharmacokinetic principles to urea kinetics provides an estimate of the elimination constant for urea (K/V), which is essentially the slope of the decrease in concentration expressed on a logarithmic scale, as shown in Fig. 63.19. K is the urea clearance and V is the volume of urea distribution, which is the patient's total body water volume. If one incorporates the treatment time element and ignores fluid removal and the generation of urea during dialysis, the log of the ratio of predialysis to postdialysis BUN values can be simply translated to Kt/V (see Fig. 63.19).

Because substantial fluid volume is removed as part of therapeutic HD, and a significant amount of urea is generated, especially during longer treatments, a more formal model of urea mass balance, shown in Fig. 63.20, is required to measure Kt/V accurately. In addition to the change in urea volume and urea generation, this model can be extended to include the interdialysis interval and the effects of residual kidney function (residual kidney urea clearance, K_{ru}). The latter, in contrast to the dialyzer clearance, is a continuous clearance that has minimal effect on urea removal during intermittent dialysis but provides a marked benefit between treatments when the dialyzer clearance is zero. In addition,

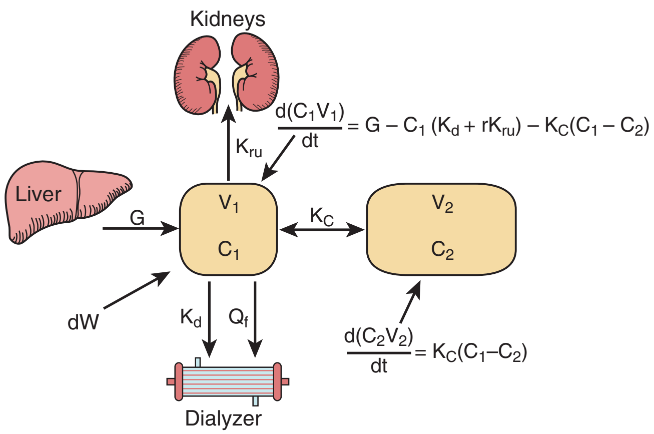


Fig. 63.21 Double-pool model of urea mass balance in a hemodialysis patient. Addition of a second compartment to the diffusion model of urea mass balance shown in Fig. 63.20 accounts for the postdialysis rebound in urea concentration shown in Fig. 63.16 and, in general, is considered a more accurate model. K_c is the coefficient of mass transfer between compartments, analogous to dialyzer K_0A. Solution of the differential equation requires numeric analysis and is not commonly applied in dialysis clinics.

solute sequestration or delayed transport of dialyzed solutes in the patient during dialysis, as discussed before, causes postdialysis rebound (see Fig. 63.17) and can be incorporated in the model if a second compartment is included, as shown in Fig. 63.21. The single-compartment model shown in Fig. 63.20, however, remains the standard for measuring dialysis in most dialysis clinics, primarily because of the complexities of the two-compartment model, which requires numeric analysis to solve, but also because the errors in the single-compartment model caused by ignoring two-compartment effects tend to cancel out.[185] The two models give similar results for Kt/V in the usual clinical range of Kt/V when dialysis is provided three times weekly.[185,375] A two-compartment model with formal numeric analysis is available on the Internet and may become useful for measuring nonstandard dialysis schedules, such as daily or nocturnal dialysis or for more prolonged treatments given three times weekly.[376,377]

IMPORTANCE OF THE POSTDIALYSIS CONCENTRATION

Accurate measures of predialysis and postdialysis BUN are required to measure the delivered dose of dialysis reliably (see Fig. 63.19). Predialysis sampling of blood is straightforward, but the postdialysis BUN can vary significantly, depending on when it is drawn (see Fig. 63.16), and measurement errors are more significant when the BUN level is low.[378,379] Although it decreases more slowly toward the end of dialysis, the BUN level rebounds rapidly as soon as the blood pump is stopped (see Fig. 63.16). The early rapid phase of upward rebound is determined by both access recirculation and cardiopulmonary recirculation. Efforts should be made to sample after access-related rebound is complete but before cardiopulmonary rebound begins. The Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines recommend slowing the blood pump to 100 mL/min for 15 seconds (to permit access rebound) or stopping the dialysate flow for 3 minutes and then drawing the sample from the dialyzer inflow port.[380] Access recirculation dilutes the postdialysis BUN level, causing a falsely high Kt/V, which can endanger the patient because of inadequate dialysis. Sampling after cardiopulmonary rebound has begun gives a falsely low Kt/V.

SOLUTE GENERATION

In addition to measuring the dose of dialysis, urea modeling allows the measurement of two patient parameters that independently influence the patient's risk of mortality—urea generation (G) and the patient's volume of urea distribution (V). Accumulation of urea in the patient results from both amino acid catabolism, a measure of protein nutrition, and failure of renal excretion. Although these dual effects on urea concentrations complicate the interpretation of any single measured level, mathematical modeling of urea mass balance allows separation of the two and an estimate of urea distribution volume. Both higher urea generation rates and higher urea volumes have been associated with lower patient mortality.[381,382] For patients dialyzed three times weekly, diurnal variations in urea generation have little effect but, for nocturnal dialysis, the reduction in urea generation at night can cause a significant error, an overestimation of Kt/V and underestimation of V if G is modeled as a constant.[383]

Blood concentrations are the net effect of solute generation and elimination. If one attributes uremic toxicity to the concentrations of accumulated solutes (concentration-dependent toxicity), it might seem logical that the clearance (Kt/V) should sufficiently balance the generation rate to maintain a safe low concentration. However, during the NCDS, attempts to demonstrate this relationship by reducing the dose of dialysis in patients who ate poorly caused an unfortunate vicious cycle of uremia-induced anorexia and malnutrition that eventually led to early discontinuation of the study.[384,385] Similarly, observational studies have shown consistently enhanced survival in patients who eat more, even when Kt/V is held constant, and patients who generate more creatinine have a similar higher rate of survival.[183,386] It appears that the relationship between diet and uremic toxin generation and elimination is complex and poorly understood. Control of solute concentrations by dialysis clearly improves outcomes, but control by limiting dietary intake is often ill-advised. Consuming a vegetarian or high-fiber diet may alter intestinal flora and reduce the generation of gut-derived toxins[387–392]; many of these purported uremic toxins are not removed well with dialysis. Although altering gut microbiome or diet may benefit the patient and reduce the residual syndrome, it does not obviate the importance of assessing what the dialysis treatment is accomplishing in terms of quantification of urea clearance as a surrogate for small, water-soluble toxin removal.

UREA VOLUME

Urea modeling essentially provides a measure of the urea elimination constant, which can be considered as the fractional rate of urea disappearance during HD (K/V). To calculate K, one must know V or vice versa. Because the prescribed K should be the same as the delivered K, and prescribed K can be determined from Eq. 5, modeled V is easily determined. By convention, V is expressed after dialysis because it is less variable. Comparison of modeled V from dialysis to dialysis can be used as a quality assurance measure, and values should not differ by more than 15%.[185,393] Causes of a discrepancy include access recirculation, dialyzer malfunction (e.g., from clotting or fouling of hollow fibers), blood pump variances, and blood sampling and measurement errors.[394,395]

Several studies have shown that various measures of body size, including V, are associated independently with mortality (Fig. 63.22).[381,382,396,397] Survival rates in larger patients are higher than in smaller patients for reasons that are not entirely clear but may be related to nutrition and the caloric buffer afforded by muscle and fat. Because body size expressed as V is the size-normalizing factor for urea clearance in the Kt/V expression, larger patients require higher clearances and are therefore at higher risk for underdialysis. However, correction for the favorable influence of large size on mortality tends to mitigate this risk, as shown in Fig. 63.23A.[382] Kt/V is a more powerful predictor of mortality than body size, and correction of Kt/V for the independent (and opposing) risk associated with body size renders it an even more powerful predictor of mortality (see Fig. 63.23B).

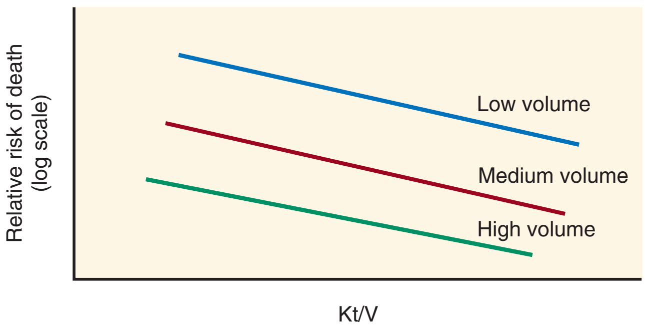


Fig. 63.22 Risk of death as a function of dialysis dose and body size. The risk of death in hemodialysis patients decreases with increased dialysis dose *(Kt/V)* and may be further stratified by urea volume as a measure of body size. Larger patients, in general, have a lower death risk.

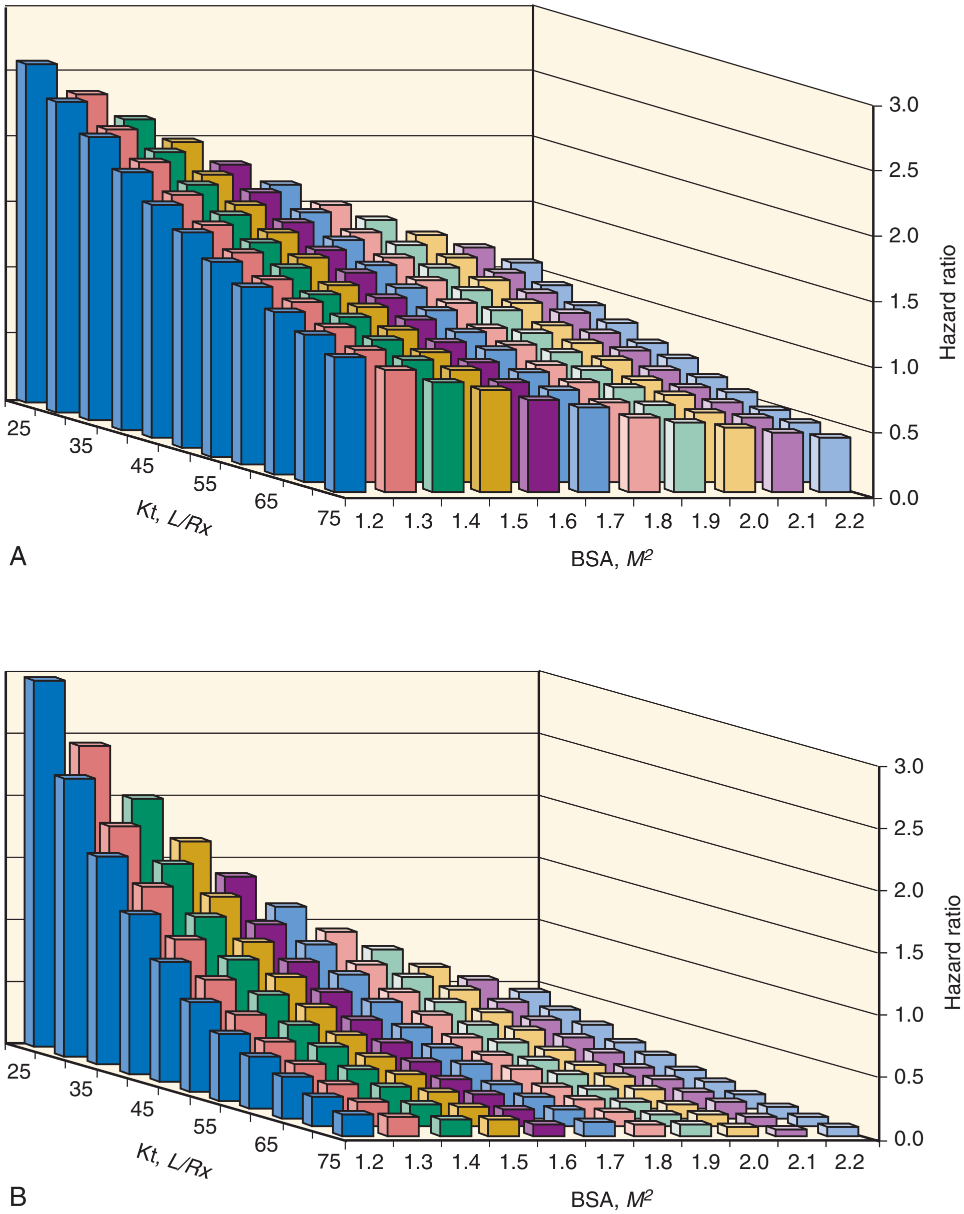


Fig. 63.23 Risk of mortality related to body size and dialysis dose. These data were obtained from a large observational study of 43,334 patients. (A) Hazard ratio analysis was adjusted for case mix. (B) Hazard ratio analysis included an interaction term between Kt and body mass index and was adjusted for case mix. *BSA,* Body surface area; *L/Rx,* liters per treatment. (From Lowrie EG, Li Z, Ofsthun N, Lazarus JM. Body size, dialysis dose and death risk relationships among hemodialysis patients. *Kidney Int.* 2002;62:1891–1897.)

The HEMO study has uncovered a potentially size-independent effect of sex on the response to higher doses of Kt/V. Although mortality was not affected by administering a higher dialysis dose for the 1846 randomized patients as a whole, when women were analyzed separately, a borderline significant improvement in outcomes was seen at the higher dose.[398] The counterbalancing effect was a nonsignificant higher mortality in men, especially African-American men. However, sex was difficult to separate from size because the two are so closely linked, especially with regard to V. If body surface area is considered to be the more appropriate denominator for dosing dialysis,[399] women, and perhaps smaller men, would clearly require more dialysis than larger men when the dose is measured as Kt/V (Fig. 63.24).[400–403] Similarly, malnourished patients who lose weight have an automatic increase in Kt/V unrelated to the effort of dialysis, but simply because the denominator in the Kt/V expression decreases. This dose increase in patients at higher risk of death may explain the reverse J-shaped relationship between Kt/V and survival in observational studies.[404] Although the latest update of the KDOQI guidelines for dialysis adequacy raises the idea of normalizing Kt by body surface area, use of body weight (V) to normalize dialysis dose still predominates currently.[380]

TREATMENT TIME

Attempts by patients to shorten their HD treatment times reflect the discomfort they experience toward the end of

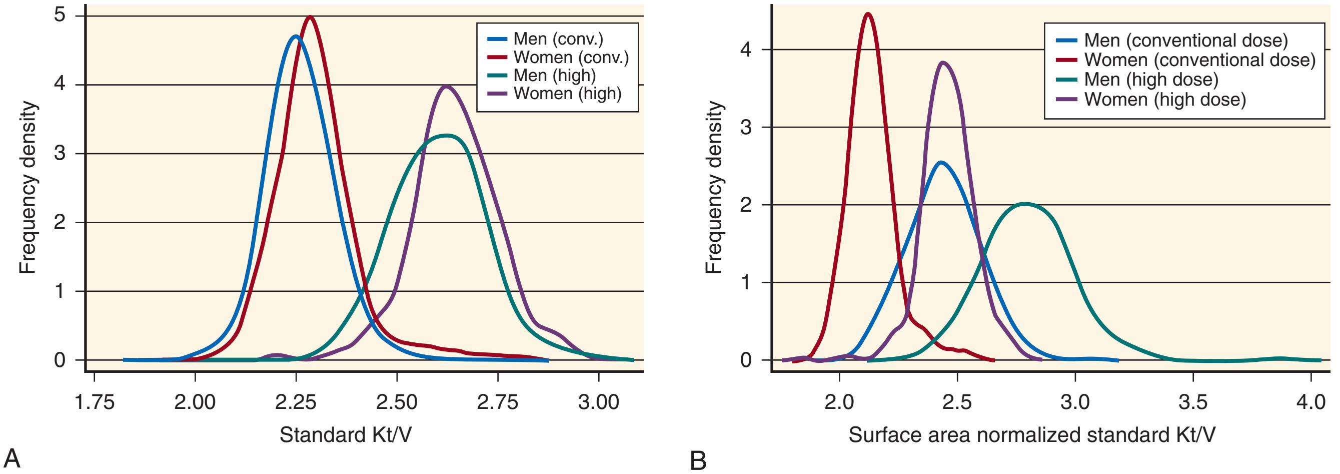


Fig. 63.24 (A) Standard Kt/V in the conventional and high-dose HEMO study subjects, by gender. (B) Surface area normalized standard Kt/V in the conventional and high-dose HEMO study subjects, by gender. Conversion to surface area was based on an anthropometric estimate of V in each patient.[403,417] *V,* Urea distribution volume. (From Daugirdas JT, Greene T, Chertow GM, Depner TA: Can rescaling dose of dialysis to body surface area in the HEMO study explain the different responses to dose in women versus men? *Clin J Am Soc Nephrol.* 2010;5(9):1628–1636.)

the procedure. Muscle cramps, fatigue, and general malaise increase in intensity as more fluid and solute are removed. Paradoxically, shortening the treatment typically accentuates these symptoms because the rate of removal must increase if the patient is to remain in solute and water balance. Extending treatment time (Td) or increasing the dialysis frequency tends to alleviate these symptoms. Many hours have been spent by nephrologists and dialysis nurses in attempts, often unsuccessful, to convince patients that extending Td would be beneficial. Sometimes a temporary trial of either an extended Td or increased frequency is sufficient to persuade the patient.

Although the NCDS has shown only a borderline significant effect of Td, most population studies have shown that a longer Td is associated with enhanced survival.[405–408] Like the NCDS, the HEMO study failed to show a significant benefit from a longer Td but it also did not specifically target Td, and the range of treatment times in the study was limited.[241] No clinical trials specifically targeting long versus short conventional treatments has been done, but prospective observational studies and clinical experience favor prolonged treatments, including slower ultrafiltration rates.[369,407,409–411]

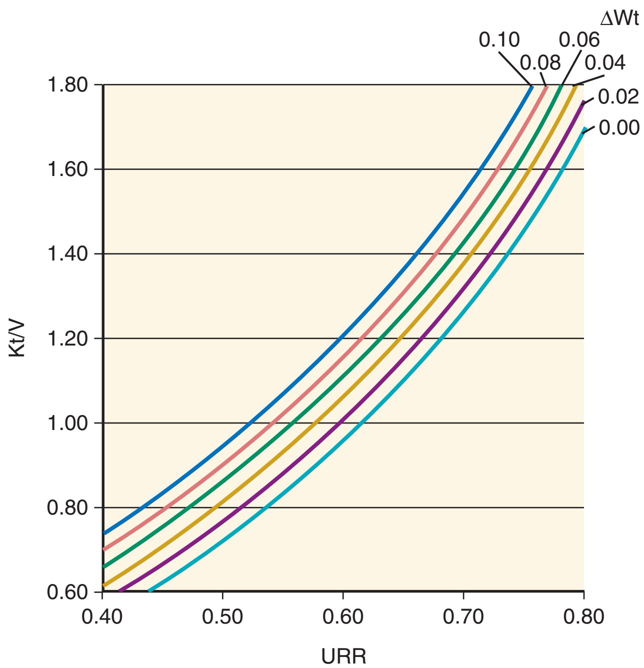


Fig. 63.25 Curvilinear relationship between urea reduction ratio *(URR)* and Kt/V, stratified by degree of ultrafiltration during dialysis. Whereas the URR (see text) falls with increasing fluid removal during dialysis (ΔWt) from 0% to 10% of body weight, Kt/V increases. The latter more appropriately accounts for the increase in clearance caused by ultrafiltration. Curves are derived from formal urea modeling.

ALTERNATIVE MEASURES OF DIALYSIS

UREA REDUCTION RATIO

In the past, the US Centers for Medicare and Medicaid Services required participating dialysis clinics to report the monthly urea reduction ratio (URR), as defined here, for each patient:

$$URR = (C_0 - C)/C_0 \qquad [10]$$

where C_0 is the predialysis BUN and C the postdialysis BUN level. Currently, Kt/V (using formal kinetic urea modeling or the Daugirdas II equation) is the standard reporting metric.

URR has the advantage of simplicity, but it is the least accurate measure of HD. For example, as the frequency of dialysis increases, and presumably its efficiency improves, URR decreases. It is not possible to add URR values to show a cumulative weekly effect and, as the frequency extends to continuous dialysis, URR extinguishes to zero. URR is also reduced by interdialysis fluid accumulation, urea generation, and residual renal function, and the additional clearance afforded by fluid removal during dialysis is not incorporated in URR. On the positive side, however, in addition to its simplicity, URR has a curvilinear relationship with Kt/V, as shown in Fig. 63.25, paralleling the relationship between outcome and Kt/V. For example, if Kt/V doubles from 1.5 to 3.0/dialysis three times a week, URR increases only from 0.75 to 0.85.

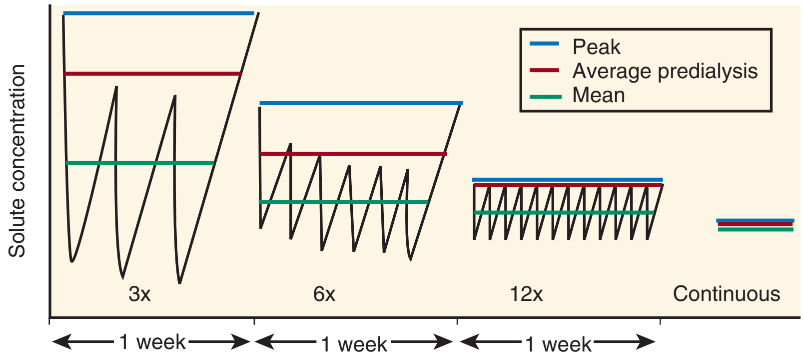


Fig. 63.26 Effect of frequency on peak and average solute concentrations. Two-compartment formal kinetic modeling of a solute with low K_c predicts that both peak and mean concentrations decrease significantly as the frequency of treatments increases, despite no change in the weekly dialyzer clearance × treatment time (Kt).

Although efforts have been made to convert Kt/V to a URR equivalent[412] or to use the solute removal index, a more reliable index of dialysis dose,[413] these approaches have not been popular. Other efforts to report the reciprocal of Kt/V as a concentration equivalent,[414] targeting low concentrations instead of high clearances, has not been applied, partly because Kt/V has become ingrained in the practice of dialysis quantification. URR is mathematically inexact because it is the ratio of postdialysis to predialysis BUN values (URR = $1 - C_{post}/C_{pre}$) instead of predialysis to postdialysis values. However, it correlates better than Kt/V with small (toxic) solute concentrations, and its relationship with the outcome is more linear.

CONDUCTIVITY CLEARANCE

The average clearance of small (dialyzable) solutes is easily derived from measurement of the predialysis and postdialysis BUN levels, as explained earlier. The instantaneous clearance of small solutes is also easily measured from the dialyzer inlet and outlet BUN, but it can also be derived by measuring the change in electrical conductivity of dialysate before and after an abrupt change in the dialysate concentration (conductivity clearance).[415,416] Because the major electrically conducting ion in the dialysate is sodium, conductivity clearance is primarily a measure of sodium clearance, which is nearly equivalent to urea clearance. This method requires multiple measurements during dialysis to obtain the treatment average, as well as an adjustment for cardiopulmonary recirculation, but it has the advantage that no blood specimens are required, and the result is available immediately because all measurements are done via the conductivity monitor on the dialysate side of the machine. The clearance is expressed in milliliters per minute and must be adjusted to body size using an estimate of V for comparison with Kt/V or an estimate of surface area.[399,400,417,418] Surface area as a denominator is more consistent with measures of native kidney clearance and may reduce or eliminate the potential gender error discussed previously.[397,401,403]

COMPARISON OF HEMODIALYSIS AND PERITONEAL DIALYSIS DOSES

The minimum recommended weekly dose of peritoneal dialysis (PD) expressed as Kt/V is 1.7 (see Chapter 64 for discussion of PD adequacy). This dose compares with a cumulative 3.6/week for HD (1.2/dialysis three times weekly). Although the minimum HD dose is more than twice the minimum PD dose, outcomes are similar or better, even when adjusted for the lower average comorbidity in PD patients.[419–421] Furthermore, solute kinetic analyses have shown that the dialysis efficiency improves with increased frequency of treatments (Fig. 63.26). These observations, together with acknowledgment of little or no benefit from more intense or more prolonged intermittent dialysis[241] have led to the conclusion that intermittent treatments are less efficient than continuous treatments and have stimulated efforts to define a continuous equivalent clearance expression for HD.

STANDARD CLEARANCE AND STANDARD KT/V

These efforts have produced a continuous equivalent expression for urea clearance as G/TAC,[185,422] where TAC is the time-averaged urea concentration and a more profound adjustment defined as “standard K” and “standard Kt/V.”[413,423] The latter redefines clearance as the removal rate factored for the predialysis concentration, placing more emphasis on the predialysis BUN level as a risk factor for uremia. Because the predialysis BUN level is always higher than the mean BUN, the standard Kt/V is always lower than the continuous urea clearance and is comparable to fractional clearances achieved with continuous PD. Despite its somewhat arbitrary definition, the matching of doses with PD has generated interest in standard Kt/V (stdKt/V) as an expression of dialysis dose that is independent of frequency. Increasing interest in clinical applications of HD given more frequently than three times/week has generated a need for quantification that accounts for the improved efficiency of more frequent treatments. For patients in a steady state of urea, mass balance in which generation equals removal and dialyzed according to any schedule of treatments, stdKt/V is defined as follows:

$$\begin{aligned} \text{stdKt/V} &= \text{(urea removal rate)/(peak concentration)} \\ &= \text{G/(average predialysis BUN)} \end{aligned} \quad [11]$$

where G is the patient’s urea generation rate derived from formal urea modeling. KDOQI guidelines call for a minimum stdKt/V of 2.0/week, significantly higher than the minimum PD dose but considered safe in the absence of controlled

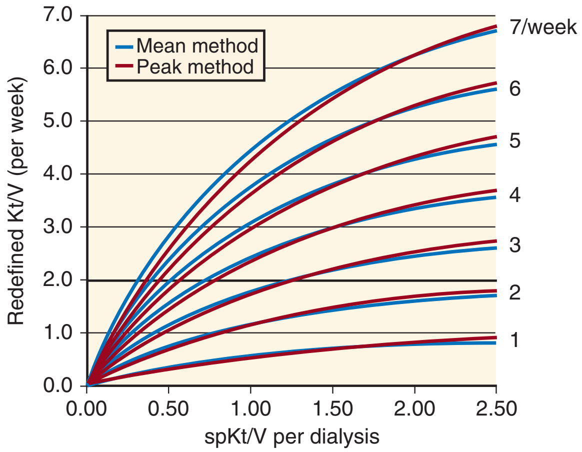


Fig. 63.27 Effect of increased dose versus increased frequency on effective clearance. The effective clearance, expressed as "standard Kt/V" on the vertical axis, tends to plateau, despite increases in dialyzer clearance, expressed as single-pool Kt/V (spKt/V) on the horizontal axis. Two different models of solute kinetics show similar diminishing returns as the delivered dose increases. A marked increase in effective clearance can be achieved only by increasing the frequency of treatments.

trials of dialysis frequency.[181] An explicit mathematical formula for calculating standard Kt/V based on spKt/V[424] has greatly simplified the calculation, and subsequent refinements have allowed inclusion of the effects of ultrafiltration during dialysis and the patient's residual native kidney clearance[425]:

$$\text{stdKt/V} = \frac{10080([1-e^{-\text{eKt/V}}]/\text{t})}{([1-e^{-\text{eKt/V}}]/\text{eKtV}]+[10080/\text{Nt}]-1)} \qquad [12]$$

where Nt is the number of dialysis treatments per week.

Because urea is relatively nontoxic, peak levels probably do not mediate uremic toxicity, so an alternative explanation for the inefficiency of infrequent HD was developed based on the sequestration of compartmentalized solutes other than urea. A two-compartment model that accounts for sequestration gives a pattern of average clearances that closely match stdKt/V values,[426] as shown in Fig. 63.27, providing further theoretic support for the clinical application of stdKt/V in recipients of frequent dialysis.

NOCTURNAL HEMODIALYSIS AND HOME HEMODIALYSIS

An alternative to thrice-weekly, in-center HD is nocturnal dialysis. As the term implies, dialysis is performed during the night, usually while the patient sleeps. Nocturnal dialysis can be performed in-center, three times weekly, with the major advantage of increased dialysis treatment duration of 6 to 8 hours. Observational studies have found regression of left ventricular mass, better bone mineral indices, and improved quality of life with this approach to HD.[427,428]

Home HD, which can be applied during the waking hours for 2 to 4 hours a treatment, five or six times weekly, provides more frequent HD treatments while controlling costs and patient burden.[429–433] Nocturnal home HD is a variant of this, with longer treatment times and perhaps only three or four treatments/week. The advantage of more frequent and/or longer treatments includes patient freedom during the day to conduct normal life activities unfettered by dialysis and associated symptoms. Several studies have shown improvements in blood pressure, nutrition, stamina, and health-related quality of life, which are presumably responsible for patient acceptance of a procedure that requires considerably more patient effort and time than standard in-center HD.[430,434,435] Controlled trials have confirmed improvements in left ventricular mass, blood pressure, need for phosphate binders, and some aspects of quality of life.[434,436–438] However, recruitment of patients for home training was difficult, more vascular access interventions were required, and the nocturnal group of patients in one study experienced a more rapid decline in residual kidney function.[431,437–439] Observational studies have reported similar patient and treatment survival rates, regardless of the home modality.[440]

SHORT DAILY HEMODIALYSIS

The incentive to shorten the Td is often patient-generated, as noted but, when combined with an increase in frequency, outcomes might be improved, as suggested in Fig. 63.27. Controlled studies of short daily dialysis have shown improvements in cardiac hypertrophy, health-related quality of life, and cardiac function.[434,436,437] All studies of short daily dialysis have shown decreases in interdialysis weight gain and predialysis blood pressure, but such changes are to be expected when the interdialytic interval is shortened to nearly half that of thrice-weekly dialysis. The NIH-sponsored daily in-center HD study, the largest randomized and controlled trial to date, has shown a reduction in left ventricular mass, an improvement in self-reported physical health, a reduction in predialysis serum phosphate concentration, and a reduction in predialysis systolic blood pressure relative to conventional thrice-weekly dialysis.[437] However, the trial was unable to demonstrate improvements in survival, hospitalization rates, predialysis serum albumin level (a marker of nutrition and inflammation), or a reduction in the dose of erythropoiesis-stimulating agents required to control anemia.

ACCOUNTING FOR NATIVE KIDNEY FUNCTION

Clearance of small solutes, the principal metric of dialyzer function, is augmented by clearance of the same solutes by the patient's remnant native kidney. Addition of the two urea clearances (K_d and K_{ru}) seems reasonable except for intermittently dialyzed patients in whom the two clearances do not occur simultaneously. Because, as noted previously, continuous clearances are more efficient than intermittent clearances, an adjustment is required before the two clearances can be summed. This adjustment consists either of inflating K_{ru} or deflating K_d. As noted previously, conversion to stdKt/V effectively deflates K_d to the equivalent of a continuous clearance, allowing simple addition. For example, if the patient's stdKt/V determined by Eq. 12 is 2.2/week, the native kidney urea clearance is 4 mL/min, and the patient's urea volume is 35 L, the two can be added to yield a continuous urea clearance:

$$\begin{aligned} &2.2/\text{week} \times (35{,}000\ \text{mL}/10{,}080\ \text{min/week}) + 4\ \text{mL/min} \\ &\quad = 11.6\ \text{mL/min} \end{aligned} \qquad [13]$$

or

$$4\ \text{mL/min} \times (10{,}080\ \text{min/week}/35{,}000\ \text{mL}) + 2.2/\text{week} = 3.4\ \text{weekly stdKt/V} \quad [14]$$

If the dialyzer stdKt/V is determined by formal urea modeling, care must be taken to avoid inappropriate deflation of K_{ru} when calculating stdKt/V from the predialysis and postdialysis BUN values.[425] Alternatively, K_{ru} can be inflated to add to K_d as originally described by Gotch and associates[440a] and later outlined in the KDOQI guidelines.[181]

Residual kidney function confers a survival advantage far in excess of that associated with the dialyzer's urea clearance.[440b] Despite near-complete destruction by the patient's disease, the native kidney continues to eliminate by secretion solutes that are eliminated poorly, if at all, by dialysis, helps maintain salt and water balance, and may supply synthetic functions.[178,440c,440d] Recent studies have suggested that residual kidney function may last longer than realized, prescribing HD to complement the amount of K_{ru} may slow its decline,[440e] and incorporation of K_{ru} in the prescription of dialysis is underused in many incident HD patients.[440f] Preservation of K_{ru} is therefore an important goal of kidney replacement therapy.

THE DIALYSIS PRESCRIPTION

GOALS OF HEMODIALYSIS

The goal of HD is to replace the kidneys' excretory function. To accomplish this goal, blood and dialysate are circulated in opposite directions (countercurrent) on opposite sides of a semipermeable membrane in the dialyzer (see Fig. 63.12), allowing unwanted solutes such as potassium, urea, and phosphorus to diffuse from the blood into the dialysate and permitting addition of solutes such as bicarbonate and calcium from the dialysate into blood. The concentrations of the solutes added to the dialysate mirror those normally maintained in the body by the native kidneys (see Table 63.6). An additional goal is the elimination of excess volume via ultrafiltration, accomplished by controlling the hydrostatic pressure gradient across the semipermeable membrane (see "Dialysate Circuit").

The rate of accumulation of solutes and fluid in each patient varies and depends on his or her nutritional and metabolic status and adherence to dietary restrictions. Thus, the HD prescription must be individualized to achieve these goals for each patient. The separate components of the HD prescription that may be individualized after clinical assessment are listed in Box 63.1.

Box 63.1 Components of the Dialysis Prescription

- Duration
- Frequency
- Vascular access
- Dialyzer (membrane, configuration, surface area, sterilization method)
- Blood flow rate
- Dialysate flow rate
- Ultrafiltration rate
- Dialysate composition (see Table 63.7)
- Anticoagulation
- Dialysate temperature
- Intradialytic medications

HEMODIALYSIS SESSION LENGTH AND FREQUENCY

The clearance of any solute, such as urea, can be increased by lengthening the session length or increasing its frequency. After optimizing blood and dialysate flows and selecting a dialyzer with a large mass transfer coefficient, the time during which the patient receives dialysis can be lengthened to augment solute clearance. However, because diffusive solute clearance depends on solute concentration on the blood side, the efficiency of solute removal declines over the course of the dialysis procedure, leading to diminishing returns for total solute removal, as suggested by urea concentrations with dialysis treatments longer than 4 to 5 hours (see "Hemodialysis Adequacy"). Conversely, reducing session length below 3 hours accentuates the effects of intermittence, exacerbates solute disequilibrium (see "Dialyzer Clearance Versus Whole-Body Clearance" and Fig. 63.18), reduces the clearance of larger molecules such as beta-2 microglobulin, for which removal is more time-dependent, increases the ultrafiltration (UF) rate, and increases the potential for hypotension and myocardial stunning.[12,211,212,441–446] Providing HD more frequently lessens the impact of declining solute concentrations and improves clearance but incurs added expense, resources, vascular access dysfunction, and potentially patient and caregiver burnout.[12,32,441,442,445–449]

Additional benefits of longer or more frequent HD sessions include optimal volume homeostasis and enhanced removal of high-molecular-weight, sequestered, or protein-bound solutes (see "Hemodialysis Adequacy").[32,441,442,448,450–456] Longer or more frequent HD allows the accumulated fluid to be removed over a longer period of time and may reduce intradialytic symptoms such as nausea, vomiting, cramping, and hypotension (see "Complications for Patients on Maintenance Hemodialysis"), decrease postdialysis fatigue, improve blood pressure control, and ameliorate myocardial stunning.[436–438,457,458] Sequestered solutes such as phosphate have more time to equilibrate among the various volume compartments, leading to improved total removal and lower serum concentrations. More frequent HD also may mitigate the higher cardiovascular morbidity and mortality rate observed at the end of the long interdialytic interval in patients receiving conventional thrice weekly dialysis.[451,458–461] The Frequent Hemodialysis Network (FHN) Daily Trial, the largest randomized controlled study to compare thrice-weekly, in-center HD with 6 days/week, in-center HD, has confirmed that more frequent HD improves blood pressure and phosphate control, decreases left ventricular mass and left ventricular end-diastolic volume, and results in favorable effects on the composite endpoints of death or change in left ventricular mass and death or change in self-reported physical health.[437,457] However, more frequent HD did not significantly improve several tests of cognitive function, measures of depressive symptoms, serum albumin level, or utilization of ESAs. The FHN and Canadian studies comparing frequent nocturnal HD with thrice-weekly HD, hampered by the small number of trial participants, only demonstrated beneficial effects on blood pressure and phosphate control and possibly left ventricular mass.[436,438] Because of the increased burden and cost of longer or more frequent HD sessions, along with concerns regarding the adverse effects of more frequent HD on the provision of vascular accesses procedures and (in the smaller, nocturnal trial) on residual kidney function,[439,447,449] more frequent HD has not been routinely adopted. The current conventional practice in the United States is to prescribe HD three times/week for 3 to 4 hours each session.[441] Longer duration or more frequent dialysis is often used for larger patients, patients with severe hypertension not responding to maximal antihypertensive treatment, or those with volume overload and intradialytic hypotension preventing fluid removal. Current session lengths average about 3.5 hours in the United States, about 3.25 hours for women and 3.75 hours for men on average, depending on body size. Prolonging session length and/or including one additional session of HD or UF to avoid the long interdialytic interval can help patients who are struggling to maintain their health and well-being on a conventional thrice-weekly HD schedule, possibly mitigating risks that might develop with daily or near-daily (five or six times weekly) therapy.[441,442,449]

DIALYZER CHOICE

In choosing a dialyzer, the most critical determinants are as follows: (1) its capacity for solute clearance; (2) its capacity for fluid removal; and (3) the potential of the dialyzer membrane to interact with components of the blood or the degree of biocompatibility.[209] The ideal hemodialyzer membrane would have high clearance of low-molecular-weight and mid-molecular-weight uremic toxins, adequate UF, high biocompatibility, and low blood volume compartment to maximize the efficiency of and reduce the adverse metabolic and hemodynamic effects from the HD procedure.

Urea is the solute most often used to evaluate dialyzer solute clearance characteristics because of its relevance to kinetic models of dialysis adequacy (see earlier). In clinical practice, physicians rely on industry-derived determinations of in vitro dialyzer clearance of low-molecular-weight and mid-molecular-weight solutes. Gibbs-Donnan effects, membrane adsorption of solute, protein binding of solute, and solute aggregation are not taken into account in determining in vitro dialyzer clearances and will reduce in vivo clearances. The variable relation between the diffusive and convective clearance of a solute further complicates the determination of solute clearance of different dialyzers. Solutes larger than 300 Da have lower diffusive clearance than smaller solutes such as urea and potassium and may rely primarily on convective clearance. For patients with large interdialytic weight gains requiring more UF during each dialysis session, simple comparisons of the in vitro diffusive solute clearances may be misleading.

Another factor in the selection of a hemodialyzer is its UF coefficient, which describes the capacity of a dialyzer to remove fluid (in units of mL/min/mm Hg). As with solute clearances, the manufacturer performs in vitro tests to determine the UF coefficient of each dialyzer model. In vivo values may vary by as much as 10% to 20%.

As already discussed, dialyzer membranes vary in their capacity to activate the coagulation cascade and formed blood elements, with synthetic membranes, in general, being the most inert and hence the most biocompatible,[210,213–216] but even synthetic membranes vary in the degree of biocompatibility.[217,462,463] In addition to the issues of biocompatibility and cytokine release, activated thrombin is adsorbed on the dialyzer membrane, creating a nidus for platelet adhesion and further thrombin deposition.[464] The propensity of a dialyzer for thrombogenesis may be another important factor in selecting a dialyzer, especially when anticoagulation during dialysis is not feasible. It is unclear whether dialyzers bonded with heparin will reduce the incidence of thrombosis during heparin-free dialysis. A randomized crossover study has suggested that such dialyzers are superior to both saline flushes and infusion in preventing intradialytic thrombosis,[465] whereas others have found comparable thrombotic rates with the use of saline flushes or a polysulfone memebrane.[466,467]

An additional consideration in hemodialyzer selection is whether it will be reused for subsequent dialysis sessions because the chemicals used in reprocessing dialyzers may damage some of the membranes.[207] Bleach, which is commonly used to strip protein off the membrane and improve the appearance of the dialyzer, may increase the pore size of some synthetic membranes after repeated use. This results in the loss of plasma proteins during each dialysis session, rivaling that seen in nephrotic patients. Heat disinfection may result in cracks in the headers of the dialyzers.

BLOOD AND DIALYSATE FLOW RATES

Configuring the dialysate flow countercurrent to blood flow maximizes the concentration gradient between the two throughout the length of the dialyzer (see Fig. 63.12A and Table 63.5). When flows are in the same direction (cocurrent), small solute clearance decreases by about 10%. Increasing the dialysate flow (Q_d) reduces the accumulation of waste products in the dialysate and provides a higher solute gradient between blood and dialysate for optimal diffusion. Higher Q_d also decreases boundary layers and streaming effects in the dialysate. Dialysate flowing along the membrane tends to adhere to it to create an unstirred layer, or boundary layer, which reduces the rate of diffusion across the membrane.[214,468] Dialysate also tends to move along the path of least resistance or channel (streaming effect), resulting in nonuniform flow and bypassing some of the membrane area. As dialysate flow increases or turbulence is produced at the membrane surface, the unstirred layer becomes thinner, channeling is minimized,[469] and K_0A increases,[470] although the effect is less in vivo than in vitro.[471,472] These findings prompted an increase in Q_d from 500 to 800 mL/min when the dialyzer blood flow rate (Q_b) was prescribed at 350 to 500 mL/min. Advances in hemodialyzer technology have led to modification of the hollow fiber shape and insertion of inert spacer yarns, reducing channeling and unstirred layers and further improving dialyzer performance.[205,473] With these new dialyzers, increasing Q_d above 600 mL/min has yielded minimal increases in urea, phosphate, and beta-2 microglobulin clearance,[474,475] but may still have a significant impact on the clearance of protein-bound solutes.[476,477]

Dialyzer blood flow (Q_b) is driven by a roller pump and usually ranges from 200 to 500 mL/min, depending on the type of vascular access. Blood flow influences the efficiency of solute removal (see Table 63.5). As Q_b increases, more solute is presented per minute to the membrane, and solute removal increases. Urea removal increases steeply as Q_b increases to 300 mL/min, but the rate of increase is less steep as Q_b approaches 400 to 500 mL/min because of increased resistance to and turbulence of flow within the hollow fibers, resulting in nonlinear flow and reduced clearance. Unlike dialysate, the boundary layer and streaming effects are less prominent on the blood side of hollow fibers because of the geometric advantages of flow in hollow fibers, the scrubbing effects of RBCs, and less variance in Q_b. For larger molecules, sequestered solutes and protein-bound solutes, removal is slower and more time-dependent rather than flow-dependent because of limited diffusion across the membrane and protein binding, as discussed previously.[440d,441,442,454–456]

ANTICOAGULATION

Blood clotting during dialysis results in patient blood loss and reduces solute clearance through decreasing the dialyzer surface area.[479,480] To prevent clotting, an anticoagulant is usually delivered into the blood circuit before the dialyzer via a peristaltic pump or syringe pump.

Heparin, the most commonly used anticoagulant, may be given as a bolus at the start of dialysis (fixed dose 1000–5000 U or weight-based dose 50-U/kg bolus) followed by a continuous infusion (1000–1500 U/hr) until 15 to 60 minutes before the end of dialysis or as intermittent boluses as needed during dialysis.[479–483] Disadvantages of the bolus method include an increase in nursing time and episodic over- and underanticoagulation. In patients at risk of bleeding (Table 63.8), low-dose heparin (500–1000 U bolus followed by 500–750 U/hr), regional anticoagulation, dialyzers coated with heparin, or no anticoagulation may be appropriate.[479,482,483]

In regional anticoagulation, the anticoagulant is infused into the blood circuit (arterial line) before the hemodialyzer, followed by infusion of a neutralizing agent into the venous line (after the dialyzer). Regional citrate anticoagulation, a common strategy in the acute dialysis setting, uses citrate as the anticoagulant and calcium as the neutralizing agent, with the dialysate being calcium-free.[482–485] Citrate binds calcium in the blood, an important cofactor in the coagulation cascade, thereby inhibiting clotting in the dialyzer. Infusion of calcium after the dialyzer restores the ability of blood to clot. Regional anticoagulation also may be accomplished with heparin as the anticoagulant and protamine as the reversing agent.[486,487] Both methods are labor-intensive and prone to error in inexperienced hands, requiring frequent monitoring of ionized calcium or partial thromboplastin time, respectively, when using the citrate-calcium or heparin-protamine combination. Citrate anticoagulation also may result in hypocalcemia if calcium replacement is inadequate and metabolic alkalosis as citrate is metabolized.[483–485] However, in intermittent short-duration HD, metabolic alkalosis may not be an issue.[484] Rebound in anticoagulation may be seen after the completion of dialysis with regional heparinization because heparin has a longer half-life than protamine. Because of the close monitoring required and the risk of serious complications, regional anticoagulation is not commonly used in the outpatient dialysis setting and is used more in the intensive care unit for continuous renal replacement therapy. However, if a simplified treatment protocol can be perfected, regional citrate anticoagulation may become more feasible and desirable in the outpatient setting because citrate may reduce inflammation, lower bleeding risk, and improve clearance from less dialyzer clotting when compared with heparin.[485,486,489–492] Currently, low-dose heparin and anticoagulation-free HD remain more commonly used strategies in outpatients.

During anticoagulation-free dialysis, several strategies may help prevent clotting, such as the following: (1) rinse the circuit before dialysis with heparinized saline; (2) use a less thrombogenic dialyzer; (3) flush the circuit with 100 to 200 mL of 0.9% sodium chloride every 30 minutes during dialysis; (4) avoid blood or platelet transfusions through the circuit; (5) maintain a high blood flow rate to decrease sludging of blood in the hollow fibers; and (6) limit UF as feasible because hemoconcentration in the hollow fibers increases thrombotic risk. In a patient with a hypercoagulable state or the situation in which higher blood flow rates and limited UF are not possible, these measures are unlikely to prevent clotting. Remaining options, then, are regional citrate anticoagulation or the use of heparin-coated dialyzers,[483,493] although heparin-coated dialyzers may be inferior to citrate in reducing dialyzer clotting.[467,493]

Alternative anticoagulants include low-molecular-weight heparin (LMWH), hirudin, prostacyclin, dermatan sulfate, and argatroban.[480–483,494–501] Of these, LMWH is more widely used in Europe,[483,496–498] but the complexity of use, expense, lack of sufficient experience, and equivalency to heparin have deterred the widespread use of the other anticoagulants. For the rare patient with confirmed heparin-induced thrombocytopenia, lepirudin, bivalirudin, argatroban, and citrate anticoagulation are viable alternatives.[483,501–503] Finally, substituting citric acid for acetic acid in the dialysate may augment the effect of heparin use, improve clearance, and increase dialyzer reuse, presumably because of decreased clotting, but may increase cramps and hypotension.[504–507]

Table 63.8 Guidelines for Anticoagulation in Hemodialysis Patients at High Risk for Serious Bleeding

Anticoagulation for Hemodialysis	Clinical Condition
No anticoagulation or regional anticoagulation	Actively bleeding
	Significant risk for bleeding
	Major thrombostatic defect
	Major surgery within 7 days
	Intracranial surgery within 14 days
	Biopsy of visceral organ within 72 hours
	Pericarditis
Low-dose heparin	Major surgery beyond 7 days
	Biopsy of visceral organ beyond 72 hours
	Minor surgery 8 hours prior
	Minor surgery within 72 hours
Low-dose heparin or no anticoagulation	Major surgery 8 hours prior

DIALYSATE COMPOSITION

The composition of dialysate is crucial to attaining the desired blood purification and to achieving body fluid and electrolyte homeostasis.[508–511] To reach these endpoints, dialysate contains the solutes listed in Box 63.1 in concentrations comparable to those of plasma. Addition of electrolytes and glucose to the dialysate reduces or eliminates their concentration gradients and prevents excessive removal during dialysis. Potassium is nearly always individualized; sodium, calcium, and bicarbonate concentrations may also be individualized, although they may be standardized for most patients in a facility where centralized dialysate production and distribution are used (see Table 63.7). Because the dialysate glucose concentration is comparable to plasma, osmotic forces do not drive fluid removal, unlike PD.

SODIUM

Because sodium is the major determinant of tonicity of extracellular fluids, the dialysate sodium concentration influences cardiovascular stability during HD. Historically, the dialysate sodium concentration was kept lower than blood sodium concentration (130–135 mEq/L) to facilitate diffusive sodium loss during dialysis and prevent interdialytic hypertension, exaggerated thirst, and excessive interdialytic weight gain.[297–299] However, with the advent of high-flux dialyzers and more efficient solute removal, headaches, nausea, vomiting, seizures, hypotension, and cramps became more common and were attributed to the hyponatric dialysate[297–299,310,508] but were more likely caused by the use of acetate as a source of base (see later). This development prompted the progressive increase in dialysate sodium concentration—first to that of plasma and subsequently to higher than plasma—with an improvement in symptoms.[297–299,310,508] The pendulum now has swung back, with some, but not all studies, demonstrating that a high-sodium dialysate leads to thirst with polydipsia, increased weight gains, and hypertension, leading to a resurgent interest in reducing dialysate sodium concentrations and abandoning the use of sodium modeling or "ramping" (see previously) in most patients. [297–299,301,302,306,313,508,512,513]

However, some studies have suggested that the use of a high-sodium dialysate may reduce mortality in a subset of patients whose predialysis serum levels are low.[301,302] The ongoing debate and uncertainty regarding the optimal dialysate sodium concentration are due to multiple confounding factors, such as lack of randomized and controlled studies, inaccuracy of dialysate sodium delivery compared with prescribed, differences in dietary sodium intake and urinary sodium excretion, varying tissue sodium stores, which affect vascular stiffness, and disparate predialysis plasma osmolarity not due to blood sodium concentrations that results in hypotension during dialysis as the osmolarity decreases.[297–299,307–310,514] The conflicting data have suggested that one size does not fit all, leading to a growing interest for an individualized approach. Although a computer-controlled biofeedback system using conductivity to lower the plasma sodium level to 135 mEq/L may offer the added benefits of decreased extracellular water, improved blood pressure control, and lower interdialytic weight gains without sacrificing hemodynamic stability, these methods add complexity and/or increase demand on staff time.[297–300,310,513,515] A reasonable alternative may be to apply a constant dialysate sodium of 136 to 138 mEq/L empirically or align the dialysate sodium with the average predialysis serum sodium concentration to achieve the same ends, because many dialysis patients tend to be hyponatremic.[297–299,305,515]

POTASSIUM

Unlike sodium, only 2% of the 3000 to 3500 mEq of potassium is distributed in the extracellular space. Although colonic elimination of potassium increases threefold in patients with ESKD and accounts for about 30% of dietary intake,[511] the remaining potassium accumulates in between dialysis treatments and can become life-threatening. By using a dialysate potassium concentration lower than that of plasma, excess potassium is removed during dialysis, mainly through diffusion down its concentration gradient.[269,508,510,515,516] However, potassium flux from the intracellular to extracellular compartment is usually slower than the efflux of potassium into the dialysate, and use of a high dialysate bicarbonate concentration and/or creating a large bicarbonate gradient during HD may enhance potassium shift into the intracellular compartment, potentially creating significant intradialytic hypokalemia; this is followed by a 30% rebound in the potassium concentration 3 to 4 hours after completion of dialysis.[511,516–518] Life-threatening levels of hypokalemia typically occur during the first 2 hours of dialysis, when a high predialysis potassium level favors efficient removal and a precipitous decline in its concentration, leading to arrhythmias through hyperpolarization of cardiac membrane potential, QT prolongation, and increase in ventricular late potentials.[508,510,511,516,518]

Minimizing the risk for intradialytic hypokalemia and postdialysis rebound is made even more complex by the highly variable efficiency of potassium removal among patients (≤70% variability) and between treatments for each patient (≤20%), despite an identical dialysis prescription.[519] The intracellular distribution of potassium leads to a variable volume of distribution such that the greater the total body potassium content, the lower the volume of distribution and the higher the fractional decline in potassium concentration during dialysis. During dialysis, amelioration of acidosis, stimulation of insulin release by dialysate glucose, release of catecholamines in response to hemodynamic events, and decline of plasma tonicity all favor the shift of potassium into cells, thus reducing the gradient for its removal.[508,510,511,516,518] The degree to which each of these factors is present varies considerably among patients.

Several large epidemiologic studies have sought to clarify the risks associated with varying dialysate potassium concentrations.[461,520–522] The optimal predialysis serum potassium level appears to be 4.6 to 5.3 mEq/L,[521] with poor nutritional status contributing to death at lower potassium levels and fatal arrhythmias at predialysis potassium levels higher than 5.6 to 6 mEq/L. Dialyzing against a dialysate potassium concentration lower than 2 or 3 mEq/L appears to increase the risk for sudden death, especially in patients with a predialysis potassium level less than 5 mEq/L.[461,522] A more recent DOPPS study did not find a difference in risk for sudden death between dialysate potassium concentrations of 2 mEq/L versus 3 mEq/L and was unable to interpret data for dialysate potassium lower than 2 mEq/L because of the small number of patients in this category, likely reflecting changes in practice since the previous studies.[520] Although some studies have suggested a survival benefit in hyperkalemic patients dialyzed against a low-potassium dialysate,[461,521] other studies have found no difference in survival, even in patients with predialysis serum potassium levels higher than 6.5 mEq/L.[520,522] In addition, lower dialysate potassium concentrations seem to have little effect on predialysis serum potassium, given the 2- to 3-day interval in between dialysis.[520] Individualizing the dialysate potassium concentration, depending on the unique situation of each patient, may be crucial in navigating between increased mortality from predialysis hyperkalemia and sudden death from hypokalemia during and after each dialysis session.[510,511,516]

The prescribed dialysate potassium concentration is guided by the serum concentration before dialysis and the previous considerations.[510,511,516,518] Because of the increased incidence of sudden death during and after dialysis, presumably due to a rapid decline in serum potassium levels, dialysate potassium concentration of 0 mEq/L should be abandoned. Most patients should dialyze against a dialysate potassium of 2 to 3 mEq/L. Patients with increased total body potassium from diet, medications, hemolysis, tissue breakdown, catabolism, or gastrointestinal bleeding may require a lower dialysate potassium concentration. However, concentrations of 1 mEq/L should be used only when a compelling reason exists because of the higher risk for arrhythmias and death and only after exhausting all efforts targeting dietary potassium restriction and discontinuing medications that interfere with aldosterone production and gastrointestinal elimination of potassium (e.g., ACE inhibitor, angiotensin receptor blocker [ARB], aldosterone antagonist). In particular, patients on digoxin must dialyze against a dialysate potassium level of at least 2 mEq/L because of the greater propensity for digoxin toxicity and death with predialysis potassium levels less than 4.3 mEq/L[523] and intradialytic hypokalemia. Potassium modeling, with a gradual stepdown in dialysate potassium concentration, thus keeping the blood to dialysate potassium gradient constant, during each dialysis session may optimize potassium removal and minimize the risk for arrhythmias.[508,510,511,516,524] However, experience with and data for this approach are scant and consist of small studies using electrocardiography to detect repolarization abnormalities (e.g., prolonged QT interval or QT dispersion) as surrogate markers for sudden death. The validity of these tools as surrogate markers has been questioned in the cardiology literature.[525,526] Randomized trials are needed to inform this dilemma.

The use of oral sodium polystyrene sulfonate to control hyperkalemia remains controversial, although data from DOPPS have suggested that it may be safe and effective, associated only with increased IDWG and higher serum bicarbonate and phosphorus levels.[527] New oral agents to control hyperkalemia—patiromer and sodium zirconium cyclosilicate—are now available and may help solve the potassium conundrum.[528,529]

CALCIUM

Historically, patients with ESKD were dialyzed against a higher calcium (Ca) concentration (3–3.5 mEq/L [1.5–1.75 mmol/L]) to help control hyperparathyroidism and prevent Ca and subsequent bone mineral loss.[510,511] However, increased use of Ca-containing phosphate binders and vitamin D analogs has led to more frequent hypercalcemia and concern for accelerated vascular calcification in HD patients, prompting the KDOQI committee to recommend reducing the dialysate Ca concentration to 2.5 mEq/L (1.25 mmol/L).[530] In contrast, international guidelines have recommended a concentration of 2.5 to 3 mEq/L (1.25–1.5 mmol/L).[530] The differing opinions reflect limited understanding of Ca mass balance in patients undergoing dialysis.[510,511,530–532] Data from retrospective and small randomized studies have suggested that 2.5 mEq/L is the fulcrum for dialysate Ca, below which Ca is removed from the patient and above which Ca diffuses into the patient during dialysis, although there is wide variability among patients.[510,511,531,532] Patients whose dialysate Ca concentration was reduced from 3 to 3.5 mEq/L to 2.5 mEq/L or lower had lower serum Ca and higher parathyroid hormone concentrations, correction of low bone turnover (adynamic bone disease), and improvement in vascular calcification, although other studies have reported unchanged or worsening vascular calcification with lower dialysate Ca concentration.[532–540] Vascular calcification was assessed variably with measures of aortic calcification, coronary artery calcification, arterial stiffness, carotid intimal media thickness, and carotid femoral pulse wave pressures. None of the studies addressed interdialytic Ca mass balance, which may be variable, given the selection of available phosphorus binders, vitamin D analogues, and calcimimetic agents, and its contribution to vascular calcification.

Dialysate Ca concentration may also affect hemodynamic stability during dialysis through lowering ionized Ca concentrations, resulting in impaired left ventricular contractility and peripheral vasoconstriction.[508,532,541] Levels less than 2.5 mEq/L or a higher serum to dialysate Ca gradient are associated with an increased risk of intradialytic hypotension,[510,511,532,533,541] potentially placing patients at risk of myocardial stunning and increased risk of sudden death. Using a higher dialysate bicarbonate concentration may exacerbate intradialytic hypotension because an increase in pH during dialysis may further reduce ionized Ca concentration, as well as affect the proportioning of dialysate Ca from concentrate.[510,511,532]

Whether or how vascular calcification and intradialytic hypotension relate to morbidity and mortality is not clear. Reports from studies of patients treated with lower dialysate Ca concentration (≤2.5 mEq/L) have ranged from an increased risk for hospitalization for congestive heart failure, enhanced survival, and no difference in survival to an increased risk for sudden death.[533,535,537,541] Most of these studies were small; the largest study involved over 43,000 patients but used a case-control methodology using retrospective data, with potential for residual confounding.[541] Phosphorus (P) is a potential confounder that likely contributes to vascular calcification, yet has not been adequately addressed in studies on dialysate calcium. Rat aortic rings incubated with the Ca and P concentrations found in patients before (Ca, 2.1 mmol/L; P, 2.5 mmol/L) and after hemodialysis (Ca, 2.4 mmol/L; P, 1.5 mmol/L) against a 3-mEq/L Ca dialysate concentration demonstrated increased calcification. This was not seen when aortic rings were incubated with Ca and P concentrations found in postdialysis serum with neutral Ca flux (Ca, 2.1 mmol/L; P, 1.5 mmol/L; Ca is unchanged but P is reduced).[542]

Given the complexities discussed and the observed wide variations in predialysis Ca concentration, individualizing dialysate Ca concentration may be ideal. Patients prone to hypotension or at risk for sudden death may benefit from increasing the dialysate Ca concentration to 3 to 3.5 mEq/L at the expense of increased risk for hypercalcemia, vascular calcification, and low bone turnover.[510,511,532] Until a Ca kinetic model accounting for the various factors discussed is available to guide an optimal dialysate Ca prescription, one rational approach would be to maintain near-neutral Ca mass balance during dialysis, which can be accomplished with a 2.5-mEq/L dialysate Ca in patients with predialysis Ca less than 8.75 mEq/L and a 3-mEq/L dialysate Ca in those with predialysis Ca more than 9.15 mEq/L.[542]

MAGNESIUM

As observed with potassium, only 1% to 2% of magnesium (Mg) is in the extracellular compartment.[543] Because two-thirds is in bone, Mg flux during HD is difficult to predict. With current practice in the United States of using a dialysate Mg concentration of 0.5 to 1 mEq/L, up to one-third of HD patients have low serum Mg concentrations, sometimes exacerbated by concurrent proton pump inhibitor use.[543] Raising the dialysate Mg to 1.5 mEq/L will normalize predialysis Mg levels in most patients and will make some mildly hypermagnesemic.[543,544] Traditionally, nephrology practice has been to avoid Mg supplementation in patients with advanced CKD because of their reduced ability to eliminate Mg, placing them at risk of life-threatening hypermagnesemia with attendant bradyarrhythmias and neurotoxicity (decreased deep tendon reflexes and muscle weakness). However, higher Mg levels may also inhibit vascular calcification, suppress parathyroid hormone concentrations, and reduce bone mineralization; some of these may be beneficial.[543] Low serum Mg concentrations may predispose to intradialytic hypotension, increase osteoclast activation, and enhance the propensity for ventricular arrhythmias through depolarizing resting membrane potentials.[543] Epidemiologic studies have suggested that low serum Mg concentrations are associated with mortality, but the absolute level differed among the studies and ranged from less than 1.6 to less than 2 mg/dL, with one study suggesting that the optimal serum Mg level may be between 2.7 and 3.1 mg/dL.[544–546] Adjusting for comorbidities and indices of malnutrition, inflammation, and atherosclerosis syndrome markedly attenuated the risk conferred by low serum Mg levels, but residual increased risk persisted in malnourished and/or inflamed HD patients.[544–546] Only one study found an increased risk of death with high serum Mg levels, more than 3.1 mg/dL.[546] Whether supplementing the dialysate with higher levels of Mg would reduce mortality risk is unclear. For now, consider the use of higher dialysate Mg concentration for patients with persistent intradialytic hypotension and/or at risk for cardiac arrhythmias or vascular calcification and lower dialysate Mg if adynamic bone disease is a concern.[543]

BICARBONATE

Correction of metabolic acidosis during dialysis is achieved through increasing the dialysate concentration of a base equivalent to promoting its diffusion into the blood.[510,547,548] Historically, bicarbonate was introduced into the dialysate by bubbling carbon dioxide through it to lower the pH and prevent the precipitation of Ca and Mg salts. In the 1960s, acetate was introduced as a source of bicarbonate and became the standard for 2 decades. Acetate offered the advantages of a low incidence of bacterial contamination, lack of precipitation with Ca and Mg, and ease of storage. However, it became a hemodynamic stressor when high-efficiency and high-flux dialysis was introduced in the 1980s because the higher rate of acetate diffusion into blood exceeded the metabolic capacity of the liver and skeletal muscle. Acetate accumulation led to acidosis, vasodilation, and hypotension. These complications prompted a resurgence of bicarbonate-based dialysate, which has been sustained.

The major complications of bicarbonate dialysate are bacterial contamination and the precipitation of Ca and Mg salts.[510,547,548] Gram-negative halophilic rods thrive in bicarbonate dialysate because they require sodium chloride or sodium bicarbonate to grow.[549,550] With regular disinfection of bicarbonate containers, these bacteria have a latency period of 3 to 5 days, an exponential growth phase at 5 to 8 days, and maximal growth rate at 10 days,[549] which compare favorably with a latency of 1 day, exponential growth at 2 to 3 days, and maximal growth by 4 days in a contaminated container. Thus, disinfecting the containers and mixing the bicarbonate daily help prevent bacterial contamination. The use of commercially available dry powder cartridges offers an alternative solution to this problem.[547–550]

To minimize the formation of insoluble Ca and Mg salts with bicarbonate, bicarbonate and the acid concentrate, which contains all solutes other than bicarbonate, are separated until use.[510,547,548] The acid concentrate derives its name from the small amount of acetic acid (4–8 mEq/L in the final dilution, depending on the preparation) used to ensure the solubility of divalent cations. The dialysate delivery system draws up the two components separately and mixes them proportionately with purified water to form the final dialysate. This technologic advance allowed the widespread reintroduction of bicarbonate as a dialysate buffer in the 1970s. Because some precipitation of Ca and Mg salts still occur, the dialysate delivery system must be rinsed periodically with an acid solution to eliminate any buildup.

In many dialysis centers, the bicarbonate concentration is fixed at 32, 35, or 38 mEq/L to accommodate the use of a central bicarbonate delivery system in which the bicarbonate concentrate is piped from a centrally located tank to the individual patient stations. The advantage of a centralized delivery system is fewer back injuries in the dialysis personnel, but a major disadvantage is the inability to individualize the dialysate bicarbonate concentration. As noted, dry powder cartridges placed in line at each patient station or individual bicarbonate containers at each station will allow individualized dialysate bicarbonate prescriptions.

Although correction of metabolic acidosis is desirable to reduce protein catabolism, bone demineralization, inflammation, and insulin resistance, overcorrection to generate metabolic alkalosis during dialysis may predispose patients to hemodynamic instability, reduced cerebral blood flow, paresthesias, muscle twitching, and cramping, possibly through alkalosis-induced lowering of the serum potassium and ionized Ca levels, as well as increased tissue calcium phosphate deposition.[547,548,551–553] Several large observational studies have reported that very low (<17 mEq/L) and very high (>27 mEq/L) predialysis bicarbonate levels were associated with mortality and hospitalization but, after adjusting for case mix and markers of inflammation and malnutrition, only the association between very low bicarbonate levels and adverse outcomes remained.[553–555] A higher predialysis bicarbonate level is likely a marker for poorer nutritional status. Patients with mild to moderate acidosis (bicarbonate, 20–23 mEq/L) appear to have the best survival, perhaps reflecting more robust dietary protein intake, although more recent studies have found no association between serum bicarbonate levels and mortality.[551–556] Instead, a predialysis serum pH of 7.40 or higher and a higher dialysate bicarbonate concentration are associated with death from cardiovascular causes for pH and, unexpectedly, infectious causes for the latter.[547,551,553,556] It is difficult to reconcile these findings, and no large-scale, event-driven randomized trials have compared higher versus lower dialysate bicarbonate concentrations or oral bicarbonate supplementation in ESKD.

Although lowering dialysate bicarbonate levels for patients with predialysis hyperbicarbonatemia is prudent, especially if postdialysis metabolic alkalosis is present, it may not improve outcomes. Instead, causes of malnutrition and inflammation should be identified and corrected if possible.[552,553] Whether patients with very low serum bicarbonate concentrations would benefit from raising dialysate bicarbonate levels is unclear.[551] In addition, kinetic studies have indicated that raising dialysate bicarbonate levels increases serum bicarbonate levels at the end of and 2 hours after HD, but do not affect predialysis levels.[557] Postdialysis metabolic alkalosis and its effect on serum potassium and ionized calcium levels, however, could explain a fraction of the excess mortality observed after the first HD treatment of the week.[547,553]

Residual confounding from inaccurate reporting of comorbidities, variability of serum bicarbonate measurements, disparity between prescribed and delivered dialysate bicarbonate concentration, and incomplete accounting of total base (acetate or citrate in the acid bath are converted to bicarbonate in the body) may have contributed to the disparate findings discussed previously.[547,553] Recent data have suggested that the base equivalent in the acid bath has negligible influence on the serum bicarbonate level.[557–559] If increased mortality risk is related to a high dialysate-to-blood bicarbonate gradient and abrupt changes in serum bicarbonate levels, bicarbonate modeling and individualizing dialysate bicarbonate may be of benefit, in theory.[558,560] Until a definitive answer is available, oral bicarbonate supplementation may be preferable to raising the dialysate bicarbonate to correct very low predialysis serum bicarbonate levels (<18 mEq/L).

Glucose

Historically, dialysate glucose concentrations were high (>320 mg/dL) to provide osmotic pressure for fluid removal and prevent hypoglycemia. However, a high dialysate glucose concentration can lead to hyperglycemia and reduce potassium removal through stimulation of insulin production and a consequent potassium shift to the intracellular space.[508] With technologic advances to allow alteration of hydrostatic pressure to enhance UF, using a glucose-free or lower glucose dialysate concentration of 100 to 200 mg/dL has become the current standard.[508,509] Glucose-free dialysis resulted in more episodes of hypoglycemia in diabetic patients during HD, especially in those with better glycemic control.[561] However, patients with diabetes undergoing HD against a dialysate glucose level of 200 mg/dL had more frequent episodes of hyperglycemia and evidence for increased vagal tone, which may increase the risk for intradialytic hypotension (IDH).[562] Given the paucity of available data regarding the optimal dialysate glucose concentration, continued use of physiologic dialysate glucose concentrations (100–200 mg/dL) seems reasonable.[508,563]

DIALYSATE TEMPERATURE

The dialysate temperature is generally maintained between 35° and 37°C at the inlet of the dialyzer (see "Dialysate Circuit"). If the dialysate temperature is kept constant at 37°C or at the patient's core body temperature at the start of dialysis (thermoneutral dialysis, where no heat energy is transferred through the dialysis circuit), the patient's core temperature increases during dialysis. The average predialysis core body temperature in patients undergoing dialysis is around 36.6°C[564] and increases by 0.7°C during HD when the dialysate temperature is set at 37°C.[565] The pathogenesis is incompletely understood, but may in part be due to vasoconstriction in response to fluid removal and blood volume contraction and consequent reduced heat loss from the skin.[315,564] With progressive heat accumulation, a reflex dilation of the peripheral blood vessels occurs and leads to intradialytic hypotension.

Isothermic dialysis (no intradialytic change in core temperature) using a blood temperature monitor with computer-controlled modulation of dialysate temperature improves hemodynamic stability in hypotension-prone patients.[315,318,319,564,566] If such technology is not available, cooling the dialysate temperature empirically to 35° to 36° C or 0.5° to 1°C below predialysis core body temperature also alleviates IDH but may increase patient discomfort from cold and reduce clearance through inducing flow disequilibrium and impeding solute diffusion across membranes. The hemodynamic effects of dialysate cooling appear similar to those of sodium modeling and high dialysate sodium levels but without the undesirable side effects of positive sodium balance.[267,270,318,319,564,565,567]

However, a small randomized crossover trial in 80 patients has suggested that dialysate cooling may be less effective than sodium modeling in ameliorating IDH.[568]

Small mechanistic studies have suggested that dialysate cooling ameliorates IDH through increased baroreceptor variability, with an attendant rise in systemic vascular resistance, cardiac output, and stroke volume during HD.[564,565,567] Dialysate cooling does not seem to improve vascular refilling.[569] Regardless of the mechanism, dialysate cooling in hypotension-prone patients is associated with a lower risk for cardiovascular mortality, preserved left ventricular function, and reduced injury of brain white matter, but does not appear to affect risks for hospitalization, cardiovascular events, and all-cause mortality.[320,321,564,570]

ULTRAFILTRATION RATE AND DRY WEIGHT

Another main goal of HD is to maintain fluid balance, accomplished through establishing a dry weight and applying UF during each dialysis to remove the interdialytic weight gain. UF, or fluid removal, occurs when hydrostatic pressure in the blood compartment is higher than that in the dialysate compartment, accomplished through a combination of positive pressure in the blood compartment and "negative" pressure created in the dialysate compartment. Adjusting the amount of negative pressure within the dialysate compartment controls the total level of TMP (the difference between the two pressures); the higher the TMP, the greater the UF rate.

Traditionally, dry weight is defined as the lowest body weight a patient can tolerate without becoming hypotensive, using clinical examination and evaluation as a crude estimate.[282,446,571] A more rigorous definition is the body weight at which extracellular volume is physiologic because both volume depletion and volume overload are associated with significant morbidity and mortality.[282,446,571] However, a physiologically appropriate extracellular volume and body weight are difficult to assess clinically, especially because patients undergoing dialysis vary widely in their response to fluid removal.

Although healthy individuals can tolerate a loss of 20% of their circulating blood volume before becoming hypotensive, HD patients are highly variable; some become symptomatic with as little as a 2% decline of their blood volume.[315] This wide patient variability likely results from the differing response to blood volume depletion and the disparate rate of vascular refilling from the interstitial and intracellular spaces, as well as a myriad of other factors.[315,446,572–574] Autonomic dysfunction, diastolic dysfunction, increased core temperature, and intradialytic hypocalcemia, hypokalemia, alkalosis (see previously), and myocardial stunning may all lead to an impaired cardiac response and impaired constriction of resistance and capacitance vessels during volume depletion. Dialytic removal of solutes, malnutrition, and inflammation may retard vascular refilling through decreased osmotic pressure, reduced oncotic pressure, and increased vascular permeability. Hence, patients on HD may become symptomatic before their physiologic weight is reached, and clinically determined dry weight is an unreliable measure of physiologic weight. Even the presence or absence of pedal edema and hypertension are unreliable tools to assess dry weight because they correlate poorly with volume status estimated by multifrequency bioimpedance.[282,571,575]

Newer technologies to help determine the optimal dry weight and improve tolerance of dialysis include continuous online blood volume determination during dialysis coupled with computer-controlled UF rates and UF modeling (see "Online Monitoring") and bioimpedance analysis (BIA).[575–577] Although continuous blood volume determination may reduce hypotensive episodes during dialysis, it is unable to assess the extracellular volume compartment accurately and detect patients with impaired vascular refilling; therefore, it is less useful for determining an optimal dry weight.[282,578,579]

BIA shows promise in establishing dry weight and in reducing intradialytic symptoms but is not widely used because of the underlying complex principles and lack of a gold standard method for determining dry weight to allow full validation.[576,577,580,581] Briefly, an electrical current is applied to the body, and the resistance (opposition to flow of the current) and reactance (opposition to passage of the current) are measured. The resistance is used to estimate the volume of extracellular fluid; the reactance is used to estimate the volume of intracellular compartments. Two small randomized controlled studies have suggested that multifrequency bioimpedance spectroscopy is superior to clinical evaluation in determining physiologic dry weight, as evidenced by improvements in blood pressure control, left ventricular mass index and arterial stiffness, and lower mortality.[582,583] Cases of intradialytic hypotension and access thrombosis were comparable, but the percentage of patients with residual kidney function declined from 20% to 10% in the bioimpedance group.[582] Although promising, the small number of patients studied and the potentially deleterious effects from loss of residual kidney function require further study to help define the role of BIA in optimizing volume status.[576,577]

Recent data have linked a greater rate of fluid removal during HD (UF rate [UFR]) with higher mortality, although the threshold has been debated; it ranges from more than 10 to 13 mL/kg/hour.[311,446,572,584,585] Some studies have suggested that a continuum exists with incremental risk, starting at 6 mL/kg/hour.[584,585] Strategies to reduce the UFR include reducing interdialytic weight gain through dietary sodium restriction and isonatric to hyponatric dialysis, judicious use of diuretics in patients with residual kidney function, extending session length, and/or increasing frequency of dialysis.[446,572,586] Although a higher UFR, greater interdialytic weight gain, and shortened session length are closely intertwined, each may independently confer mortality risk through mechanisms that are poorly understood.[446,587] The HD treatment itself may induce left ventricular dysfunction, independently from the hemodynamic effects of UF.[588]

Current strategies to determine dry weight rely on clinical evaluation and maintaining a high index of suspicion for volume overload, with periodic empiric challenges of the patient's end-dialysis weight by 0.2 to 0.3 kg/session when excess volume is suspected. Subtle clinical indicators include persistence of hypertension despite escalation of antihypertensive medications, reduced appetite, and very low intradialytic weight gains.[446,571] In hypotension-prone dialysis patients, UF modeling, avoiding intradialytic hypocalcemia, hypomagnesemia, and alkalosis, lowering dialysate temperature, increasing the duration or frequency of dialysis, reducing dietary sodium intake, and possibly separating UF from diffusive clearance during dialysis may be of benefit.[510,572–575] Sequential UF and diffusive clearance provides initial isolated UF with iso-osmotic removal of fluid, followed by diffusive clearance with or without additional fluid removal. Maintaining constant plasma osmolarity during UF prevents further depletion of the blood volume from fluid shifts into the interstitial and intracellular spaces, although sequential UF was found to be inferior to sodium modeling and dialysate cooling in preventing intradialytic hypotension.[589]

REUSE

Hemodialyzer reuse peaked in the 1990s, primarily because of the potential benefits of improved biocompatibility and reduced cost.[207,590,591] Reprocessing dialyzers for repeated use took advantage of dialyzer membrane coating with plasma proteins that occurs during the first treatment, effectively camouflaging hydroxyl groups that can activate complements, white blood cells, and platelets. It also reduced the incidence of first-use syndrome, thought to be mediated in part by ethylene oxide, a commonly used sterilant that is absorbed by the potting compound of the dialyzer that can induce an immunoglobulin E (IgE)-mediated anaphylactic reaction (see "Hemodialyzers").

Automated devices that reprocess the dialyzers are safer and result in lower incidences of febrile reactions compared with manual reprocessing.[207] During the cleaning process, bleach or hydrogen peroxide is used to improve the aesthetics. However, bleach also strips proteins off the membrane and can damage the membrane, thereby negating the improved biocompatibility afforded by the protein-coated membrane, reducing phosphorus clearance through increased negative membrane charge, and increasing loss of albumin.[592,593] After cleaning, dialyzer integrity is assessed by measuring the volume of the fiber bundle in the blood compartment (fiber bundle volume) and by pressurizing the dialyzer to ensure that the fibers are structurally intact (pressure test). For a dialyzer to be accepted for reuse, the fiber bundle volume must be greater than 80% of the initial value, and the dialyzer should hold more than 80% of the maximal operating pressure. The dialyzer is then packed with chemical disinfectants such as peracetic acid or formaldehyde or subject to heat disinfection, with or without citrate. Over the decades, peracetic acid has gained popularity over formaldehyde, increasing from 5% in 1983 to 72% in 2002 of centers that practice reuse; formaldehyde use fell from 95% to 20% by 2002.[594]

Close scrutiny of the safety of dialyzer reuse practices has yielded conflicting results when comparing reuse with nonreuse and when comparing the various disinfectants, largely because the studies were nonrandomized and uncontrolled.[207,591,595–597] Overall, however, data have suggested that when reuse is applied meticulously and complies with AAMI standards, risk-adjusted mortality rates are similar, and the various disinfectants are comparable.

Since the peak of 80% in 1997, the prevalence of reuse declined to 40% in 2005, with an even lower rate of 24% in 2012.[598,599] This sharp decline is largely due to a change in practice patterns in some large dialysis chain providers favoring single use, the wide availability and lower cost of dialyzers constructed with synthetic and more biocompatible membranes, lingering concerns that long-term exposure to chemical disinfectants may be detrimental to the health of patients and health care staff, and the potential for infectious or pyrogenic reactions from flawed reuse practices.[207,590,591,595,598,599]

However, abandoning reuse would result in an estimated 903,000 tons of dialyzer-related polymer waste worldwide, compared with 45,000 tons if each dialyzer were reused 20 times.[591,600–602] Research on best management of the medical waste associated with dialysis is needed, with the decline in reuse and potential rise in more frequent dialysis.

MANAGEMENT OF PATIENTS ON MAINTENANCE HEMODIALYSIS

END-STAGE KIDNEY DISEASE

ESKD is considered the level of kidney function that requires renal replacement therapy to sustain a patient's life. If HD is the therapy of choice, the patient begins a therapy that removes numerous solutes and water, as described in detail in this chapter. In some respects, the process of HD can be viewed as the glomerular component of kidney function, during which water and smaller solutes cross a membrane, limited in part by its molecular size. Unfortunately, the analogy ends because the filtering surface in HD is an artificial membrane without biologic function. For example, dialysis lacks an active tubular transport component to reclaim or further eliminate specific solutes. Beneficial solutes such as amino acids may cross the dialysis membrane and are "excreted," but larger or protein-bound substances that may be toxic are not removed at all. Furthermore, HD does not replace the metabolic capacity of the kidneys to synthesize critical proteins. Some of the renally synthesized proteins can be exogenously provided (e.g., erythropoietin), but others may not be recognized. HD remains a lifesaving therapy but is far from full renal replacement, and thus it is not surprising the general care of an HD patient is broad and complicated. This section provides an overview of the medical management of patients undergoing HD.

ANEMIA

The kidney is the major source of endogenous erythropoietin. (See Chapter 55 for a full discussion of anemia in advancing CKD and its management; in this chapter, we focus on the management of anemia in patients undergoing maintenance HD.) Before the advent of recombinant erythropoietin in 1989, severe anemia with hemoglobin (Hgb) levels below 7 g/dL was common in patients undergoing HD, leading to frequent transfusions and iron overload in many patients. HD by itself only partially corrected the anemia, presumably by improving erythrocyte survival and reducing erythropoietin resistance. Currently, with erythropoietin and optimal management of CKD, patients generally begin and undergo HD therapy with Hgb concentrations in an acceptable range.

PRINCIPLES OF ERYTHROPOIESIS-STIMULATING AGENT USE

Recombinant human erythropoietin (EPO) is extremely effective in increasing Hgb concentrations in the vast majority of patients.[603] The positive clinical effects are numerous—enhanced exercise capability, presumably in part from improved cardiac function with reduction in ventricular hypertrophy, a better quality of life, with improved physical performance, work capacity, and cognitive capacity, improved sexual function and, because of fewer transfusions, reduced rates of hepatitis, iron overload, and alloantibody production precluding transplant.[604–614]

Several preparations of EPO are produced and available worldwide (e.g., epoetin alpha, epoetin beta, epoetin omega, epoetin zeta). They differ from each other and the native hormone with respect to production methods and glycosylation. These differences may contribute to the development of neutralizing anti-EPO antibodies and pure red cell aplasia (PRCA), characterized by severe anemia due to the absence of bone marrow erythroid precursors. Reports of PRCA first surfaced in 1998, peaked in 2002, and continue to appear periodically.[615,616] Potential mechanisms for anti-EPO antibody development include immunogenic aggregation of inappropriately stored EPO—for example, at a high temperature, various organic compounds are leached from syringe components by polysorbate 80, a stabilizer substituted for albumin. Unfortunately, these neutralizing antibodies appear to react to native EPO and all forms of EPO, including epoetin zeta and darbepoetin.[617,618] Diagnosis should include a bone marrow biopsy and evaluation for anti- EPO antibodies. Affected patients are treated with withdrawal of EPO and immunosuppression, yet they largely remain transfusion-dependent. Fortunately, the incidence of PRCA appears to be on the decline. PRCA has predominantly been reported with subcutaneous administration of an older formulation of Eprex, an epoetin alfa product produced outside the United States. Newer preparations of EPO appear to have less risk of PRCA,[619] although a small number of cases may still occur.[620]

At present, HD patients with anemia are generally treated with EPO or one of its longer acting analogues such as darbepoetin (created by the addition of *N*-linked carbohydrates) or methoxy polyethylene glycol-epoetin beta (formed by adducts with polyethylene glycol). However, as discussed in Chapter 55 and later in this section, controversy exists about the safety of EPO because high doses or higher Hgb targets are associated with adverse effects, including increased risks of cardiovascular and cerebrovascular events. Thus, several novel drugs that act on alternate pathways of red cell production have recently been studied or are under development. EPO and these newer agents are collectively referred to as erythropoiesis-stimulating agents (ESAs). Further research is needed to determine the efficacy and safety of the different ESAs. As a cautionary example, peginesatide, a synthetic EPO receptor agonist, corrected anemia as well as EPO, and was approved for use by the US Food and Drug Administration (FDA) in 2012.[619] Unfortunately, postmarketing reports of serious hypersensitivity reactions, including anaphylaxis causing death, resulted in its withdrawal from the market in 2014. Prolyl hydroxylase inhibitors that increase erythropoiesis through the transcription pathway of hypoxia-inducible factor have been shown to be efficacious in anemia management,[621–624] although they remain in development, with ongoing trials necessary to determine their safety and efficacy relative to ESAs.

The optimal route of administration of ESAs remains controversial. Studies have demonstrated that subcutaneous dosing may reduce EPO requirements by approximately 30% and, possibly because of reduced EPO exposure, may be associated with less cardiovascular and cerebrovascular events.[625,626,630,631] However, intravenous administration has been favored historically in HD due to less patient discomfort, readily available blood access, and concern about the increased risk of anti-EPO antibodies with subcutaneous administration.

HEMOGLOBIN TARGET

The optimal goal for Hgb concentrations in patients with ESKD has received substantial attention since the advent of ESAs. Goals of therapy should be to minimize symptoms attributable to anemia while also reducing the need for transfusion. ESAs may mitigate symptoms at lower Hgb concentrations than formerly thought. A systematic review of randomized controlled trials up to 2015 comparing higher versus lower Hgb concentrations has shown no substantial improvement in health-related quality of life with the higher targets.[627] More importantly, multiple clinical trials have shown that Hgb targets more than 13 g/dL, in attempts to "normalize" the Hgb level in patients with CKD, are associated with adverse outcomes. The best data in ESKD have come from a large randomized trial of more than 1200 HD recipients with underlying cardiovascular disease comparing a higher versus lower hematocrit target (42% vs. 30%).[628] The trial was terminated early because of safety concerns about a trend toward worse mortality and more vascular access thrombosis in the higher hematocrit target group. Three large randomized controlled studies in the predialysis CKD population have reported that compared with an Hgb level less than 11.5 g/dL, normalization of Hgb increases risks for cardiovascular and cerebrovascular events, hypertension, and headaches.[629–631] Careful reviews have concluded that in patients undergoing HD, no data support an Hgb goal above 12 g/dL due to these risks.[632–634] These conclusions led the Kidney Disease: Improving Global Outcomes (KDIGO) group to recommend initiating an ESA when the Hgb level is less than 10 g/dL while avoiding a target Hgb greater than 11.5 g/dL.[634] A large meta-regression analysis of more than 12,000 patients has suggested that higher doses of ESAs may increase mortality risk independently of the Hgb concentration.[635]

IRON THERAPY

The role of iron in the management of anemia for patients on maintenance HD has been a fascinating saga. Before ESAs, frequent transfusion requirements in severely anemic patients resulted in iron overload and hemosiderosis, requiring iron chelation with deferoxamine to prevent hemochromatosis and other complications. The complex protocols required to ensure intradialytic removal of the chelated iron and the increased risk of infection further complicated the delivery of dialysis.[636] However, the widespread use of ESAs has completely transformed the problem from a risk of iron overload to one of insufficient iron stores limiting response to ESAs.

Patients on HD who are receiving ESAs have substantial requirements for iron, typically 1 to 2 g/year, due to loss of blood from dialysis, laboratory testing, vascular procedures, and gastrointestinal bleeding.[634] The challenge is to identify the markers that would indicate the need for iron replacement most accurately.[637] The most commonly used markers to define iron status are serum ferritin, which mirrors iron stores, and transferrin saturation, which is a surrogate for iron availability. However, these markers may have limited sensitivity and specificity, particularly because the ferritin level is frequently elevated in the setting of inflammation, independently of iron stores. A multicenter trial of patients with serum ferritin concentrations higher than 500 ng/mL, transferrin saturation less than 25%, and anemia despite ESAs has demonstrated that empiric intravenous iron replacement increases Hgb.[638] These patients are thought to have functional iron deficiency, recognition of which may be as important to anemia management as identifying absolute iron deficiency, especially with the current interest in avoiding high ESA doses. However, the long-term clinical benefits and safety of intravenous iron dosing in patients with high ferritin levels is unknown (see later).

Among patients undergoing hemodialysis, intravenous iron is more effective than oral iron in increasing Hgb and markers of iron availability.[639] Multiple formulations exist, with varying potential for side effects, including allergic reactions, such as rash, flushing, hypotension, and anaphylaxis. A large retrospective study using Medicare claims data has found the highest anaphylactic risk with iron dextran and substantially lower risk with agents such as iron sucrose and sodium ferric gluconate.[640] Newer forms of supplemental iron have garnered interest. Ferric pyrophosphate citrate was approved by the FDA in 2015 as a water-soluble iron salt that can be administered via dialysate. Small randomized trials have suggested that this form of delivery can help maintain Hgb levels.[641,642] Additionally, novel iron-based phosphorus binders may also allow for the maintenance of iron stores, reducing or eliminating the need for intravenous iron.[643,644]

Theoretically, excess iron may increase the risk of cardiovascular disease and infection by exacerbating oxidative stress, enhancing bacterial growth, and impairing phagocytic cell function.[645–647] However, whether these effects translate into clinically relevant events is unknown. Observational data have been conflicting, with some studies suggesting increased hospitalization and mortality associated with high doses of intravenous iron[648] and others finding no significant associations.[649–651] Of note, these studies were not randomized or controlled, and the dosing of iron may be affected by various confounders. Comparative studies of intravenous iron sucrose, ferric gluconate, and ferumoxytol also have shown mixed results, with mostly small observational differences between the formulations.[652–654]

Hyporesponsiveness to Erythropoiesis-Stimulating Agent Therapy

ESA hyporesponsiveness describes patients who do not achieve or maintain a target Hgb concentration, despite high and/or increasing ESA doses. Resistance to ESAs may be due to inadequate iron stores, presence of inflammation, hyperparathyroidism, inadequate dialysis, nutrient deficiencies, or underlying bone marrow diseases.[655,656] Inflammatory states are common in patients undergoing HD and may interfere with erythropoiesis and the response to ESAs beyond their effect on iron availability.[657] The use of hemodialysis catheters is associated with chronic inflammation,[658] even when not overtly infected. Occult infections in thrombosed AV grafts and inflammation from a failed kidney transplantation also should be considered in patients with ESA hyporesponsiveness. Maneuvers that reduce inflammation, such as the use of ultrapure water, may improve the response to ESAs.[656] Although more frequent dialysis was thought to improve ESA response in early trials, a randomized trial did not confirm this finding.[659]

NUTRITION

The complex relationship between protein metabolism and body composition in kidney disease is discussed elsewhere (see Chapter 60). Nutrition in HD patients is influenced by factors related to kidney failure and the HD treatment itself (Box 63.2). There is strong evidence that nutrition affects overall morbidity and mortality in the dialysis population, highlighting the need to look for and treat patients with malnutrition. Growing evidence has suggested that the problem is not simply protein malnutrition but protein energy wasting (PEW), similar to the cachexia seen with inflammation.[660–663] Hence, simply adding protein supplements will not reverse the wasting.

Box 63.2 Factors Causing Malnutrition in Hemodialysis Patients[a]

- Inadequate protein or calorie intake
- Increased energy expenditure
- Metabolic acidosis
- Hormonal alterations
- Comorbidities or hospitalizations
- Dialytic nutrient losses
- Dialysis-induced catabolism
- Infection

[a]For a comprehensive discussion on the factors causing malnutrition, see Chapter 60.

MARKERS OF NUTRITION

A global approach to nutritional assessment should include an evaluation of dietary history, physical findings, and laboratory testing. Patients should be questioned monthly about their appetite, gastrointestinal symptoms, and weight changes. Monthly trends of postdialysis dry weight should be monitored. Unintentional weight loss over time, particularly in patients with low body weight or body mass index (BMI), likely reflects PEW and is associated with increased mortality.[664,665]

No single laboratory assessment of nutrition is adequate. Low predialysis BUN concentrations are associated with mortality. However, the predialysis BUN level is also affected by residual kidney function and adequacy of dialysis. The normalized protein catabolic rate (nPCR), calculated using predialysis and postdialysis BUN levels adjusted for interdialytic interval and body habitus, is likely a better surrogate for dietary protein intake.[666] However, the nPCR calculation assumes a steady state from day to day, which may overestimate nutritional status in patients with active catabolism such as inflammation or infection and underestimate nutrition in anabolic states such as a maturing child or pregnancy. The association of nPCR with serum albumin levels and survival may be strengthened after accounting for residual renal urea clearance,[667] an important factor to consider when assessing nPCR values.

The serum albumin concentration is a strong predictor of mortality in observational studies,[668] yet it is difficult to determine whether the nutritional component or other factors are responsible for the association. Albumin is increased by nutritional protein intake but also may fall due to nonnutritional factors such as acute illness, inflammation, acidemia, urinary loss, and volume overload.[669–671] Because of its shorter half-life, serum prealbumin (transthyretin) has been studied as another nutritional marker. However, the kidneys are the primary source of clearance for prealbumin, rendering interpretation of its level difficult in the setting of a low GFR.[662] The serum creatinine level also may reflect dietary protein intake and muscle mass, but is unreliable as a marker for PEW because other factors may affect its level. Other laboratory markers such as insulin-like growth factor 1 (IGF-1) and ghrelin have been studied but are neither readily available nor validated.[672–675]

Body composition techniques may be useful in assessing nutrition. Anthropometric measurements, such as midarm muscle circumference and skinfold thickness, appear deceptively easy to perform but must be taken carefully by a trained technician. They also may be inaccurate because of varying tissue hydration and because standardization was undertaken in healthy volunteers.[676,677] Therefore, changes over time may be more reliable than isolated values. Additional methodologies to estimate body composition, such as dual-energy x-ray absorptiometry (DEXA) and BIA are available.[678–680] However, DEXA does not differentiate intracellular from extracellular water, and extracellular fluid is counted as part of the lean (nonfat) body mass, potentially overestimating lean body mass in edematous patients. In addition, DEXA is relatively expensive and requires travel to a facility with dedicated equipment.

Although BIA has been available for years, it remains mainly a research tool. As the technology has progressed from single-frequency to multifrequency, its accuracy in estimating total body water and lean body mass has improved, but equipment cost and complexity have increased as well, making multifrequency BIA less practical.[679] Results obtained with BIA correlate with those from DEXA[680] but, as with other body composition technologies, are most reliable when completed after dialysis.[681] The increased use of BIA in clinical trials may eventually inform its optimal use in the clinical setting.[583,682] Protocols using multifrequency BIA to estimate changes in extracellular fluid in the calf during HD may become a tool for establishing an optimal dry weight (see also "Ultrafiltration and Dry Weight" section).[581,683]

MANAGEMENT OF PROTEIN ENERGY WASTING

Several criteria have been proposed for diagnosing PEW. Each assigns a score to multiple nutritional measures and designates a threshold above which the diagnosis of PEW is made.[684–687] Each of these PEW criteria correlates strongly with clinical outcomes, including mortality, yet it is not clear how these criteria, and which one, will be incorporated into future clinical management strategies across wide groups of patients. At present, the diagnosis of PEW is suspected clinically in patients with a progressive trend of weight loss, low albumin level, and declining nPCR.

If PEW is present, increased nutritional support is critical. Use of the enteral route in patients with adequate gastrointestinal function can improve serum albumin and prealbumin concentrations and other measures of nutritional status such as the subjective global assessment,[663] even when oral supplements are only provided during dialysis.[688,689] In fact, intradialytic oral protein supplementation in patients with low albumin levels has been associated with reduced hospital stays and better survival in observational studies.[690–692] When oral supplementation is unsuccessful, tube feeding has been attempted in some cases, especially in young children for whom nutrition is critical for growth and development.[689,693] Parenteral nutrition has been considered when enteral feeding is not possible, mainly during each HD session to ameliorate the challenge of large volume infusions. Providing intradialytic parenteral nutrition (IDPN) to patients with hypoalbuminemia has increased serum albumin concentrations.[689,694] However, the only randomized controlled trials evaluating the efficacy of IDPN was in comparison to oral protein supplementation rather than placebo[695] and did not find a difference between the two interventions with respect to serum albumin, weight, or mortality. Thus, IDPN is generally reserved for patients with intestinal malabsorption or other types of dysfunction, in which oral or enteral supplementation is unsafe or poorly tolerated, particularly because of its expense.

Pharmacologic intervention to reverse the anorexia and muscle loss often seen in patients on maintenance HD has received increased attention.[696] Providing specific replacement of deficient branched-chain amino acids or ghrelin may improve appetite and nutritional status.[674,697,698] Agents such as megestrol acetate, used in other chronic diseases, have led to weight gain and an improved ability to exercise in a small pilot study.[699] Other hormonal therapies such as human growth hormone, IGF-1, and anabolic steroids have also been used and are associated with increased lean muscle mass.[696] Further investigation is needed to determine if these interventions are sufficiently safe to warrant administration.

VITAMINS AND TRACE ELEMENTS

Water-soluble vitamins are recommended for patients on HD because intake of these vitamins is low as a result of dietary restriction and because the dialysis process itself removes these vitamins, especially folic acid, pyridoxine, and vitamin C.[700] Vitamin supplementation has been associated with lower mortality risk in observational studies,[701,702] although the direct mechanism of benefit is unclear, and these analyses suffered from confounding by indication. B vitamins play important diverse roles as coenzymes and cofactors in various cellular metabolic processes, although vitamin B supplementation alone does not alter cardiovascular risk or mortality in randomized trials.[703,704] Vitamin C is an antioxidant involved in collagen formation, wound healing, hormone regulation, resistance to infections, and iron metabolism.[700] Intravenous vitamin C supplementation may reduce ESA dose requirements while improving transferrin saturation without changes in ferritin,[705] although its use has been limited due to lack of long-term safety data related to oxalic acid production and effects on oxidative stress. Regardless, prescribing oral, water-soluble vitamin supplementation is advocated to prevent nutritional deficiency and its adverse effects.

Vitamin D is covered in detail in Chapter 53, along with the complex interactions among the kidney, bone, and Ca and phosphorus metabolism. Retrospective studies have suggested that vitamin 25(OH)D deficiency may be associated with cardiovascular mortality and is a growing problem in the general population, as well as in ESKD.[706,707] However, two small trials evaluating ergocalciferol supplementation in patients undergoing HD with vitamin D deficiency have found no difference in effects on Ca, phosphorus, parathyroid hormone, other bone medication use, ESA use, hospitalizations, or death.[708,709]

Because most trace elements are excreted by the kidneys, their levels rise as kidney function declines and may contribute to uremic toxicity. Exceptions are selenium and zinc, which are the most common elements found at low serum concentrations.[710–714] Selenium is an important cofactor in certain antioxidant enzymes, and its deficiency is associated with cardiovascular disease in patients without kidney disease. However, selenium supplementation is controversial because it has a narrow therapeutic index and may result in selenosis, with nausea, vomiting, peripheral neuropathy, and loss of hair or nails.[710,711,714] Zinc deficiency is associated with immunodeficiency, ESA resistance, anorexia, dysgeusia, and impotence. Limited data have shown that zinc supplementation in deficient HD patients may improve appetite, nutritional status, and ESA responsiveness.[715–717]

MINERAL METABOLISM–RELATED ISSUES

The complex influence of kidney disease on bone and mineral metabolism is covered in detail in Chapter 53. Although maintaining a normal or near-normal serum phosphorus concentration may be one of the most challenging goals for patients on maintenance HD, growing evidence has suggested its importance because serum phosphorus concentrations correlate directly with mortality.[718] Several complications thought to be related to Ca and phosphorus perturbations in kidney failure present challenging management issues in patients undergoing HD and are highlighted here briefly.

VASCULAR CALCIFICATION

Although atherosclerosis is a major cause of cardiovascular mortality in the general population, growing evidence has suggested that diffuse vascular calcification may be an equally or more important contributor to cardiovascular mortality and morbidity in the HD population. Vascular calcification in advanced CKD appears to differ from "garden variety" atherosclerosis and disproportionately involves the medial portion of the vessel in association with disorganized vascular smooth muscle cells (VSMCs) and expression of bone matrix protein.[719–721] This process is evident even in children and adolescents.[722] The molecular mechanisms underpinning vascular calcification involve a complex balance among inhibitors and potentiators of calcification.[720,723–727] Inhibitors of calcification, such as fetuin-A and matrix Gla protein, are reduced or inactivated in vessel walls.[723,728] A careful study of blood vessels in patients with CKD and ESKD has suggested that although significant Ca loading of vessels is present, dialysis appears to trigger an increased rate of apoptosis of VSMCs, which then leads to vascular calcification.[729]

The exact signals responsible for inducing VSMC apoptosis have yet to be identified. Serum phosphorus, Ca, and parathyroid hormone (PTH) concentrations correlate with the degree of vascular calcification, making control of phosphorus and PTH and avoidance of Ca logical targets for reducing cardiovascular risk.[721,725] To date, studies have yielded conflicting results. Use of Ca-free phosphorus binders has ameliorated vascular calcification in some studies but not in others and did not appear to be associated with lower rates of all-cause or cause-specific (infectious or cardiovascular) mortality in HD patients compared with calcium-containing binders.[730–735] A prospective cohort study of more than 10,000 patients has found that those who received phosphorus binders during the first 90 days of HD therapy had a lower 1-year all-cause mortality than those not taking binders,[736] regardless of the type of binder and serum phosphorus concentration. However, improved control of hyperparathyroidism with cinacalcet in patients undergoing dialysis[737] and with paracalcitol in patients with nondialysis-requiring CKD[738] did not reduce mortality or morbidity, although off-protocol use of the study drug in control subjects may have reduced the power of these studies to detect significant differences.[739] Additional controlled trials addressing the effect of phosphorus and/or PTH control with different classes of phosphorus binders and PTH-lowering agents on hard clinical endpoints are needed.

CALCIFIC UREMIC ARTERIOLOPATHY

Calcific uremic arteriolopathy (CUA), previously referred to as *calciphylaxis*, is a devastating condition characterized by painful ischemia of the skin and subcutaneous tissues, manifesting as symmetric, violaceous patches early on and usually progressing to subcutaneous plaques as a result of infarction[740] and less so to necrotizing, nonhealing skin ulcers.[741–743] The name suggests a connection to substantial kidney impairment but, rarely, cases are seen in nonuremic patients.[744] Lesions tend to occur in areas of high adipose content and usually involve the abdominal wall, breasts, buttocks, and thighs, but they can also develop at more distal sites such as the calves, forearms, hands, feet, and face. Pathologically, the presence of diffuse calcification of the media and internal elastic lamina of dermal and subcutaneous small arteries and arterioles, with atrophy of smooth muscle cells leading to cutaneous ischemia, is an important diagnostic finding.[742,743] Imaging modalities such as plain radiography, computed tomography (CT), mammography, and bone scanning may provide supportive information, although none of these modalities has been systematically evaluated.[745–747] Skin biopsy provides a definitive diagnosis, allowing the exclusion of other potential causes of skin necrosis with secondary calcification, although it requires extreme care because of the increased risk of poor healing and infection associated with CUA.[740,743,748]

The pathogenesis of CUA is controversial.[741,742,747] Recognized risk factors include diabetes mellitus, obesity, female gender, white race, hypercalcemia, hyperphosphatemia, use of Ca-containing phosphorus binders and vitamin D, hyperparathyroidism, chronic inflammatory states, and the use of vitamin K antagonists (e.g., warfarin). Skin trauma may provide an inciting event because diabetics requiring insulin injections appear more likely to develop CUA than those not on insulin injections.[749] The extent to which disorders of mineral metabolism contribute to CUA is unclear because severe CUA cases appeared to resolve after parathyroidectomy in some reports but not in others,[750–754] and hyperparathyroidism did not appear to be critical for CUA development in some observational reports.[755] Analyses of pathologic specimens from a small number of patients with CUA have demonstrated deposits of iron and aluminum in the lesions but none in adjacent normal tissue or in tissue from CKD controls, suggesting a potential role for metal deposition in the pathogenesis.[756,757] Histologic studies have also revealed the presence of an active osteogenic process in CUA lesions, including increased expression of bone morphogenic protein 2 and osteopontin.[719,758]

Therapy for CUA is focused on reducing Ca and phosphorus concentrations and controlling high PTH levels,[741,742,747,759] although the role of parathyroidectomy remains controversial.[750–754] Discontinuing or avoiding Ca-containing phosphorus binders and vitamin D, along with a prescription of a physiologic or slightly lower calcium dialysate, seem prudent. When ulcerations occur, aggressive local wound care and antibiotics to prevent wound infections are important. Case reports and small case series have introduced novel therapies that may aid in healing—hyperbaric oxygen, cinacalcet, bisphosphonates, and sodium thiosulfate, with many authors advocating a multimodal treatment approach to include optimization of and potentially more frequent dialysis.[747,748,760–770] Sodium thiosulfate therapy appears to be the most promising and may offer benefit through both chelation of Ca and amelioration of inflammation.[761,764,770] Reports of intralesional sodium thiosulfate have also shown promise for localized CUA.[771]

CARDIOVASCULAR DISEASE

Cardiovascular disease is the leading cause of mortality in patients undergoing HD, accounting for roughly half of known causes of death.[10] The complex relationships between CKD and cardiovascular biology are described in detail in Chapter 54. In general, the interplay of traditional cardiac risk factors with nontraditional cardiac risk factors unique to patients undergoing HD, such as inflammation, oxidative stress, retained metabolites (e.g., advanced glycation end products), and PEW combine to lead ultimately to severe cardiomyopathy, ischemic heart disease, and death.[772]

TRADITIONAL CARDIOVASCULAR RISK FACTORS

Atherosclerotic cardiovascular risk factors such as diabetes, hypertension, low physical activity, left ventricular hypertrophy, and low high-density lipoprotein (HDL) levels are substantially more common in HD patients than in the general population.[773] Risk factors such as obesity and high low-density lipoprotein (LDL) cholesterol levels are less frequent. When present, obesity seems to confer a paradoxically protective effect and elevated serum concentrations of LDL are not associated with mortality.[774–776] These findings suggest that the pathogenic relationship of traditional cardiovascular risk factors and adverse cardiovascular outcomes differs in ESKD compared with the general population.

One possible explanation is that lipoproteins are modified in the setting of ESKD. LDL appears to undergo uremia-induced modification and carbamylation, resulting in a more dense form that better predicts cardiac events than total LDL.[777,778] In particular, elevated levels of lipoprotein(a)—Lp(a)—a modified LDL that has a highly glycosylated apolipoprotein A (apo A) linked to apolipoprotein B100 (apo B100), appears to be a key predictor.[779–782] Lp(a) has atherogenic properties, accumulates in the wall of atherosclerotic lesions, and is increased in HD possibly because the kidney degrades Lp(a). An in vivo turnover study has suggested that elevated levels of Lp(a) in the HD population are caused by decreased clearance rather than overproduction.[783] Moreover, low levels of HDL exist, and those particles have lost their antiinflammatory and antiatherogenic components, potentially due to reduced activity of HDL-associated enzymes—paraoxonase 1, nitric oxide synthase, and lecithin-cholesterol acyltransferase.[777,784]

Left ventricular hypertrophy (LVH) is a risk factor for cardiovascular events in the general population.[785] It is prevalent in most patients on HD and may represent a major risk factor for cardiac events, not specifically related to traditional atherosclerosis.[786] Multiple factors have been invoked for the high prevalence of LVH in the HD population, including volume overload, hypertension, age, anemia, and hypoalbuminemia. AV HD access with prolonged high cardiac output may also contribute to the frequency of left ventricular disease,[787,788] although this has been debated.[789] Assessment of LVH by echocardiography is optimal when the patient is at her or his dry weight. Cardiac magnetic resonance imaging (CMRI) may be better for measuring left ventricular mass than traditional echocardiography because it abrogates confounding by volume overload,[790,791] and LVH demonstrated by CMRI has been associated with higher systolic blood pressure, predialysis pulse pressure, and calcium × phosphorus products.[791]

NONTRADITIONAL CARDIOVASCULAR RISK FACTORS

Inflammation has gained appreciation as an important factor in cardiovascular disease in both the CKD and general populations.[792–794] Plasma levels of C-reactive protein (a marker of inflammation) and interleukin-6 (a proinflammatory cytokine) are strong predictors of cardiovascular and all-cause mortality in the ESKD population.[774,775,795] Numerous factors predispose patients undergoing HD to the burden of endogenous inflammation compared with the general population. Kidney failure, the dialysis process, genetic predisposition, chronic periodontitis, type of vascular access, and other as yet unidentified factors act in concert to contribute to oxidative stress, endothelial dysfunction, carbonyl stress, and accumulation of advanced glycation end products.[110,774,792,796] The critical role of inflammation in cardiovascular disease has led to the hope that statins, which lower lipid levels and reduce vascular inflammation,[797] would ameliorate cardiovascular risk in dialysis patients, but several large randomized controlled studies have yielded disappointing results (see later, "Diagnosis and Treatment").[798]

Homocysteine levels are elevated in ESKD and correlate with cardiovascular risk,[799–801] presumably through inducing endothelial dysfunction and a prothrombotic state. Because homocysteine is largely protein-bound, poor nutritional status may confound the relationship between homocysteine concentrations and cardiovascular disease.[801] Although small studies in the HD population have suggested a potential benefit from folic acid and methylcobalamin administration to lower homocysteine levels and cardiovascular risk, most data in the general population and patients with CKD and ESKD have suggested that lowering homocysteine levels with folic acid and B vitamins does not reduce the risk of cardiovascular events.[703,704,800,802–806] Nonetheless, oral supplementation with folic acid and B vitamins is recommended to prevent nutritional deficiencies and support hematopoiesis, as discussed earlier.

Oxidative stress from kidney failure or the dialysis process itself may contribute to the high risk of cardiovascular disease.[807,808] Biomarkers of oxidative stress in HD patients are difficult to quantify accurately but seem to correlate with carotid intimal thickness (CIMT), a surrogate marker for cardiovascular disease.[809] Strategies to reduce oxidative stress include oral administration of antioxidants such as vitamin E, which reduce the composite rate of cardiac events but not individual endpoints or all-cause mortality in small controlled trials.[807,810–812] These findings were not replicated in larger trials of antioxidant therapy. One randomized trial of oral tocopherols and alpha-lipoic acid that included 353 patients on HD has shown no effect on markers of inflammation (e.g., high-sensitivity C-reactive protein. [hsCRP], interleukin-6 [IL-6], F2 isoprostane, and isofuran concentrations), EPO responsiveness, hospitalization, or mortality.[813] A large randomized trial of vitamin E–impregnated membranes, a novel method targeting intradialytic oxidative stress, did not improve response to ESAs,[814] although smaller studies have suggested that these dialyzers lower biomarkers of oxidative stress and reduce CIMT.[221,815,816] Coenzyme Q10 appears to reduce plasma isoprostanes and isofurans, both markers of oxidative stress and endothelial dysfunction.[817,818] Use of ultrapure dialysate produced by filtering standard dialysis fluid through an added bacteria- and endotoxin-retentive filter reduced markers of inflammation and oxidative stress.[354,819,820] Whether reducing inflammatory mediators and oxidative stress translates to improved cardiovascular outcomes is unclear.

Novel biomarkers such as the proteins paraoxonase (PON1), fibroblast growth factor 23 (FGF-23), and indoxyl sulfate also are associated with cardiovascular disease. HDL-associated PON1 protects LDLs against oxidation. Reduced

PON1 activity, determined by genetic factors in concert with several environmental factors, including smoking, correlates inversely with markers of inflammation and cardiovascular mortality in patients on HD.[821,822] Elevated levels of FGF-23 in CKD may directly influence LVH and cardiovascular risk, independent of its relationship to phosphorus.[823,824] Indoxyl sulfate appears to be a powerful inducer of oxidative stress. It is derived from dietary protein metabolism by intestinal bacteria, accumulates in ESKD, and is poorly removed, even with more frequent dialysis, due to protein binding.[825,826] Heightened interest in extradialytic methods of indoxyl sulfate removal has included intensification of dietary fiber to alter the gut microbiome.[390]

DIAGNOSIS AND TREATMENT

Unique factors in the HD population complicate the diagnosis and treatment of cardiovascular disease. A classic example is the reduced rate of symptoms, especially in patients with diabetes, despite substantial angiographic evidence for coronary disease.[827,828] There are conflicting data regarding the utility of the various screening tests for coronary artery disease in HD patients.[828–830] In general, tests that require the patient to exercise or that rely on electrocardiographic (ECG) findings for diagnosis are less reliable because of reduced exercise tolerance and abnormal resting ECG findings in HD patients. Dobutamine stress echocardiography and pharmacologic nuclear stress imaging have better sensitivity and specificity in HD patients,[831] although their predictive value remains lower than for the general population. Angiography remains the gold standard for the detection of coronary artery disease and should be considered in patients with positive stress tests or acute coronary syndromes.

Studies have suggested that patients on HD with acute coronary syndromes are managed less aggressively than the general population. They are less likely to receive thrombolytic therapy[832,833] or undergo diagnostic coronary angiography and revascularization,[834] despite epidemiologic data suggesting that revascularization improves survival.[833,835] Part of the reluctance in pursuing aggressive interventions in HD patients may stem from the five- to sixfold increase in early mortality (9%–12% vs. 2%–3%) and 5-year mortality (>50% vs. 10%) after coronary revascularization compared with the general population.[836–840] Optimal management of HD recipients with coronary artery disease remains controversial[839,841] because no randomized trials have been published that compare interventional strategies.[842] Several epidemiologic studies and meta-analyses have reported better long-term outcomes—lower risks of late cardiac deaths, sudden death, myocardial infarction, and repeat coronary revascularization—with coronary artery bypass grafting (CABG) compared with percutaneous intervention, but at the cost of a higher risk of mortality and stroke within the initial 3 months.[837,838,840,843–847] In the absence of definitive data, the decision to select CABG versus percutaneous intervention may be influenced by individual patient characteristics and center-specific factors. If a percutaneous approach is chosen, drug-eluting stents appear to be superior to bare metal stents.[845,848,849]

Strategies for the secondary prevention of cardiovascular disease in ESKD are largely based on evidence from the general population. Randomized controlled trials in ESKD have been limited in size and number, and those that have been undertaken have often yielded disappointing conclusions. For example, despite the promise of statins to reduce LDL, atherosclerotic burden, and perhaps inflammation in the HD population, two large randomized, placebo-controlled trials of statins in HD patients have reported no improvement in rates of cardiovascular deaths and nonfatal cardiovascular events despite a greater than 40% reduction in LDL cholesterol levels.[850–853] Reviews of these trials have highlighted the presence of underlying cardiomyopathy, increased incidence of sudden death, and altered lipid profile with higher levels of Lp(a) and modified LDL particles, all of which may, in part, explain the lack of benefit from statins in this population.[850,854] A subsequent trial randomizing patients to simvastatin plus ezetimibe has demonstrated that statins reduce the risk of major atherosclerotic events in the total CKD study population, but not in the subgroup of over 3000 patients with ESKD, although the study was not specifically powered for this subgroup.[855]

The growing emphasis on health has focused attention on behavioral interventions that may reduce cardiovascular risk by modifying traditional risk factors. For example, smoking is a risk predictor even in the HD population.[512,856] However, little is known regarding the impact of smoking cessation on cardiovascular risk in patients on HD because the only data available in this patient population only explored the pharmacokinetics of various smoking cessation aids in the setting of kidney impairment.[857] Exercise has received growing attention. Intradialytic exercise may improve adherence and self-reported physical function measures,[858] as well as cardiac functional measures such as aerobic capacity, heart rate variability, late potentials, and T wave alternans.[859] A home-based exercise program managed by the dialysis staff is another example of a simple personalized intervention that improves physical performance and measures of quality of life.[860] Although physical activity is associated with survival in observational studies,[861,862] evidence that increasing exercise lowers mortality is still lacking. However, for patients on HD who have undergone a cardiac intervention such as CABG, exercise through a cardiac rehabilitation program is beneficial and cost-effective.[863] Finally, substantial epidemiologic data have linked poor oral health and the presence of periodontal disease to systemic inflammation and cardiovascular risk in the general population,[864] and some data for the link exist in patients on HD,[796,865,866] but causality has not been established. Whether diminishing the periodontal disease burden would reduce mortality remains to be seen.[865]

As mentioned previously, sudden cardiac death appears to be the leading cause of mortality among HD patients.[867–869] A myriad of factors may increase the risk for arrhythmias and sudden death in patients on dialysis, with the presence of LVH being one of the most prominent and coronary artery diseases playing a minor role.[289,450,870–873] Reducing LVH remains a major therapeutic focus, but without clear evidence that doing so will improve survival. The dialysis prescription itself poses additional risks for sudden death (see also "Dialysis Duration and Frequency" and "Dialysate Composition"), including short treatment time, large UF volume, and low dialysate potassium, which are potentially modifiable practices that may improve outcomes.[461] The timing of sudden death appears to cluster around the long interdialytic interval in patients on three times/week dialysis or less (either Monday or Tuesday),[460,874] a pattern not seen with more frequent therapy or PD.[875] The high incidence of sudden death has focused attention on the role of implantable defribrillators to improve survival in the dialysis population, but indications for implantation may differ from those in the general public, risks may be higher, and benefit is uncertain.[867,876,877]

In the general population, LVH is a cardiovascular risk factor that may be modified by treatment to reduce ventricular mass, such as with an ACE inhibitor.[878,879] For the HD population, an extensive review has found very limited evidence to support the use of pharmacologic treatment for cardioprotection.[880] The only reported controlled trial for ACE inhibitors and ARBs in patients on HD has demonstrated improved blood pressure control but did not reduce cardiovascular risk.[881,882] Patients with a recent MI may benefit more specifically from classic cardioprotective medications such as ACE inhibitors and beta-blockers.[880] More frequent HD reduces LVH significantly,[436,457] but whether this will translate into lower mortality is not clear.

Cardiovascular disease remains the number one cause of death in patients on HD, despite substantial progress in understanding the natural history of this disease through all the stages of CKD. Part of the reason for the lack of success in preventing cardiovascular deaths in this population may be that the vast burden of disease already exists at the initiation of HD. Another contribution is the complex interplay among inflammation, oxidative stress, malnutrition, retention of uremic solutes, the dialysis process itself, and cardiovascular disease that remains poorly understood, but renders traditional cardiac risk factors and treatment less effective in predicting and ameliorating cardiovascular mortality and morbidity in the HD population.[839,850,852,883] Finally, the transition from the high incidence of atherosclerosis-induced cardiovascular events in CKD patients (for whom traditional therapies such as statins do improve outcomes) to the predominance of sudden cardiac deaths in dialysis patients likely accounts in part for the persistently high cardiovascular mortality once patients start dialysis. For now, applying established therapeutic approaches to ameliorating traditional risk factors earlier in the course of CKD while awaiting more evidence to support altering HD management and treating nontraditional risk factors in the HD population would seem prudent. Increasing the frequency of dialysis, prolonging treatment times, limiting rapid ultrafiltration, and modifying dialysate potassium, bicarbonate, and calcium concentrations may reduce the rate of intradialytic myocardial stunning and the risk for sudden death. [436,444,445,457,843,884]

HYPERTENSION

The complex relationship between blood pressure (BP) and the kidneys is discussed in detail in Chapters 46 and 54. Here we provide a brief discussion of the unique challenges in patients receiving HD.

The evaluation and management of BP in the HD population are complicated by the many variations in predialysis, intradialytic, postdialysis, and interdialytic BPs in any given patient and their clinical significance.[885–887] Combining intradialytic BP values with both predialysis and postdialysis BP measures have traditionally been used as surrogates of overall BP status.[888] However, peridialysis BPs are influenced significantly by volume status and correlate less well with mortality than interdialytic BPs.[889] Ambulatory BP monitoring may yield a more realistic picture of BP control and offer more prognostic information,[890,891] but its use is limited by cost and availability to patients.

Complicating matters further, several studies have shown that excessively low BPs rather than high BPs measured in the dialysis clinic are associated with mortality.[892,893] An explanation for this U-shaped curve may be that low BP acts as a potential indicator of underlying cardiac disease, poor nutritional status, and risk for dialysis-associated events. This phenomenon was illustrated in a recent prospective study that compared the utility of dialysis clinic BPs with home BPs.[894] Whereas a U-shaped association of dialysis clinic systolic BP was shown to nadir from 140 to 170 mm Hg, home BP readings showed a linear association with cardiovascular risk and death, suggesting that BPs outside the dialysis clinic may be a better measure to target.

The pharmacologic approach to managing BP has been presented (see Chapter 49) and applies to the HD population, with several unique caveats. Volume expansion is a major cause of hypertension in HD patients, with high BPs that persist, even posttreatment.[895] The challenge for nephrologists is to determine the optimal dry weight for each patient at which the BP is controlled with the fewest medications and the patient can tolerate the UF required to achieve that goal weight.[896] Achieving optimal dry weight through more frequent HD treatments per week[437] or through a gradual, protocol-driven reduction in weight for patients receiving conventional thrice-weekly HD[897] markedly improves BP control and supports volume regulation as a critical element in BP management (see also "Dialysis Duration and Frequency" section). The optimal antihypertensive regimen in ESKD has not been established.[887] The only controlled trial comparing antihypertensive agents in an HD population randomized 200 patients with LVH to atenolol versus lisinopril, with each administered thrice weekly. The findings have suggested that atenolol is superior to lisinopril in reducing cardiovascular morbidity and all-cause hospitalization.[898] Other studies have suggested that the use of ACE inhibitors and ARBs may reduce left ventricular mass[899,900] but does not seem to have an effect on cardiovascular events or mortality.[882,900,901] Drugs that are removed poorly with HD (e.g., losartan, fosinopril, ramipril, carvedilol, bisoprolol, propranolol) may improve BP control, but appear to have variable effects on cardiovascular mortality, likely through the competing effects of improved overall BP control but higher risk for intradialytic hypotension.[902–906] No definitive conclusions are possible because the studies have been few, retrospective and, when randomized, small. BPcontrol continues to be important, although the best method for establishing dry weight, optimal use of antihypertensives, target for BP control, and observed BP paradox in dialysis require further research.[885–887,896]

IMMUNE DISORDERS AND INFECTION

Infection is the second leading cause of death and hospitalization for patients on maintenance HD after cardiovascular disease, accounting for nearly 25% of deaths and one-third of hospitalizations.[907,908] The mortality rate from sepsis may be as high as 300 times that of the general population.[909,910] Of these infections, 20% are access-related, with pulmonary, soft tissue, and genitourinary infections accounting for the remainder.[907,908] Likely reasons for the increased infection risk can be divided into two categories—those related to the

HD treatment itself and those endogenous to the patient and the uremic milieu.

ROLE OF THE VASCULAR ACCESS AND HEMODIALYSIS PROCEDURE

A leading factor related to the HD treatment is the universal presence of some form of vascular access. As discussed earlier, the use of catheters is a significant cause of sepsis in the HD population (see "Vascular Access" section).[147,907] Biofilm, which seems to develop on all indwelling artificial surfaces, appears to play an important role in the pathogenesis of these infections.[911] Although indwelling catheters are clearly the major source of vascular access infection, the repeated percutaneous needle insertions required with AV access also contribute, with AV grafts implicated more frequently than AVF.[110] Infections in AV grafts often require removal of the graft.

The HD treatment itself presents a risk of bacterial or viral infection from the dialysis machine, dialyzer, or dialysate because of exposure to nonsterile water in the dialysate and through dialyzer reuse (see "Water Treatment"). In addition, microbe-generated impurities such as endotoxins in water may cross the dialyzer membrane and stimulate an endogenous inflammatory response. As noted, strict standards have been established for purifying water, disinfecting HD machines, and handling dialyzers, particularly if reuse is practiced. Growing evidence has suggested that impurities in the water may be a significant cause of the inflammatory responses in patients on HD,[330] and efforts are underway to find more sensitive assays and improved methods of treating water to allow for the detection and removal of these impurities.[345,354,361]

UREMIA-INDUCED IMMUNE DISORDER

The uremic milieu gives rise to both immune activation and immunodeficiency, which contribute to the high risk and severity of infections, the reduced response to vaccines, a state of chronic inflammation and, indirectly, to cardiovascular disease.[912–914] Both innate and adaptive immune systems are altered. Monocytes, neutrophils, and dendritic cells exhibit decreased endocytosis and impaired maturation but enhanced production of interleukin-12p70 (a T cell–stimulating factor) and allogeneic T-cell proliferation.[915] The expression of Toll-like receptor 4, which detects lipopolysaccharides (LPSs) from gram-negative bacteria and leads to activation of the innate immune system, is reduced constitutively in infection-prone CKD patients not yet on HD.[916] Monocytes from these patients, when challenged with LPS, demonstrate reduced synthesis of tumor necrosis factor-α and several interleukins. This relative acquired immunodeficiency state from chronic immune activation leads to inflammation and may explain the high failure rates for vaccinations in the HD population, as well as the viral infections endemic in this population.[913,914,917]

INFECTIONS AND RESPONSE TO VACCINATION

Hepatitis viruses have presented management challenges for the HD population since maintenance therapy became available. The risk of hepatitis B transmission has been greatly reduced by a lesser need for blood transfusions (through the use of ESAs and intravenous iron), strict isolation procedures, and vaccination, although the hepatitis vaccine is less effective than in the general population, especially if administration is delayed until after the initiation of dialysis.[913,914,918] With the declining incidence of hepatitis B, hepatitis C has become a major concern in HD clinics since the discovery that this virus explained much of the non-A, non-B hepatitis in patients on HD. The prevalence of hepatitis C is extremely variable, depending on the location of the dialysis clinic, and ranges from 4% to 70% worldwide.[919] The natural history of the disease is also variable and depends on the severity of underlying liver disease and the presence or absence of known complications, such as hepatocellular carcinoma. In a large meta-analysis, the presence of anti–hepatitis C antibodies was associated independently with a 34% increase in all-cause mortality.[920] Current recommendations to control hepatitis C transmission in the dialysis clinic include strict adherence to universal precautions, careful attention to hygiene and sterilization of dialysis machines, and routine serologic testing and surveillance for hepatitis C infection, but they do not require isolation of the affected patient or dialysis machine.[921] In the past, hepatitis C treatment in HD patients with a pegylated interferon–based regimen was complicated by lower response rates and more side effects, particularly because the use of ribavirin was limited due to the risk of drug accumulation and hemolysis.[922–924] Fortunately, newer direct-acting antivirals have become available, which have already begun to transform hepatitis C treatment in patients with CKD and ESKD.[925] These drugs are safer, better tolerated, and provide a dramatic virologic response, particularly for genotype 1, the most prevalent genotype in the United States.[926,927]

The prevalence of human immunodeficiency virus (HIV) in the HD population remains unknown because routine screening is not practiced. Estimates by the CDC of rates of HIV infection and acquired immunodeficiency syndrome (AIDS) in HD patients were about 1.5% and 0.4%, respectively.[928] The survival of HD patients infected with HIV improved significantly during the latter half of the 1990s as experience treating HIV and AIDS in the HD population increased.[929,930]

Pneumonia is a particularly common infection in the HD population. A recent observational study of Medicare patients found an incidence rate of 21.4 events/100 patient-years, with over 90% requiring hospitalization and more than 10% dying within 1 month of the event.[931] Hospital stays were almost five times longer compared with nondialysis patients and were associated with substantial cost.[931,932] Administering the pneumococcal vaccine to patients undergoing HD is recommended and is associated with decreased mortality.[933] Because of the reduced acquired immunity, patients on HD may have an impaired antibody response to vaccines, and measurable antibody levels may decrease quickly,[934] raising questions about the best approach for revaccination.

Current recommendations are for all patients on dialysis to receive the full vaccine series for hepatitis B, pneumococcal vaccine, and the appropriate seasonal influenza vaccine. The observed poor response rate to hepatitis B vaccine in the HD population mandates vaccination as early in the course of CKD as possible, careful follow-up, and revaccination when indicated. Appropriate surveillance for hepatitis B should continue, with booster vaccination for low titers of anti–hepatitis B surface antigen. Caution should be given to avoid surveillance of hepatitis B surface antigen within 3 weeks of vaccination because false- positives may occur.[935] Vaccination for herpes zoster infection is now available to minimize the impact of zoster infections in adults older than 60 years, and it appears to be most effective when given near the time of dialysis initiation.[936] Special consideration should be given to provide zoster vaccination in older pretransplantation candidates because posttransplantation immunosuppression precludes live vaccination.

PRIMARY CARE MANAGEMENT

Patients undergoing HD still need routine health care for nondialysis problems, just like the general population. Although opinions differ widely on its practicality and effectiveness,[937] the nephrologist will frequently find himself or herself as the sole care provider for a patient on HD, particularly those receiving in-center treatment. Of 158 patients surveyed in a suburban HD clinic, only 56 had a primary care physician. Patients just starting HD were more likely to have a primary care physician than those on HD for more than 1 year.[938] A Canadian survey has demonstrated the importance of communication among family physicians, nephrologists, and patients because excess use, duplication, and/or omission of important health care services was common.[939]

Based on limited survival benefit, routine preventive cancer screening may not be cost-effective in the average HD patient.[940,941] Instead, efforts should be made to direct age-appropriate tests to HD patients with a longer expected survival or are transplantation candidates while limiting screening in patients with significant baseline comorbidities. A recent study of colon cancer screening of Medicare beneficiaries from 2007 to 2012 reflected this, although high usage rates across all risk groups suggested persistent overscreening in the HD population.[942]

Serious clinical depression is an underrecognized illness in HD patients and correlates strongly with mortality and morbidity (see also "Transition from Chronic Kidney Disease Stage V").[30,37–39] Hence, depression is another important health care issue that should be addressed by the primary care physician, nephrologist or, optimally, both. In a review of a national Taiwanese database, it was found that HD patients are three times more likely to commit suicide than controls, with many events occurring within the first 3 months of starting dialysis.[943] It is paramount that nephrologists know if each patient has a primary care physician and communicate regularly with this physician.

COMPLICATIONS FOR PATIENTS ON MAINTENANCE HEMODIALYSIS

Hemodialysis is a complex, life-sustaining procedure provided to patients with comorbidities. Despite the significant physiologic events that occur during the process, HD has become a relatively safe procedure. Advances in water treatment, more physiologic dialysate, technologic advances, and detailed treatment policies and protocols have contributed to its safety. However, with more than 400,000 patients undeergoing over 60 million HD sessions/year in the United States alone,[10] complications are rare but inevitable. This section will highlight important complications of maintenance HD that a clinician may encounter.

HYPOTENSION

KDOQI guidelines define intradialytic hypotension (IDH) as a decrease in systolic BP of 20 mm Hg or more or a decrease in mean arterial pressure of 10 mm Hg or more, provided the decrease is associated with clinical events and a need for staff intervention.[944] IDH occurs in 15% to 30% of all HD sessions, with subsets of patients experiencing IDH in more than 50% of their sessions, strongly associated with large interdialytic weight gains.[575,945] Interdialytic weight gain depends principally on the difference between residual urine output and intake of sodium and fluid. Excess fluid must be removed during HD to maintain balance. Large weight gains contribute to the frequency of IDH, particularly in female patients and those with diabetes, longer dialysis vintage, older age, and underlying heart disease.[946]

ULTRAFILTRATION

For successful UF, refilling the intravascular compartment from the extravascular space must occur. Heart rates and systemic vascular resistance increase during UF to help compensate for the decline in vascular volume. IDH occurs when these physiologic responses are unable to compensate for imbalances in vascular refilling, particularly in the setting of aggressive UF. Importantly, high UF rates (e.g., >10–13 mL/kg per hour) may increase cardiovascular morbidity and mortality.[311,572,947]

Complicating matters, solute diffusion during HD will transfer heat from dialysate to the patient (see later). A physiologic response to heat transfer is vasodilation, which counteracts the corrective response to intravascular volume reduction. Furthermore, solute diffusion will reduce intravascular osmolality and can therefore reduce the osmotic drive for vascular refilling.

DIALYSATE FACTORS

During HD, solute removal decreases plasma osmolality and favors a shift in volume from the intravascular to extravascular space. This osmolar shift may be reduced by using a high dialysate sodium of more than 140 mEq/L. Unfortunately, the consequence is a sodium gain in most patients because predialysis serum sodium levels tend to be less than 140 mEq/L, leading to thirst, worsening of hypertension, and larger interdialytic weight gains. For these reasons, high fixed dialysate sodium concentrations are discouraged. Sodium modeling protocols, with the dialysate sodium programmed to start at a high level and decrease during treatment, were designed to reduce positive sodium balance. However, they still resulted in higher interdialytic weight gains and hypertension.[303–305,948] Individualized dialysis prescriptions with dialysate sodium approximating the endogenous sodium level may be preferred.[297–300,305]

Calcium plays a crucial role in cardiac myocyte and vascular smooth muscle contractility. A high dialysate calcium concentration of 3.5 mEq/L or higher can reduce the incidence of IDH but may accelerate vascular calcification.[510,511] In the United States, dialysate calcium concentration is typically set between 2.25 and 2.5 mEq/L to minimize calcium loading. Dialysate calcium concentrations lower than this range are associated with more frequent hypotensive episodes and cardiac arrhythmia.[541]

Although infrequently discussed, intracellular magnesium is also important for myocardial and vascular stability. Lower dialysate magnesium concentrations and hypomagnesemia may result in IDH.[543] Dialysate magnesium concentrations in the United States are usually from 0.5 to 1 mEq/L to prevent intradialytic hypomagnesemia.

As discussed in the "Dialysate Composition" section, replacement of acetate with bicarbonate as base in the dialysate may greatly reduce the incidence of IDH.

MANAGEMENT

The first approach to managing hypotension during HD is to establish the pattern of the hypotensive episodes. Isolated episodes of hypotension may not require any alteration in the dialysis prescription other than treating the acute event. Modest declines in BP are addressed by temporary reduction in the UF rate alone or in combination with placing the patient in a supine or Trendelenburg position. For significant hypotension, these maneuvers should be combined with fluid replacement, usually 100 to 250 mL of normal saline every few minutes. On occasion, albumin or mannitol is used, although availability of these agents is limited. Saline is preferred and may be just as effective, at less expense.[949] When an episode does not reverse quickly, the dialysis staff should administer oxygen, reduce the blood flow rate, and consider more complex causes such as myocardial injury or pericardial disease. Persistent hypotension or suspicion of a dangerous clinical event requires discontinuation of the HD treatment.

Preventing further episodes of hypotension becomes the goal after the acute episode has resolved (Box 63.3). The timing of the episode within the treatment may provide some insight into a prevention strategy. Hypotension that occurs late in the treatment course may reflect a fluid removal goal that is incorrect or excessive, and patients should be carefully evaluated to determine whether postdialysis weight should be increased. There is no simple method to determine the ideal dry weight, and physical examination findings such as edema may not correlate well with intravascular volume.[575,950] Thus, careful and frequent adjustments while monitoring hemodynamics are required when reestablishing goal weight. Promising technical tools such as bioelectrical impedance analysis and lung ultrasound to estimate fluid status are under review,[582,951–953] although are not readily available at this time.

Box 63.3 Interventions to Consider in Recurrent Intradialytic Hypotension

- Reassess dry weight.
- Reduce interdialytic sodium gain.
- Reduce intradialytic sodium gain.
- Assess intradialytic hypocalcemia, hypokalemia, and hypomagnesemia.
- Avoid food intake during dialysis.
- Adjust antihypertensive medications and their timing.
- Assess cardiac function.
- Cool dialysate.
- Extend dialysis time or add sessions per week.
- Use sequential ultrafiltration or ultrafiltration modeling.
- Prescribe midodrine.

Meal ingestion during dialysis may also make patients prone to IDH by lowering the peripheral vascular resistance.[954,955] Susceptible patients, particularly diabetics, should be asked to eat meals after dialysis. Antihypertensive medications also render patients more susceptible to hypotension during HD.[575] If possible, shorter acting antihypertensive medications should be avoided before HD. Longer acting antihypertensives are preferred and should be taken daily at the same time, selecting a time that would allow the scheduled antihypertensives to be given after dialysis on treatment days.

Reducing the dialysate temperature may improve hemodynamic stability by increasing vascular resistance and cardiac contractility via cold-induced sympathetic nervous system activation and may ameliorate dialysis-induced left ventricular dysfunction, myocardial stunning, and brain ischemia.[316,317,319–321,956,957] Empiric reductions of dialysate temperature by 0.5° to 1°C below body temperature, staying above 35°C, may be considered while monitoring the patient for cold-induced symptoms that could limit this intervention.

Often, the target weight may be correct but the interdialytic fluid gain is too large to remove during a single HD treatment. In this case, IDH may occur earlier in the treatment because vascular refilling cannot match the required UF rate. The focus for these patients must be on the reduction of sodium and water intake. Lowering the dialysate sodium concentration as previously described to match the patient's serum sodium level may prevent a positive sodium balance and help limit interdialytic weight gains.[297–300,305,515] In some cases, prolonging the treatment duration or providing more frequent dialysis[32,958] is required to ameliorate IDH, mainly by increasing the time available for UF in patients with large interdialytic fluid gains. Other tools to improve UF tolerance include sequential UF[589,959] and computer-controlled UF modeling. Separating filtration from diffusion may improve hemodynamic tolerability[959] but is difficult to achieve in the outpatient setting because of time constraints. Advances in technology now allow continuous monitoring of intravascular volume and/or sodium concentration with automated control of the UF rate but as yet have not solved this complication in HD. This is because most methods monitor relative and not absolute blood volume (see earlier discussion).[279,280,284,291,960]

Assessment of primary cardiac disease should also be considered. IDH may be a manifestation of heart failure, ischemic heart disease, or a pericardial effusion. Echocardiography or stress testing may help elucidate any potential underlying cardiac factors.

A meta-analysis of the literature concluded that patients with resistant IDH despite the interventions described or with autonomic insufficiency may respond to midodrine, an oral α_1-adrenergic agonist, at a dose of 2.5 to 10 mg given 15 to 30 minutes before HD.[961] Side effects include supine hypertension, urinary retention, pilomotor reactions, and gastrointestinal complaints.[956,962] Peak levels occur at 60 minutes; it is partially removed by HD, so a second smaller dose may be required partly through the HD session.

MUSCLE CRAMPS

Muscle cramps occur during or after approximately 60% of HD treatments, are quite painful, reduce quality of life, and are a common reason for early session termination.[963]

Pathogenic mechanisms may relate to vasoconstriction and impaired oxygen delivery to muscle in the setting of hypotension, as well as muscle cell osmotic and fluid shifts during HD. The accumulation of as yet unidentified uremic solutes may predispose to interdialytic muscle cramps, along with a deficiency of various nutritional substances.

When muscle cramps accompany hypotensive episodes on HD, giving careful attention to UF rates, reassessing dry weight, increasing the frequency or duration of dialysis, and educating the patient to reduce interdialytic fluid gains can prevent cramps in many cases. The strategies described earlier to prevent IDH may also be applicable to dialysis-associated cramps. Acute management may include cessation of UF in conjunction with small boluses of normal saline to treat both cramps and IDH. Hypertonic (23%) saline and 50% dextrose solution have occasionally been used in small volumes for acute management of cramps,[964] although they will predispose to interdialytic weight gain and hyperglycemia, respectively, and are thus less desirable agents.

No agent is universally effective in preventing HD-associated cramps. Vitamin E before bedtime may be considered, although its effectiveness has only been examined in a few small studies.[965–968] L-Carnitine deficiency has been proposed as a cause of muscle cramping, but a meta-analysis of L-carnitine supplementation was inconclusive as to its benefit.[969] In the past, quinine was prescribed to treat intradialytic and nocturnal cramps. However, due to controversy over its efficacy and adverse events such as drug-induced thrombotic microangiopathy, QT prolongation, and hypersensitivity reactions, its use has largely been abandoned. The FDA has placed a black box warning against quinine use for muscle cramps and has removed it from the over-the-counter market.

DIALYSIS DISEQUILIBRIUM SYNDROME

Severe dialysis disequilibrium syndrome (DDS) is characterized by mental status changes, generalized seizures, and coma. Its occurrence has declined, in part because of increased awareness and earlier initiation of maintenance HD.[970,971] A milder form of DDS, manifesting with nausea and vomiting, headaches, fatigue, and restlessness, is still evident. Risk factors include first dialysis treatment, a markedly elevated predialysis BUN level, severe metabolic acidosis, extremes of age, and preexisting neurologic conditions.[970] However, there is no set BUN value above which patients predictably develop DDS nor a BUN value below which uremic patients are safe from developing DDS.

The pathogenesis for DDS is likely multifactorial, but the major suspect is the rapid reduction in solute levels over a relatively short time frame.[971,972] Animal studies have suggested that a transient urea concentration gradient may be created between plasma and the cerebrospinal fluid (CSF) during dialysis.[970,971] Uremia may potentiate this gradient by reducing urea transporter expression and increasing aquaporin expression in the brain.[973] Cerebral edema during dialysis may occur due to the resulting delay in urea egress and enhancement of water uptake into brain cells. In addition, provision of bicarbonate in the dialysate may lead to paradoxic acidosis in the CSF through diffusion of carbon dioxide across the blood-brain barrier, further compromising the ability of the brain to regulate solute and water transport.[970,971]

Timely initiation of dialysis and careful attention in crafting the initial dialysis prescription are the keys to preventing DDS. Several maneuvers can be used to reduce clearance and hence the risk for DDS during the first two or three HD treatments: (1) prescribing shorter treatment times of 1.5 to 2 hours; (2) lowering blood flow rates to 200 to 250 mL/min; (3) reducing dialysate flow rate and using concurrent rather than countercurrent flow; and (4) choosing a dialyzer with a small surface area. Increasing the dialysate sodium concentration or administering intravenous mannitol may also help prevent disequilibrium.[970,971] If an acute event with significant mental status changes or seizures occurs, immediate termination of treatment and the use of mannitol or hypertonic saline are recommended.

CARDIAC EVENTS

ARRHYTHMIAS, MYOCARDIAL STUNNING, AND DEATH

According to USRDS data, arrhythmia and cardiac arrest account for 40% of identifiable deaths among patients with ESKD, an alarmingly high number.[10] Underlying cardiac disease with LVH, coronary vascular disease, and disordered calcium and phosphate metabolism, with possible calcific deposits in the conduction system, predispose patients to arrhythmias and cardiac arrest during HD.[445] The high propensity for arrhythmias is partly attributable to the interdialytic accumulation and subsequent HD-induced removal of solute and fluid, especially when the removal rate exceeds diffusion out of the extravascular and intracellular compartments. Cardiac arrest events appear to cluster around the long interdialytic interval in patients undergoing three or fewer HD sessions/week (either Monday or Tuesday),[460,874] a pattern that is not seen with more frequent therapy or PD.[875]

Alterations in serum potassium during dialysis may be the major factor, although changes in Ca, magnesium, and pH may also be contributing factors (see "Dialysate Composition"). Large observational studies have shown associations between arrhythmia and low-potassium baths (<2 mEq/L).[461,522,974] Many patients in these studies were prescribed low potassium dialysate concentrations despite normal predialysis serum potassium levels, supporting the need to be vigilant in adjusting potassium prescriptions. No difference in risk was seen in a recent observational study comparing potassium baths of 2 versus 3 mEq/L.[520]

Cardiac arrest may be the most concerning occurrence in an outpatient HD clinic. One center has reported 102 cardiac arrests over a 14-year period, with the vast majority occurring during treatment; 72 of the episodes were related to ventricular tachycardia.[975] Although the availability of easy to use external defibrillators may have a positive impact on the outcomes of such arrests, the 1-year survival rate is poor at 15%.[975] An implantable cardioverter defibrillator (ICD) may reduce cardiac arrest rates in high-risk patients, but still leave HD patients with a 2.7-fold higher risk of death compared with nondialysis ICD recipients.[976,977] The use of ICDs for primary prevention does not appear to provide a survival benefit.[978,979] Furthermore, there is a risk of central venous stenosis and infectious complications from intravascular ICD leads.[980] Newer leadless devices, such as subcutaneous implantable and wearable external defibrillators, may be promising alternatives in the future.

In addition to ventricular arrhythmias, atrial arrhythmias may occur during HD. Atrial fibrillation is common in patients on HD, with a prevalence rate of over 20%.[10] Compared with the general population, patients with ESKD have a risk factor profile that could theoretically benefit from anticoagulation because of enhanced stroke risk.[981–984] However, patients on HD also experience more bleeding complications, making management decisions difficult and requiring an individualized risk-benefit analysis.[985–989] Most studies of anticoagulation in patients undergoing HD have evaluated the efficacy of warfarin and are observational.[982,983,990] The outcomes of patients treated with newer, non–vitamin K-dependent oral anticoagulants are presently being evaluated in clinical trials.

Finally, ischemia may occur during HD, even in the absence of chest pain, as demonstrated by the release of troponin during HD, the onset of regional left ventricular dysfunction on the echocardiogram, and reduced myocardial blood flow by positron emission tomography.[289,870,871,991,992] Ultimately, such changes in myocardial blood flow lead to myocardial stunning, and the HD procedure itself may contribute significantly to overall cardiac mortality.[12]

PERICARDIAL DISEASE

Pericardial disease is a well-recognized complication in patients with ESKD. Occasionally, pericarditis develops in patients with stage 5 CKD before they start dialysis, likely because of an inappropriate delay in therapy. Patients may present with typical findings of fever, pleuritic chest pain worse when recumbent, and a pericardial rub, although a fair proportion of patients may be asymptomatic and without physical findings.[993] Pericarditis that occurs within 8 weeks of initiation of dialysis is arbitrarily considered uremic disease, presumed to be due to the accumulation of as yet unidentified uremic toxins. Pericarditis in patients already on maintenance HD is likely multifactorial and not attributable to uremia because most of these patients are undergoing adequate and consistent dialysis. However, HD patients with a catheter as access, vascular access malfunction (e.g., stenosis at the venous anastomosis, with recirculation), larger body size, or poor adherence with treatment are at risk of having poor solute clearance and, hence, uremic pericarditis. Patients with uremic disease are managed with intensification of dialysis via longer or more frequent sessions.[994] Pericarditis in a well-dialyzed patient does not respond to intensifying dialysis and is likely related to another condition, such as a viral infection or autoimmune disease.[994] Complications of pericarditis in patients undergoing HD include cardiac tamponade and constrictive disease. Systemic anticoagulation, including during HD, should be avoided to reduce the risk of pericardial hemorrhage. Approaches to treating pericarditis that does not improve with intensification of dialysis range from antiinflammatory medication (both systemic and intrapericardial) to surgery when pericardial disease leads to hemodynamic compromise.[994]

REACTIONS TO DIALYZERS

Historically, reactions to hemodialyzer membranes or residual sterilants frequently occurred during the HD treatment. Dialyzer reactions were classified as types A and B.[201,995] Type A reactions occur within 5 to 20 minutes and present with pruritus, urticaria, bronchospasm, or anaphylactic shock. The classic "first use" syndrome, when a patient is exposed to a particular dialyzer for the first time, can resemble an anaphylactic episode and result in profound hypotension and death. These reactions are likely caused by IgE antibodies to membrane material or ethylene oxide, a sterilizing agent that is rarely used now (see also "Hemodialyzers" section).[996]

Type B reactions are complement-mediated, occur later in the dialysis session, and are associated with less severe symptoms, such as chest and back discomfort. The advent of more biocompatible membrane material, careful rinsing of the dialyzer before each session, and nonchemical methods such as the use of steam, gamma rays, and electron beams to sterilize dialyzers, has greatly reduced the frequency of these events. Newer synthetic membrane materials such as polyacrylonitrile and polysulfone cause less complement activation and are better tolerated than older cellulose membranes. However, one specific membrane, polyacrylonitrile (AN69), confers a unique risk when used in patients receiving ACE inhibitors. Bradykinin that is stimulated by AN69-induced activation of Hageman factor accumulates in the presence of ACE inhibitors (ACE degrades bradykinin) and predisposes to episodes of hypotension.[997,998]

The differential diagnosis for an allergic event during HD includes reactions to dialyzers, dialyzer sterilants or disinfectants (e.g., ethylene oxide, formaldehyde, bleach), or administered medications (e.g., heparin, intravenous iron).[201] Treatment of patients suspected of having a severe allergic reaction includes saline for hypotension, epinephrine for anaphylaxis or severe hypotension, urgent cessation of HD without blood return, and possibly corticosteroid use. Other complications that can mimic an allergic reaction should also be ruled out, such as exposure to contaminated water used to prepare the dialysate (see "Water Treatment" section), pyrogenic reaction, hemolysis, and air embolism.

OTHER COMPLICATIONS

When the dialyzer membrane separating blood from the dialysate is disrupted, the patient's blood is contaminated with nonsterile dialysate, and pyrogenic reactions may ensue. The HD machine is designed to detect such blood leaks, signal an alarm, and turn off the blood pump to stop dialysis. The dialysis staff must then test the dialysate directly to confirm the blood leak. Administration of cyanocobalamin may trigger a false blood leak alarm in some dialysis machines.[999,1000]

Hemolysis likely occurs to some degree during every HD treatment because of mechanical trauma to RBCs.[1001] More profound acute hemolysis during dialysis may result from mechanical problems such as defective dialysis tubing and roller pumps, with excessive RBC fragmentation, improper proportioning of the dialysate concentrate with osmotically induced hemolysis, overheating of the dialysate, and contamination of the dialysate with chemicals such as formaldehyde, bleach, chloramines, zinc, and copper.[201,1002] This degree of hemolysis will likely trigger the blood alarm, and patients may experience chest tightness, back pain, and shortness of breath, with acute pigmentation of the skin and port wine appearance of the blood in the venous line.[1003] In such an event, treatment should be discontinued immediately without returning the blood to the patient, and the serum potassium concentration should be determined. The diagnosis can be confirmed by assaying for free hemoglobin and examining the peripheral smear. The dialysate should be screened for its composition and contaminants, and the blood tubing should be inspected to determine the proximate cause.

Air embolism is another complication of the dialysis procedure that can lead to death unless quickly detected.[201] Air may enter the circuit due to a loose prepump connection or residual air left in the tubing or dialyzer from poor priming; this manifests as foam in the venous line. Fortunately, air detectors in HD machines have made air embolism a rare occurrence while on treatment, but air entry during disconnection may still occur, particularly with a catheter access. Immediate treatment of suspected air embolism includes stopping the blood pump, clamping the venous dialysis line to prevent further air entry, administering oxygen to reduce air bubble size, performing volume resuscitation, and placing the patient supine to preserve hemodynamic stability and reduce the potential for cerebral edema..[201,1004]

Hemorrhage during dialysis is an obvious risk, given the typical dialysis blood flow of 300 to 500 mL/min into an extracorporeal circuit. Mechanical events such as blood tubing disconnection and needle dislodgement are usually detected by safety technology in the HD machine (see "Components of the Extracorporeal Circuit" section). However, venous needle dislodgement may be unnoticed until significant loss of blood has occurred because venous pressure thresholds, and subsequent alarms, may vary with needle and tubing sizes, blood flow and viscosity, access characteristics, and patient positioning.[201] Proper taping of access needles, adequate tightening of connections, and loose looping of lines all help prevent such events. Exposure of the access during HD should be the practice in every clinic, increasing the likelihood that the dialysis staff will promptly identify and address bleeding from the vascular access or blood circuit.

Hypoglycemia is a rare event during HD because of the common use of dialysate with a glucose concentration of 200 mg/dL.[1005] When observed, it is often in a diabetic patient on an antiglycemic agent and warrants adjusting the dose or time of administration of the drug. Studies have suggested that dialysate glucose concentrations higher than 100 mg/dL predispose to hyperglycemia and increased vagal tone in patients with diabetes, possibly contributing to intradialytic hypotension.[562,563]

Complete reference list available at ExpertConsult.com.

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